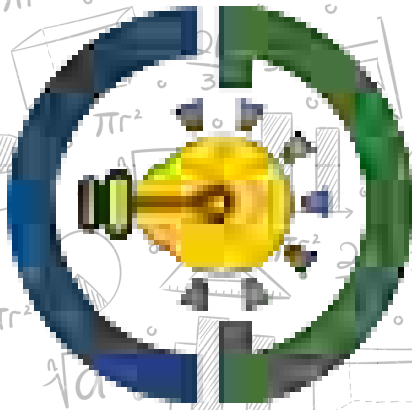


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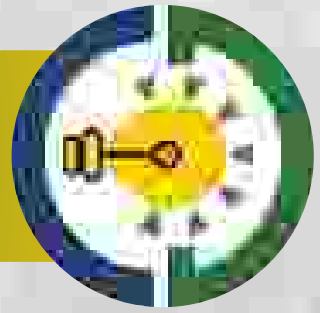
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More than 8 Years Experience

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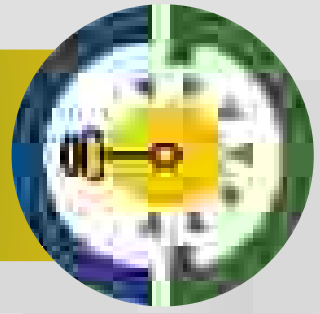


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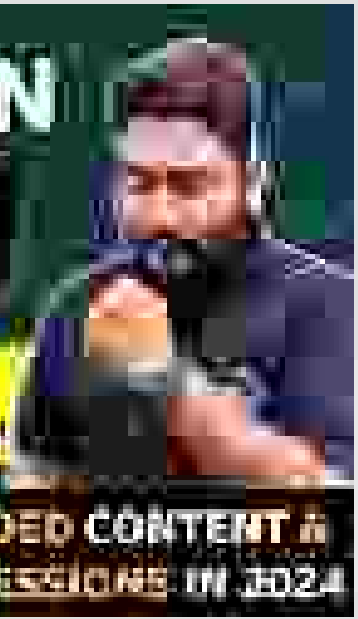
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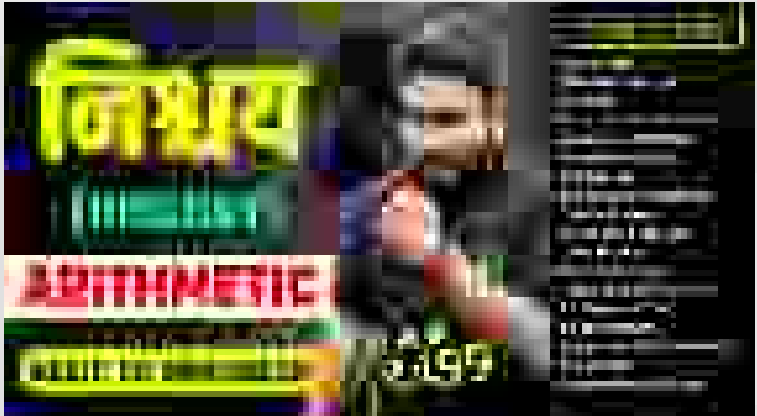


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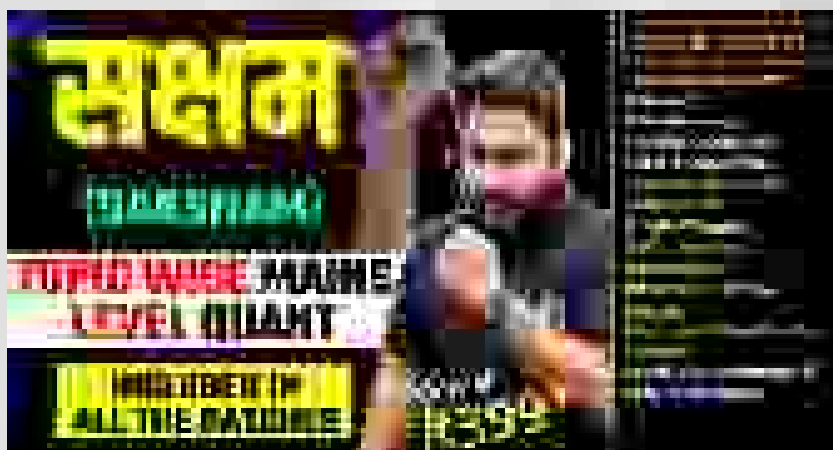
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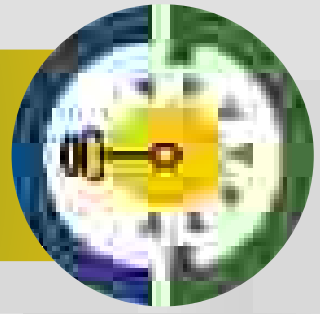
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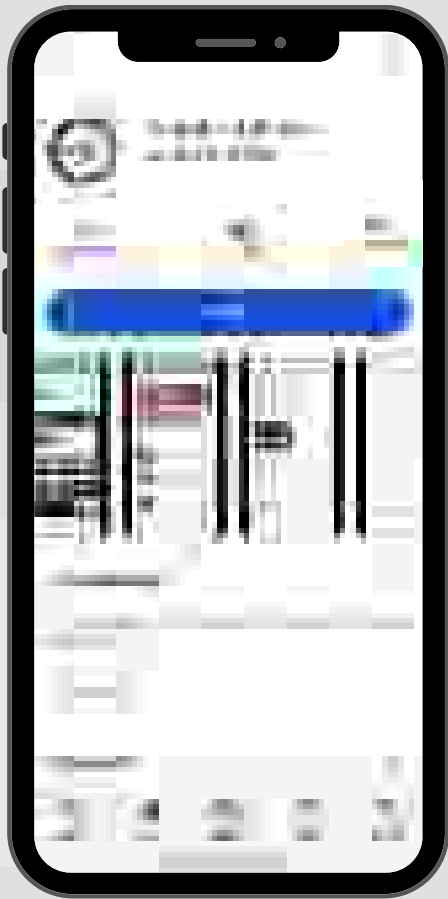


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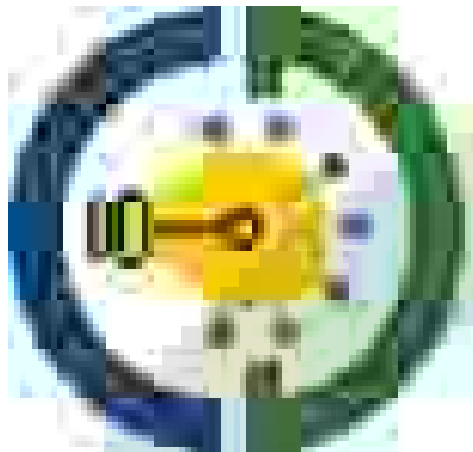


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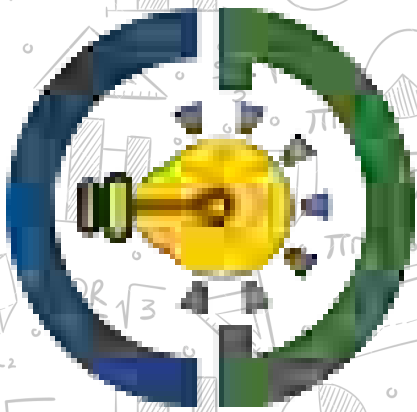


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REVISION SHEET (PART - 1)

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Time and Work



FOR SBI/IBPS/RBI/ALL OTHER BANKING EXAMS

- 1) Arjun and Priya together can complete a report in 24 days. Arjun alone can finish it in 40 days. How many days will Priya take to finish the report alone?
अर्जुन और प्रिया मिलकर एक रिपोर्ट 24 दिनों में पूरी कर सकते हैं। अर्जुन अकेले इसे 40 दिनों में पूरा कर सकता है। प्रिया को अकेले रिपोर्ट पूरी करने में कितने दिन लगेंगे?
a) 50 days
b) 60 days
c) 70 days
d) 80 days
e) None of these
- 2) Pranav and Qasim together can complete a certain project in 20 days. If Pranav alone can finish the same task in 36 days, then how many days would Qasim, working alone, take to complete the entire project?
प्रणव और कासिम मिलकर एक निश्चित प्रोजेक्ट को 20 दिनों में पूरा कर सकते हैं। यदि प्रणव अकेले उसी कार्य को 36 दिनों में पूरा कर सकता है, तो कासिम अकेले काम करते हुए पूरे प्रोजेक्ट को पूरा करने में कितने दिन लेगा?
a) 40 days
b) 48 days
c) 45 days
d) 36 days
e) None of these
- 3) In a construction project, Worker A and Worker B together complete the task in 20 days. If Worker A is 33.33% more efficient than Worker B, how many days will it take for Worker A alone to finish the work?
एक निर्माण परियोजना में, श्रमिक A और श्रमिक B मिलकर 20 दिनों में कार्य पूरा करते हैं। यदि श्रमिक A, श्रमिक B से 33.33% अधिक कुशल है, तो श्रमिक A को अकेले कार्य पूरा करने में कितने दिन लगेंगे?
a) 30 days
b) 46.67 days
c) 35 days
d) 30.33 days
e) None of these





Time and Work

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- 4) A person, Worker A, can complete 75% of a task in 24 days. Worker A is 50% more efficient than Worker B, and Worker C is 40% less efficient than Worker B. How many days will it take for Workers A and B together to finish the task?

एक व्यक्ति, कार्यकर्ता A, 24 दिनों में किसी कार्य का 75% पूरा कर सकता है। कार्यकर्ता A, कार्यकर्ता B से 50% अधिक कुशल है, और कार्यकर्ता C, कार्यकर्ता B से 40% कम कुशल है। कार्यकर्ता A और B को मिलकर कार्य पूरा करने में कितने दिन लगेंगे?

- a) $19\frac{1}{5}$ days
- b) 25 days
- c) $22\frac{1}{5}$ days
- d) 20 days
- e) None of these

- 5) Sanya and Kabir have efficiencies in the ratio 2:3. Together they complete a task in 20 days. How many days will Sanya take to do the work alone?

सान्या और कबीर की कार्यकुशलता का अनुपात 2:3 है। वे दोनों मिलकर एक काम 20 दिन में पूरा करते हैं। सान्या को अकेले उस काम को पूरा करने में कितने दिन लगेंगे?

- a) 30 days
- b) 40 days
- c) 50 days
- d) 60 days
- e) None of these

- 6) Workers A, B, and C together complete a task in 30 days, with their efficiency ratio being 3:5:6, respectively. How many days will it take for Worker B to complete double the task on their own?

श्रमिक A, B और C मिलकर एक कार्य को 30 दिनों में पूरा करते हैं, जिसमें उनकी दक्षता का अनुपात क्रमशः 3:5:6 है। श्रमिक B को अकेले दोगुना कार्य पूरा करने में कितने दिन लगेंगे?

- a) 165 days
- b) 168 days
- c) 170 days
- d) 120 days
- e) None of these





Time and Work

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- 7) Ananya, Raj, and Priya together can complete a task in 20 days. Ananya's work rate is four times Raj's, and Priya's work rate is 60% of the combined rates of Ananya and Raj. How many days will Raj alone take to complete the entire task?

अनन्या, राज और प्रिया मिलकर एक काम 20 दिन में पूरा कर सकते हैं। अनन्या की कार्य दर राज की कार्य दर से चार गुना है, और प्रिया की कार्य दर अनन्या और राज की संयुक्त दर का 60% है। राज को अकेले पूरा काम पूरा करने में कितने दिन लगेंगे?

- a) 140
- b) 160
- c) 180
- d) 200
- e) None of these

- 8) Raghav finishes half of a job in 5 days, while Deepika can complete one and a half times the same job in 10 days. If both of them work together, how many days will they take to finish 50% of the task?

राघव किसी काम का आधा हिस्सा 5 दिन में पूरा करता है, जबकि दीपिका उसी काम का डेढ़ गुना हिस्सा 10 दिन में पूरा कर सकती है। यदि वे दोनों साथ मिलकर काम करें, तो उन्हें 50% काम पूरा करने में कितने दिन लगेंगे?

- a) 1
- b) 2
- c) 3
- d) 4
- e) None of these

- 9) In 5 days, Linda finishes 30% of a task, and Mark finishes 20% of the same task. How many days will it take for them to complete the entire project if they work together?

5 दिनों में, लिंडा एक कार्य का 30% पूरा करती है, और मार्क उसी कार्य का 20% पूरा करता है। यदि वे एक साथ काम करते हैं, तो उन्हें पूरा प्रोजेक्ट पूरा करने में कितने दिन लगेंगे?

- a) 10
- b) 12
- c) 8
- d) 9
- e) None of these





Time and Work

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- 10) Aryan, Bhavesh, and Charan can finish a piece of work in 16, 12, and 15 days respectively. If Aryan and Bhavesh work on alternate days starting with Aryan, and Charan works every day, on which day will the work be completed?

आर्यन, भावेश और चरण किसी काम को क्रमशः 16, 12 और 15 दिनों में पूरा कर सकते हैं। यदि आर्यन और भावेश, आर्यन से शुरू करके वैकल्पिक दिनों पर काम करते हैं, और चरण हर दिन काम करता है, तो काम किस दिन पूरा होगा?

- a) 7th day
- b) 6th day
- c) 9th day
- d) 8th day
- e) None of these

- 11) Nakul and Rafi together can complete a task in $13\frac{1}{3}$ days, while Nakul alone can finish it in 24 days. If they work on alternate days starting with Nakul, in how many days will the work be completed?

नकुल और रफी मिलकर एक काम $13\frac{1}{3}$ दिन में पूरा कर सकते हैं, जबकि नकुल अकेले इसे 24 दिन में पूरा कर सकता है। यदि वे नकुल से शुरू करके वैकल्पिक दिनों पर काम करते हैं, तो काम कितने दिनों में पूरा होगा?

- a) 26 days
- b) 26.2 days
- c) 26.4 days
- d) 26.8 days
- e) None of these

- 12) Asha can complete a task in 12 days, Bharat can do it in 18 days, and Chitra can finish it in 24 days. If Asha and Bharat start working together and Chitra joins them every third day, how many days will it take to complete the work?

आशा एक काम को 12 दिनों में पूरा कर सकती है, भरत इसे 18 दिनों में पूरा कर सकता है, और चित्रा इसे 24 दिनों में पूरा कर सकती है। यदि आशा और भरत एक साथ काम करना शुरू करते हैं और चित्रा हर तीसरे दिन उनके साथ जुड़ती है, तो काम पूरा होने में कितने दिन लगेंगे?

- a) $(6 \frac{3}{5})$ days
- b) $(8 \frac{1}{4})$ days
- c) $(7 \frac{1}{2})$ days





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- d) 5.7 days
- e) None of these

- 13) John and Liam together can complete a project in 30 days, Liam and Sarah together can finish it in 15 days, and John and Sarah together can complete it in 40 days. How many days will it take for John, Liam, and Sarah to finish the project if they all work together?

जॉन और लियाम मिलकर एक प्रोजेक्ट को 30 दिनों में पूरा कर सकते हैं, लियाम और सारा मिलकर इसे 15 दिनों में पूरा कर सकते हैं, और जॉन और सारा मिलकर इसे 40 दिनों में पूरा कर सकते हैं। अगर जॉन, लियाम और सारा सभी एक साथ काम करें तो उन्हें प्रोजेक्ट पूरा करने में कितने दिन लगेंगे?

- a) 10 days
- b) 16 days
- c) 15 days
- d) 18 days
- e) None of these

- 14) Arjun can paint a house in 55 days, Bhavesh in 66 days, and together with Charan, they completed the task in 12 days. How many days would Charan take to paint the house alone?

अर्जुन एक घर को 55 दिनों में, भावेश 66 दिनों में पेंट कर सकता है, और चरण के साथ मिलकर वे यह काम 12 दिनों में पूरा कर लेते हैं। चरण को अकेले घर पेंट करने में कितने दिन लगेंगे?

- a) 15 days
- b) 18 days
- c) 20 days
- d) 25 days
- e) None of these

- 15) Asha and Bharat together can complete a task in 10 hours, Bharat and Chitra in 15 hours, Chitra and Deepak in 20 hours, and Deepak and Asha in 12 hours. After Asha, Bharat, Chitra, and Deepak worked together for 2.4 hours, Chitra and Deepak left. How much more time would Asha and Bharat take to finish the remaining work?





Time and Work

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आशा और भरत मिलकर एक काम 10 घंटे में पूरा कर सकते हैं, भरत और चित्रा 15 घंटे में, चित्रा और दीपक 20 घंटे में और दीपक और आशा 12 घंटे में। आशा, भरत, चित्रा और दीपक ने 2.4 घंटे साथ मिलकर काम किया, उसके बाद चित्रा और दीपक चले गए। आशा और भरत को बचा हुआ काम पूरा करने में कितना समय लगेगा?

- a) 6.4 hours
- b) 8.2 hours
- c) 10.5 hours
- d) 5.6 hours
- e) None of these

- 16) Two workers, Worker A and Worker B, together can complete a task in 25 days, and Worker B and Worker C together can complete the same task in 30 days. If Worker A is 66.67% more efficient than Worker C, how many days will it take for Workers A, B, and C together to finish the work?

दो कर्मचारी, कर्मचारी A और कर्मचारी B, मिलकर एक काम 25 दिनों में पूरा कर सकते हैं, और कर्मचारी B और कर्मचारी C मिलकर उसी काम को 30 दिनों में पूरा कर सकते हैं। यदि कर्मचारी A, कर्मचारी C से 66.67% अधिक कुशल है, तो कर्मचारी A, B और C को मिलकर काम पूरा करने में कितने दिन लगेंगे?

- a) 20 days
- b) 25 days
- c) 30 days
- d) Cannot be determined
- e) None of these

- 17) Ishita, Aditya, and Tanvi together can complete a certain task in 20 days. Ishita works four times as efficiently as Aditya, while Tanvi's work rate is 60% of the combined rate of Ishita and Aditya. Based on this information, in how many days would Aditya alone be able to complete the entire work?

इशिता, आदित्य और तन्वी मिलकर एक निश्चित कार्य को 20 दिनों में पूरा कर सकते हैं। इशिता आदित्य से चार गुना अधिक कुशलता से काम करती है, जबकि तन्वी की कार्य दर इशिता और आदित्य की संयुक्त दर का 60% है। इस जानकारी के आधार पर, आदित्य अकेले पूरे काम को कितने दिनों में पूरा कर पाएगा?

- a) 100
- b) 120
- c) 140
- d) 160





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e) None of these

- 18) Aarav can complete 50% of a manuscript in 20 days, while Aarav and Riya together finish the entire manuscript in 24 days. They work together for 12 days, then Riya alone completes the rest. How many days will Riya take?

आरव एक पांडुलिपि का 50% 20 दिनों में पूरा कर सकता है, जबकि आरव और रिया मिलकर पूरी पांडुलिपि 24 दिनों में पूरी करते हैं। वे 12 दिनों तक एक साथ काम करते हैं, फिर रिया अकेले बाकी काम पूरा करती है। रिया को कितने दिन लगेंगे?

- a) 25 days
- b) 30 days
- c) 35 days
- d) 40 days
- e) None of these

- 19) In a workshop, Anjali can complete a certain assignment in 8 days, while Bhavna can finish the same task alone in 12 days. When both of them work together along with Charu, the job gets completed in just 4 days. Further, it is known that Charu and Diya, working together, can finish the entire work in 15 days. How many days would Diya, working alone, take to complete the task?

एक कार्यशाला में, अंजलि एक निश्चित कार्य को 8 दिनों में पूरा कर सकती है, जबकि भावना अकेले उसी कार्य को 12 दिनों में पूरा कर सकती है। जब वे दोनों चारू के साथ मिलकर काम करती हैं, तो काम सिर्फ 4 दिनों में पूरा हो जाता है। इसके अलावा, यह ज्ञात है कि चारू और दीया मिलकर काम करते हुए पूरे काम को 15 दिनों में पूरा कर सकती हैं। दीया अकेले काम करते हुए उस काम को पूरा करने में कितने दिन लेगी?

- a) 30 days
- b) 36 days
- c) 40 days
- d) 38 days
- e) None of these

- 20) In a factory, Worker A can complete a task in 50 days, while Worker B can complete the same task in 20 days. When A, B, and C work together, they finish the task in 10 days. How many days will it take for Worker C to complete the task on their own?





Time and Work

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किसी फैक्ट्री में, कर्मचारी A किसी काम को 50 दिन में पूरा कर सकता है, जबकि कर्मचारी B उसी काम को 20 दिन में पूरा कर सकता है। जब A, B और C साथ मिलकर काम करते हैं, तो वे काम को 10 दिन में पूरा कर लेते हैं। कर्मचारी C को अकेले काम पूरा करने में कितने दिन लगेंगे?

- a) $33\frac{1}{3}$ days
- b) 30 days
- c) 45 days
- d) $30\frac{1}{3}$ days
- e) None of these

- 21) Anshul completes $\frac{2}{5}$ th of a job in 30 days, while Bhuvan finishes 83.33% of the same job in 30 days. Chetna can complete the entire job in x days. All three together can complete the task in 15 days. If Anshul and Chetna worked together for $(n + 7)$ days and Bhuvan worked alone for $(n + 19)$ days to complete the job, find the value of n .

अंशुल किसी काम का $\frac{2}{5}$ भाग 30 दिनों में पूरा करता है, जबकि भुवन उसी काम का 83.33% 30 दिनों में पूरा करता है। चेतना पूरा काम x दिनों में पूरा कर सकती है। तीनों मिलकर काम को 15 दिनों में पूरा कर सकते हैं। यदि अंशुल और चेतना ने मिलकर $(n + 7)$ दिन काम किया और भुवन ने अकेले काम पूरा करने के लिए $(n + 19)$ दिन काम किया, तो n का मान ज्ञात कीजिए।

- a) 6
- b) 5
- c) 4
- d) 3
- e) None of these

- 22) Mitali takes $(x + 5)$ days to finish a task, while Jay completes the same task in $2x$ days. When both worked together, they completed the task in $2\frac{2}{19}$ days. If Krish completes the task alone in $(x - 10)$ days, how long will it take for Mitali, Jay, and Krish to finish the work together?

मिताली को एक काम पूरा करने में $(x + 5)$ दिन लगते हैं, जबकि जय उसी काम को $2x$ दिनों में पूरा करता है। जब दोनों ने साथ मिलकर काम किया, तो उन्होंने काम को $2\frac{2}{19}$ दिनों में पूरा कर लिया। अगर कृष अकेले ही काम को $(x - 10)$ दिनों में पूरा करता है, तो मिताली, जय और कृष को मिलकर काम पूरा करने में कितना समय लगेगा?

- a) 12.5 days
- b) 18.5 days





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- c) 10.5 days
- d) 17.5 days
- e) None of these

23) Simran can complete a coding task alone in 50 days. Simran and Karthik together can finish the same task in 30 days. They begin the work together, but Simran leaves after 10 days. How many additional days will Karthik take to finish the remaining work alone?

सिमरन अकेले एक कोडिंग कार्य 50 दिनों में पूरा कर सकती है। सिमरन और कार्तिक मिलकर उसी कार्य को 30 दिनों में पूरा कर सकते हैं। वे साथ मिलकर काम शुरू करते हैं, लेकिन सिमरन 10 दिन बाद काम छोड़ देती है। कार्तिक को अकेले बचा हुआ काम पूरा करने में कितने अतिरिक्त दिन लगेंगे?

- a) 40 days
- b) 45 days
- c) 50 days
- d) 55 days
- e) None of these

24) Vikram can finish a project in 20 days, while Karan can do the same in 30 days. They start working on the project together, but Vikram leaves 5 days before the project is completed. How many total days will it take to complete the project?

विक्रम एक प्रोजेक्ट को 20 दिनों में पूरा कर सकता है, जबकि करण उसी प्रोजेक्ट को 30 दिनों में पूरा कर सकता है। वे दोनों एक साथ प्रोजेक्ट पर काम करना शुरू करते हैं, लेकिन विक्रम प्रोजेक्ट पूरा होने से 5 दिन पहले ही काम छोड़ देता है। प्रोजेक्ट को पूरा करने में कुल कितने दिन लगेंगे?

- a) 15
- b) 16
- c) 18
- d) 20
- e) None of these

25) Nina and Vikram together can complete a task in 12 days. However, Nina had to leave a few days before the task was completed, and the task took 16 days to finish. If Nina could complete the entire task alone in 21 days, how many days before the task was completed did Nina leave?





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नीना और विक्रम मिलकर एक काम 12 दिनों में पूरा कर सकते हैं। हालाँकि, नीना को काम पूरा होने से कुछ दिन पहले ही काम छोड़ना पड़ा और काम पूरा होने में 16 दिन लग गए। अगर नीना अकेले ही पूरा काम 21 दिनों में पूरा कर सकती है, तो नीना ने काम पूरा होने से कितने दिन पहले काम छोड़ा था?

- a) 7 days
- b) 9 days
- c) 8 days
- d) 6 days
- e) None of these

- 26) Manish can complete the entire project alone in 20 days, Tarun in 30 days, and Sneha in 60 days. All three start working together, but Manish leaves the team 8 days before the project gets completed. Tarun continues for another 5 days after Manish quits. The remaining portion is finished by Sneha alone. How many total days did it take to complete the entire project?

मनीष अकेले पूरे प्रोजेक्ट को 20 दिन में, तरुण 30 दिन में और स्नेहा 60 दिन में पूरा कर सकता है। तीनों एक साथ काम करना शुरू करते हैं, लेकिन मनीष प्रोजेक्ट पूरा होने से 8 दिन पहले टीम छोड़ देता है। मनीष के छोड़ने के बाद तरुण 5 दिन और काम जारी रखता है। शेष भाग स्नेहा अकेले पूरा करती है। पूरे प्रोजेक्ट को पूरा करने में कुल कितने दिन लगे?

- a) 18
- b) 20
- c) 15
- d) 12
- e) None of these

- 27) Aditya, Bharat, and Chandan can complete a task individually in 60, 100, and 120 days respectively. They all started the work together. Chandan left after 6 days, and 6 days later, Bharat also left. The rest of the work was completed by Aditya alone. What percentage of the total work was completed by Aditya?

आदित्य, भरत और चंदन एक काम को अलग-अलग क्रमशः 60, 100 और 120 दिनों में पूरा कर सकते हैं। वे सभी एक साथ काम शुरू करते हैं। चंदन 6 दिन बाद काम छोड़ देता है और 6 दिन बाद भरत भी काम छोड़ देता है। बाकी काम आदित्य ने अकेले पूरा किया। आदित्य ने कुल काम का कितना प्रतिशत पूरा किया?

- a) 80%





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- b) 82%
- c) 83%
- d) 85%
- e) None of these

28) Naman can complete $\frac{4}{5}$ of a task in 20 days, Raghav can complete $\frac{4}{15}$ of the task in x days, and Chetan can complete $\frac{2}{5}$ of the task in $x - 2$ days. Together, they complete $\frac{7}{10}$ of the work in 6 days. In how many days can Naman and Raghav together complete the entire work?

नमन किसी कार्य का $\frac{4}{5}$ भाग 20 दिन में पूरा कर सकता है, राघव कार्य का $\frac{4}{15}$ भाग x दिन में पूरा कर सकता है, और चेतन कार्य का $\frac{2}{5}$ भाग $x - 2$ दिन में पूरा कर सकता है। साथ में, वे 6 दिन में कार्य का $\frac{7}{10}$ भाग पूरा करते हैं। नमन और राघव मिलकर पूरा कार्य कितने दिन में पूरा कर सकते हैं?

- a) 14 days
- b) 15 days
- c) 16 days
- d) 18 days
- e) None of these

29) Rahul, Suresh, and Tapan can complete a piece of work alone in 24, 30, and 40 days respectively. All three began the work together, but Suresh left after 8 days, and Rahul left 8 days before the work was completed. How many total days did it take to finish the work?

राहुल, सुरेश और तपन अकेले एक काम को क्रमशः 24, 30 और 40 दिनों में पूरा कर सकते हैं। तीनों ने मिलकर काम शुरू किया, लेकिन सुरेश ने 8 दिन बाद काम छोड़ दिया और राहुल ने काम पूरा होने से 8 दिन पहले काम छोड़ दिया। काम पूरा करने में कुल कितने दिन लगे?

- a) 14 days
- b) 15 days
- c) 16 days
- d) 17 days
- e) None of these





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30) Neeraj, Om, and Prateek can complete a certain task in 15, 12, and 14 days respectively. They all started the work together. Neeraj left after $(x + 1)$ days, and Om left $(4x/3)$ days before the work got completed. If the entire task was finished in $(3x - 2)$ days, what is the value of $(x^2 + 25)$?

नीरज, ओम और प्रतीक एक निश्चित कार्य को क्रमशः 15, 12 और 14 दिनों में पूरा कर सकते हैं। वे सभी एक साथ काम शुरू करते हैं। नीरज $(x + 1)$ दिन बाद काम छोड़ देता है, और ओम काम पूरा होने से $(4x/3)$ दिन पहले काम छोड़ देता है। यदि पूरा काम $(3x - 2)$ दिनों में समाप्त हो जाता है, तो $(x^2 + 25)$ का मान क्या है?

- a) 36
- b) 33
- c) 34
- d) 32
- e) None of these

31) Anaya and Kabir together can complete a certain task in 7.5 days. Kabir alone can finish half of the task in 10 days. Both Anaya and Kabir worked together for 5 days, after which Kabir left the work. How many more days will Anaya take to finish the remaining part of the task alone?

अनाया और कबीर मिलकर एक निश्चित कार्य 7.5 दिनों में पूरा कर सकते हैं। कबीर अकेले कार्य का आधा भाग 10 दिनों में पूरा कर सकता है। अनाया और कबीर दोनों ने 5 दिनों तक एक साथ काम किया, जिसके बाद कबीर ने काम छोड़ दिया। अनाया को अकेले कार्य का शेष भाग पूरा करने में कितने दिन लगेंगे?

- a) 9 days
- b) 4 days
- c) 5 days
- d) 6 days
- e) None of these

32) Ritika and Neha together can finish a task in 20 days. They both worked jointly for 4 days, after which Neha had to leave. Ritika then continued the remaining work on her own and completed it in 24 more days. How many days would Neha alone take to finish the entire task if she worked independently?





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रितिका और नेहा मिलकर एक काम 20 दिन में पूरा कर सकती हैं। वे दोनों 4 दिन तक साथ-साथ काम करती रहीं, जिसके बाद नेहा को काम छोड़ना पड़ा। रितिका ने फिर बचा हुआ काम अकेले ही जारी रखा और उसे 24 दिन में पूरा कर लिया। अगर नेहा अकेले काम करे तो उसे पूरा काम पूरा करने में कितने दिन लगेंगे?

- a) 45 days
- b) 40 days
- c) 50 days
- d) 60 days
- e) None of these

- 33) Amit and Karan can finish a task in 10 days and 15 days, respectively. They both start working together, but after 4 days, Amit leaves. Karan continues to work and completes the remaining task. How many days in total will it take to complete the work?

अमित और करण किसी काम को क्रमशः 10 दिन और 15 दिन में पूरा कर सकते हैं। वे दोनों साथ मिलकर काम करना शुरू करते हैं, लेकिन 4 दिन बाद अमित काम छोड़ देता है। करण काम करना जारी रखता है और बचा हुआ काम पूरा करता है। काम पूरा करने में कुल कितने दिन लगेंगे?

- a) 8 days
- b) 9 days
- c) 10 days
- d) 11 days
- e) None of these

- 34) Raj, Priya, and Simran take time to complete a task in the ratio of 5:4:2. Raj began working alone, and after 5 days, Priya joined him for 4 days. Following that, Priya was replaced by Simran, and Raj and Simran together completed the remaining task in $3\frac{1}{7}$ days. How many days would Simran need to finish the task on her own?

राज, प्रिया और सिमरन एक काम को 5:4:2 के अनुपात में पूरा करने में समय लेते हैं। राज ने अकेले काम करना शुरू किया और 5 दिन बाद, प्रिया 4 दिन के लिए उसके साथ जुड़ गई। उसके बाद, प्रिया की जगह सिमरन ने ले ली और राज और सिमरन ने मिलकर बचा हुआ काम $3\frac{1}{7}$ दिन में पूरा कर लिया। सिमरन को अकेले काम पूरा करने में कितने दिन लगेंगे?

- a) 20 days
- b) 10 days





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- c) 25 days
- d) 5 days
- e) 15 days

35) Vikram can complete a task in 24 days, and Meera can complete it in 32 days. Vikram worked alone for 6 days, then Meera joined, and both worked together for 8 days. After that, Meera was replaced by Priya, who is twice as efficient as Meera. How many days did it take to finish the work?

विक्रम एक काम को 24 दिनों में पूरा कर सकता है, और मीरा इसे 32 दिनों में पूरा कर सकती है। विक्रम ने 6 दिनों तक अकेले काम किया, फिर मीरा शामिल हो गई, और दोनों ने 8 दिनों तक साथ काम किया। उसके बाद, मीरा की जगह प्रिया ने ले ली, जो मीरा से दोगुनी कुशल है। काम खत्म करने में कितने दिन लगे?

- a) $36/5$ days
- b) $72/5$ days
- c) $73/5$ days
- d) $78/5$ days
- e) None of these

36) Ankur, Meera, and Sunita are working together to complete a task. Ankur can finish the work in 40 days, Meera in 20 days, and Sunita in 60 days. They all started working together, but Meera left 2 days before the task was completed. How many days did it take to finish the work?

अंकुर, मीरा और सुनीता एक साथ मिलकर एक काम पूरा कर रहे हैं। अंकुर उस काम को 40 दिन में, मीरा 20 दिन में और सुनीता 60 दिन में पूरा कर सकती है। वे सभी एक साथ काम करना शुरू करते हैं, लेकिन मीरा काम पूरा होने से 2 दिन पहले ही काम छोड़कर चली जाती है। काम पूरा करने में कितने दिन लगे?

- a) 10 days
- b) 12 days
- c) 15 days
- d) 5 days
- e) None of these

37) Rajesh, Priya, and Anil are working together to complete a task. Rajesh can finish the job in 30 days, Priya in 60 days, and Anil in 90 days. They started the work together, but Anil left 9 days before the work was finished. How many days did Rajesh, Priya, and Anil work together before Anil left?





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राजेश, प्रिया और अनिल एक साथ मिलकर एक काम पूरा कर रहे हैं। राजेश उस काम को 30 दिन में, प्रिया 60 दिन में और अनिल 90 दिन में पूरा कर सकता है। उन्होंने साथ मिलकर काम शुरू किया, लेकिन अनिल काम खत्म होने से 9 दिन पहले ही काम छोड़ कर चला गया। अनिल के जाने से पहले राजेश, प्रिया और अनिल ने कितने दिन साथ मिलकर काम किया?

- a) 18 days
- b) 12 days
- c) 15 days
- d) 9 days
- e) None of these

- 38) Arun can complete a task in 24 days, while Bina can do it in 48 days. Arun and Bina worked together for 10 days, and the remaining work was finished in 3 days with the help of Chitra. Find how long it would take Chitra to finish the entire work on her own.

अरुण एक काम को 24 दिनों में पूरा कर सकता है, जबकि बीना इसे 48 दिनों में पूरा कर सकती है। अरुण और बीना ने 10 दिनों तक एक साथ काम किया, और चित्रा की मदद से शेष काम 3 दिनों में पूरा हो गया। चित्रा को अकेले पूरा काम पूरा करने में कितना समय लगेगा, यह पता करें।

- a) 36 days
- b) 12 days
- c) 15 days
- d) 16 days
- e) None of these

- 39) Pooja can complete a certain project on her own in 20 days. Her colleagues Tarun and Bhavna take time to complete the same work in the ratio 5:6, while Bhavna and Deepak take time in the ratio 1:2. All four of them — Pooja, Tarun, Bhavna, and Deepak — worked together for 5 days. After that, Pooja left the team, and the remaining work was completed in 3 more days by the others. How long would Tarun and Bhavna together take to complete the whole work?





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पूजा अकेले ही एक निश्चित प्रोजेक्ट को 20 दिनों में पूरा कर सकती है। उसके सहकर्मी तरुण और भावना उसी काम को पूरा करने में 5:6 के अनुपात में समय लेते हैं, जबकि भावना और दीपक 1:2 के अनुपात में समय लेते हैं। चारों - पूजा, तरुण, भावना और दीपक - ने 5 दिनों तक एक साथ काम किया। उसके बाद, पूजा ने टीम छोड़ दी, और शेष काम अन्य लोगों द्वारा 3 और दिनों में पूरा किया गया। तरुण और भावना को मिलकर पूरा काम पूरा करने में कितना समय लगेगा?

- a) $13 \frac{1}{11}$ days
- b) $13 \frac{1}{13}$ days
- c) $12 \frac{1}{11}$ days
- d) $11 \frac{1}{11}$ days
- e) None of these

- 40) Kavya is twice as efficient as her colleague Harsh. Another team member, Manish, takes the same amount of time to complete the work as Kavya and Harsh would take together. Harsh and Manish started doing the work together and continued for 3 days. After that, Manish left and Kavya joined Harsh to finish the remaining work. If the remaining work was completed in $3\frac{3}{4}$ days, how many days would Manish alone take to complete the entire work?

काव्या अपने सहकर्मी हर्ष से दोगुनी कुशल है। टीम का एक अन्य सदस्य मनीष, काम पूरा करने में उतना ही समय लेता है जितना काव्या और हर्ष मिलकर लेते। हर्ष और मनीष ने साथ मिलकर काम करना शुरू किया और 3 दिन तक करते रहे। उसके बाद, मनीष चला गया और काव्या हर्ष के साथ मिलकर बचा हुआ काम पूरा करने लगी। यदि बचा हुआ काम $3\frac{3}{4}$ दिन में पूरा हो जाता है, तो मनीष को अकेले पूरा काम पूरा करने में कितने दिन लगेंगे?

- a) 6 days
- b) 7.5 days
- c) 5 days
- d) 10 days
- e) None of these

- 41) A team was assigned a construction task. Nisha alone could finish the entire job in 20 days. She started working along with her teammates Mohan and Priya. The efficiency ratio of Mohan to Priya was 2:5. All three worked together for 4 days. After that, Priya left due to other commitments, and the remaining work was completed by Nisha and Mohan in $2\frac{2}{3}$ days. How many days would Mohan alone take to complete the entire work?





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एक टीम को एक निर्माण कार्य सौंपा गया था। निशा अकेले ही पूरे काम को 20 दिनों में पूरा कर सकती थी। उसने अपने साथियों मोहन और प्रिया के साथ काम करना शुरू किया। मोहन और प्रिया की दक्षता का अनुपात 2:5 था। तीनों ने 4 दिनों तक एक साथ काम किया। उसके बाद, प्रिया अन्य प्रतिबद्धताओं के कारण चली गई, और शेष काम निशा और मोहन ने $2\frac{2}{3}$ दिनों में पूरा किया। मोहन को अकेले पूरा काम पूरा करने में कितने दिन लगेंगे?

- a) 15 days
- b) 10 days
- c) 25 days
- d) 30 days
- e) None of these

42) The time taken by workers A, B, and C individually to complete a task is in the ratio of 10:6:5, respectively. How much more or less efficient is A compared to C?

श्रमिक A, B और C द्वारा किसी कार्य को पूरा करने में लिया गया समय क्रमशः 10:6:5 के अनुपात में है। C की तुलना में A कितना अधिक या कम कुशल है?

- a) 50%
- b) 20%
- c) 25%
- d) 30%
- e) None of these

43) The time taken by workers A, B, and C to complete a task is in the ratio of 10:6:5, respectively. If B alone can complete the work in 30 days, how long will it take for A, B, and C together to finish the work?

किसी काम को पूरा करने में A, B और C द्वारा लिया गया समय क्रमशः 10:6:5 के अनुपात में है। यदि B अकेले काम को 30 दिनों में पूरा कर सकता है, तो A, B और C को मिलकर काम पूरा करने में कितना समय लगेगा?

- a) 15 days
- b) $10\frac{5}{7}$ days
- c) $11\frac{1}{7}$ days
- d) 11 days
- e) None of these





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- 44) Aarti is 45% more efficient than Simran. It is given that Simran requires 63 additional days than Aarti to complete the same task. How many days would Aarti need to complete the work if she were working independently?
आरती सिमरन से 45% अधिक कुशल है। यह दिया गया है कि सिमरन को उसी कार्य को पूरा करने के लिए आरती की तुलना में 63 अतिरिक्त दिन चाहिए। यदि आरती स्वतंत्र रूप से काम कर रही होती तो उसे काम पूरा करने में कितने दिन लगते?
- a) 120 days
 - b) 130 days
 - c) 140 days
 - d) 150 days
 - e) None of these
- 45) A farmer, Anil, can plow a field in 20 days. If his helper, Bharat, is 14.28% more efficient than Anil, how many days will it take for both Anil and Bharat to finish the work together?
एक किसान, अनिल, एक खेत को 20 दिनों में जोत सकता है। यदि उसका सहायक, भरत, अनिल से 14.28% अधिक कुशल है, तो अनिल और भरत दोनों को मिलकर काम पूरा करने में कितने दिन लगेंगे?
- a) $9\frac{1}{3}$ days
 - b) $10\frac{1}{3}$ days
 - c) $11\frac{1}{3}$ days
 - d) Cannot be determined
 - e) None of these
- 46) The efficiency ratio of three workers, John, Paul, and George, is 2:3:4. John takes 2 more days than Paul to finish the same task alone. How many days will it take for John and Paul to complete the task together?
तीन श्रमिकों, जॉन, पॉल और जॉर्ज की दक्षता का अनुपात 2:3:4 है। जॉन को अकेले उसी कार्य को पूरा करने में पॉल से 2 दिन अधिक लगते हैं। जॉन और पॉल को मिलकर कार्य पूरा करने में कितने दिन लगेंगे?
- a) 2.5 days
 - b) 3 days
 - c) 2.4 days
 - d) 2.6 days
 - e) None of these





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- 47) Tarun can complete one-seventh of a project in 5 days, while Charu can finish the entire project in 21 days. They start working together, but Charu works at only 80% of her full efficiency. In how many days will they complete the entire project?

तरुण किसी प्रोजेक्ट का $\frac{1}{7}$ भाग 5 दिन में पूरा कर सकता है, जबकि चारू पूरे प्रोजेक्ट को 21 दिन में पूरा कर सकती है। वे साथ मिलकर काम करना शुरू करते हैं, लेकिन चारू अपनी पूरी क्षमता के केवल 80% पर काम करती है। वे पूरे प्रोजेक्ट को कितने दिन में पूरा करेंगे?

- a) 12 days
- b) 14 days
- c) 15 days
- d) 16 days
- e) None of these

- 48) Two engineers, Arjun and Karthik, have work efficiencies in the ratio 7:10. If Arjun alone can complete the construction of a pavilion in 84 days, how many days would Karthik take to complete $\frac{5}{7}$ of the same work on his own?

दो इंजीनियरों, अर्जुन और कार्तिक की कार्य कुशलता का अनुपात 7:10 है। यदि अर्जुन अकेले एक मंडप का निर्माण 84 दिनों में पूरा कर सकता है, तो कार्तिक को अकेले उसी कार्य का $\frac{5}{7}$ भाग पूरा करने में कितने दिन लगेंगे?

- a) 36
- b) 40
- c) 42
- d) 48
- e) None of these

- 49) A construction project can be completed in 5 days when Prateek and Karan work together. Prateek can finish the work alone in 30 days, and Karan can do it alone in 6 days. A third person, Suraj, is 20% more efficient than Prateek. How many days will Suraj alone need to complete the same project?

एक निर्माण परियोजना 5 दिनों में पूरी हो सकती है जब प्रतीक और करण एक साथ काम करते हैं। प्रतीक अकेले काम को 30 दिनों में पूरा कर सकता है, और करण अकेले इसे 6 दिनों में पूरा कर सकता है। तीसरा व्यक्ति, सूरज, प्रतीक से 20% अधिक कुशल है। सूरज को अकेले उसी परियोजना को पूरा करने में कितने दिन लगेंगे?





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- a) 24
- b) 25
- c) 28
- d) 26
- e) None of these

50) At a factory site, Ramesh observed that a man can complete twice as much work in a day as a woman. A small team consisting of 3 men and 2 women manages to complete a task in 2 days. Based on this, how many days would it take for a single woman, working alone, to finish the same task?

एक फैक्ट्री साइट पर, रमेश ने देखा कि एक पुरुष एक दिन में एक महिला की तुलना में दोगुना काम पूरा कर सकता है। 3 पुरुषों और 2 महिलाओं की एक छोटी टीम 2 दिनों में एक काम पूरा करने में कामयाब हो जाती है। इसके आधार पर, अकेले काम करने वाली एक महिला को उसी काम को पूरा करने में कितने दिन लगेंगे?

- a) 12 days
- b) 18 days
- c) 14 days
- d) 16 days
- e) None of these

51) Neeraj can complete a certain task on his own in Y days, while Shahid can complete the same task alone in $(Y + 5)$ days. It is known that Shahid is 20% less efficient than Ravindra. If Ravindra can finish the work alone in exactly 20 days, then in how many days can Neeraj, Shahid, and Ravindra together complete 60% of the entire task?

नीरज अकेले एक निश्चित कार्य को Y दिनों में पूरा कर सकता है, जबकि शाहिद अकेले उसी कार्य को $(Y + 5)$ दिनों में पूरा कर सकता है। यह ज्ञात है कि शाहिद रविंद्र से 20% कम कुशल है। यदि रविंद्र अकेले काम को ठीक 20 दिनों में पूरा कर सकता है, तो नीरज, शाहिद और रविंद्र मिलकर पूरे कार्य का 60% कितने दिनों में पूरा कर सकते हैं?

- a) 40/9 days
- b) 30/7 days
- c) 25/6 days
- d) 35/8 days
- e) None of these





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- 52) Arjun and Kabir together can complete a certain task in $3\frac{1}{13}$ days. When Arjun, Kabir, and Ritesh work together, they finish the same task in 2 and $\frac{6}{17}$ days. If Ritesh's efficiency is increased by 50%, in how many days can Ritesh alone complete the task?

अर्जुन और कबीर मिलकर एक निश्चित कार्य को 3 और $\frac{1}{13}$ दिनों में पूरा कर सकते हैं। जब अर्जुन, कबीर और रितेश साथ मिलकर काम करते हैं, तो वे उसी कार्य को 2 और $\frac{6}{17}$ दिनों में पूरा करते हैं। यदि रितेश की कार्यकुशलता 50% बढ़ जाती है, तो रितेश अकेले उस कार्य को कितने दिनों में पूरा कर सकता है?

- a) $6\frac{2}{3}$ days
- b) $7\frac{1}{3}$ days
- c) $5\frac{1}{3}$ days
- d) $8\frac{2}{3}$ days
- e) None of these

- 53) Harshil can complete $\frac{5}{6}$ of a large landscaping project in 25 days. The efficiency ratio between Harshil and Rudra is 3:10. Nakul can complete 80% of the same project in 14.4 days. How many days will Rudra and Nakul together take to complete the entire project?

हर्षिल एक बड़े लैंडस्केपिंग प्रोजेक्ट का $\frac{5}{6}$ भाग 25 दिनों में पूरा कर सकता है। हर्षिल और रुद्र के बीच दक्षता अनुपात 3:10 है। नकुल उसी प्रोजेक्ट का 80% 14.4 दिनों में पूरा कर सकता है। रुद्र और नकुल को मिलकर पूरा प्रोजेक्ट पूरा करने में कितने दिन लगेंगे?

- a) 9 days
- b) 10 days
- c) 5 days
- d) 6 days
- e) None of these

- 54) Anish can complete 75% of a task in 15 days. The efficiency ratio of Anish and Bharat is 3:2. How long will it take for Anish and Bharat to complete 25% of the task together?

अनीश किसी कार्य का 75% 15 दिनों में पूरा कर सकता है। अनीश और भरत की कार्यदक्षता का अनुपात 3:2 है। अनीश और भरत को मिलकर कार्य का 25% पूरा करने में कितना समय लगेगा?

- a) 3 days
- b) 2 days
- c) 5 days





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- d) 6 days
- e) 8 days

55) Ravi's efficiency is 25% lower than Priya's efficiency, and Karan's efficiency is half of Priya's efficiency. If Priya, Ravi, and Karan together can complete the work in 8 days, how many days will it take for Priya alone to finish the work?

रवि की कार्यकुशलता प्रिया की कार्यकुशलता से 25% कम है, तथा करण की कार्यकुशलता प्रिया की कार्यकुशलता की आधी है। यदि प्रिया, रवि और करण मिलकर काम को 8 दिनों में पूरा कर सकते हैं, तो प्रिया को अकेले काम पूरा करने में कितने दिन लगेंगे?

- a) 24 days
- b) 20 days
- c) 16 days
- d) 15 days
- e) 18 days

56) Ajay can complete $\frac{1}{4}$ of a project in 6 days. He is 50% more efficient than Bharat, and Chitra is 20% more efficient than Bharat. How long will Ajay, Bharat, and Chitra take to complete the entire project working together?

अजय किसी प्रोजेक्ट का $\frac{1}{4}$ भाग 6 दिन में पूरा कर सकता है। वह भरत से 50% अधिक कुशल है, और चित्रा भरत से 20% अधिक कुशल है। अजय, भरत और चित्रा को एक साथ काम करते हुए पूरा प्रोजेक्ट पूरा करने में कितना समय लगेगा?

- a) $\frac{120}{43}$ days
- b) $\frac{360}{37}$ days
- c) $\frac{420}{37}$ days
- d) $\frac{240}{17}$ days
- e) $\frac{180}{31}$ days

57) Aniket and Shruti can each complete a software module in teams of 9 developers or 12 testers respectively, taking 20 days. If 6 developers and 8 testers work together, how many days will they take to complete the task?

अनिकेत और श्रुति 9 डेवलपर्स या 12 टेस्टर्स की टीमों में एक सॉफ्टवेयर मॉड्यूल को 20 दिनों में पूरा कर सकते हैं। यदि 6 डेवलपर्स और 8 टेस्टर्स एक साथ काम करते हैं, तो उन्हें कार्य पूरा करने में कितने दिन लगेंगे?

- a) 12 days
- b) 14 days





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- c) 15 days
- d) 18 days
- e) None of these

58) 15 researchers are working 4 hours a day and complete an experiment in 12 days. If 5 more researchers join the team and all researchers work 3 hours a day, how many days will it take to complete the same experiment?

15 शोधकर्ता प्रतिदिन 4 घंटे काम करते हैं और 12 दिनों में एक प्रयोग पूरा करते हैं। यदि 5 और शोधकर्ता टीम में शामिल हो जाते हैं और सभी शोधकर्ता प्रतिदिन 3 घंटे काम करते हैं, तो उसी प्रयोग को पूरा करने में कितने दिन लगेंगे?

- a) 12 days
- b) 14 days
- c) 16 days
- d) 18 days
- e) None of these

59) If 12 gardeners, working 8 hours a day, can plant 46 trees in 18 days, how many trees will 9 gardeners plant in 8 days, each working 12 hours a day? यदि 12 माली, प्रतिदिन 8 घंटे काम करके, 18 दिनों में 46 पेड़ लगा सकते हैं, तो 9 माली, प्रतिदिन 12 घंटे काम करके, 8 दिनों में कितने पेड़ लगाएंगे?

- a) 22
- b) 27
- c) 23
- d) 34
- e) None of these

60) A team of 18 workers can finish a project in 48 days by working 8 hours each day. How many workers will be needed to complete half of the project in 24 days by working 12 hours per day?

18 श्रमिकों की एक टीम प्रतिदिन 8 घंटे काम करके 48 दिनों में एक परियोजना पूरी कर सकती है। प्रतिदिन 12 घंटे काम करके परियोजना का आधा हिस्सा 24 दिनों में पूरा करने के लिए कितने श्रमिकों की आवश्यकता होगी?

- a) 12
- b) 15
- c) 18
- d) 24





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e) None of these

61) A team of 6 men and 9 women can complete a certain task in 2 days. On the other hand, 3 men and 3 women can finish the same work in $24/5$ days. How much time will it take for 4 men to complete the task on their own?

6 पुरुषों और 9 महिलाओं की एक टीम एक निश्चित कार्य को 2 दिनों में पूरा कर सकती है। दूसरी ओर, 3 पुरुष और 3 महिलाएं उसी कार्य को $24/5$ दिनों में पूरा कर सकते हैं। 4 पुरुषों को अकेले उस कार्य को पूरा करने में कितना समय लगेगा?

- a) 9 days
- b) 10 days
- c) 5 days
- d) 6 days
- e) 8 days

62) A group of 10 male workers can complete a project in 20 days, and 10 female workers can finish the same project in 40 days. How many days will it take for a team consisting of 20 men and 40 women to complete the project?

10 पुरुष कर्मचारियों का एक समूह एक परियोजना को 20 दिनों में पूरा कर सकता है, और 10 महिला कर्मचारी उसी परियोजना को 40 दिनों में पूरा कर सकती हैं। 20 पुरुषों और 40 महिलाओं वाली टीम को परियोजना को पूरा करने में कितने दिन लगेंगे?

- a) 5
- b) 8
- c) 10
- d) 12
- e) None of these

63) A group of 12 workers, each working 7 hours a day, can finish a task in 5 days. How many days would it take for 14 workers, working at half their usual efficiency and 6 hours a day, to complete the same task?

12 श्रमिकों का एक समूह, जिनमें से प्रत्येक प्रतिदिन 7 घंटे काम करता है, किसी कार्य को 5 दिनों में पूरा कर सकता है। 14 श्रमिकों को, जो अपनी सामान्य क्षमता से आधी क्षमता पर तथा प्रतिदिन 6 घंटे काम करते हैं, उसी कार्य को पूरा करने में कितने दिन लगेंगे?

- a) 9 days
- b) 10 days
- c) 12 days





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- d) 15 days
- e) 8 days

64) A group of 3 men and 4 boys can complete a certain task in 6 days, while a group of 6 men and 3 boys can finish the same task in 4 days. What is the ratio of the efficiency of a man to that of a boy?

3 पुरुषों और 4 लड़कों का एक समूह एक निश्चित कार्य को 6 दिनों में पूरा कर सकता है, जबकि 6 पुरुषों और 3 लड़कों का एक समूह उसी कार्य को 4 दिनों में पूरा कर सकता है। एक पुरुष और एक लड़के की दक्षता का अनुपात क्या है?

- a) 1:2
- b) 2:1
- c) 1:1
- d) 3:1
- e) None of these

65) In a historical fort, food provisions were stocked to last 75 residents for 35 days. However, on the very first day, 15 additional visitors including Priyanka joined them for a vacation. Based on this change, for how many days will the available supplies now last for everyone?

एक ऐतिहासिक किले में 75 निवासियों के लिए 35 दिनों तक भोजन का स्टॉक रखा गया था। लेकिन पहले ही दिन प्रियंका समेत 15 अतिरिक्त आगंतुक छुट्टी मनाने उनके साथ शामिल हो गए। इस बदलाव के आधार पर, अब उपलब्ध आपूर्ति सभी के लिए कितने दिनों तक चलेगी?

- a) 28 days
- b) 30 days
- c) 32 days
- d) 33 days
- e) None of these

66) A group of 20 workers was initially assigned to complete a construction project within 48 days. However, after working for 24 days, only one-third of the total work was completed. To ensure the project is completed within the original 48-day timeline, how many additional workers need to be brought in?





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20 श्रमिकों के एक समूह को शुरू में एक निर्माण परियोजना को 48 दिनों के भीतर पूरा करने का काम सौंपा गया था। हालाँकि, 24 दिनों तक काम करने के बाद, कुल काम का केवल एक तिहाई ही पूरा हो पाया। यह सुनिश्चित करने के लिए कि परियोजना मूल 48-दिन की समयसीमा के भीतर पूरी हो जाए, कितने अतिरिक्त श्रमिकों को लाने की आवश्यकता है?

- a) 15
- b) 18
- c) 20
- d) 24
- e) None of these

- 67) A construction task was started by 20 male labourers who managed to finish half of the work in 8 days. Later, a new team comprising 8 men and 12 women was assigned to complete the entire project. Each woman is twice as efficient as a man. In how many days will this new team complete the full work?

20 पुरुष मजदूरों ने एक निर्माण कार्य शुरू किया, जिसमें से आधे काम को 8 दिन में पूरा कर लिया गया। बाद में, 8 पुरुषों और 12 महिलाओं की एक नई टीम को पूरे प्रोजेक्ट को पूरा करने के लिए नियुक्त किया गया। प्रत्येक महिला एक पुरुष से दोगुनी कुशल है। यह नई टीम कितने दिनों में पूरा काम पूरा करेगी?

- a) 14
- b) 15
- c) 10
- d) 18
- e) None of these

- 68) Ravi Construction Co. had to finish a project in 50 days using 105 workers, each putting in 8 hours daily. After 25 days, only $\frac{2}{5}$ of the work was completed. To complete the remaining work on time, they decided to increase working hours to 9 per day. How many additional workers are now required?

रवि कंस्ट्रक्शन कंपनी को 105 श्रमिकों का उपयोग करके 50 दिनों में एक परियोजना पूरी करनी थी, जिनमें से प्रत्येक को प्रतिदिन 8 घंटे काम करना था। 25 दिनों के बाद, केवल $\frac{2}{5}$ काम पूरा हुआ। शेष कार्य को समय पर पूरा करने के लिए, उन्होंने काम के घंटे बढ़ाकर प्रतिदिन 9 करने का निर्णय लिया। अब कितने अतिरिक्त श्रमिकों की आवश्यकता है?

- a) 35





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- b) 20
- c) 30
- d) 25
- e) None of these

69) Suman and her team were assigned a construction task. They worked 6 hours daily and managed to complete two-thirds of the work in 15 days. After this period, 8 members left the team. The remaining workers increased their daily working hours to 8 and finished the rest of the task in 7.5 days. How many people were originally present in the team when the work began?

सुमन और उनकी टीम को एक निर्माण कार्य सौंपा गया था। उन्होंने प्रतिदिन 6 घंटे काम किया और 15 दिनों में दो-तिहाई काम पूरा करने में सफल रहे। इस अवधि के बाद, 8 सदस्यों ने टीम छोड़ दी। शेष श्रमिकों ने अपने दैनिक कार्य घंटे बढ़ाकर 8 कर दिए और शेष कार्य 7.5 दिनों में पूरा कर लिया। काम शुरू होने पर टीम में मूल रूप से कितने लोग मौजूद थे?

- a) 24
- b) 28
- c) 32
- d) 36
- e) None of these

70) Ramesh and his team were hired to finish a large painting project. Initially, a specific number of workers could complete the task in 90 days if each worked for 8 hours daily. Later, 8 more workers joined, and the team worked 12 hours daily to finish the same task in just 45 days. Based on this, how many workers would be needed to complete the same project in 72 days if they work 6 hours each day?

रमेश और उनकी टीम को एक बड़ी पेंटिंग परियोजना को पूरा करने के लिए काम पर रखा गया था। शुरू में, यदि प्रत्येक व्यक्ति प्रतिदिन 8 घंटे काम करे तो एक निश्चित संख्या में श्रमिक 90 दिनों में कार्य पूरा कर सकते थे। बाद में, 8 और श्रमिक शामिल हो गए, और टीम ने प्रतिदिन 12 घंटे काम करके उसी कार्य को केवल 45 दिनों में पूरा कर लिया। इसके आधार पर, यदि वे प्रत्येक दिन 6 घंटे काम करते हैं तो उसी परियोजना को 72 दिनों में पूरा करने के लिए कितने श्रमिकों की आवश्यकता होगी?

- a) 40
- b) 36





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- c) 32
- d) 28
- e) None of these

71) If $(a + 5)$ gardeners working 9 hours per day can complete 75% of a garden project in 24 days, while $(a - 4)$ gardeners working 2.5 hours per day can complete the remaining work in 36 days, find the number of gardeners who completed 75% of the garden project.

यदि $(a + 5)$ माली प्रतिदिन 9 घंटे काम करके एक उद्यान परियोजना का 75% 24 दिनों में पूरा कर सकते हैं, जबकि $(a - 4)$ माली प्रतिदिन 2.5 घंटे काम करके शेष कार्य 36 दिनों में पूरा कर सकते हैं, तो उद्यान परियोजना का 75% पूरा करने वाले माली की संख्या ज्ञात कीजिए।

- a) 54
- b) 45
- c) 38
- d) 34
- e) None of these

72) A team led by Raghav comprising of $(n + 2)$ workers, each putting in 6 hours of effort daily, manages to complete 30% of a construction project within 6 days. Later, a new team under Samar with $(n + 4)$ workers, each working 4 hours a day, is assigned the remaining part of the project and finishes it in 15 days. Based on this information, what is the value of $(n + 13)$?

राघव के नेतृत्व में $(n + 2)$ श्रमिकों वाली एक टीम, जिनमें से प्रत्येक प्रतिदिन 6 घंटे काम करता है, 6 दिनों के भीतर एक निर्माण परियोजना का 30% पूरा करने में सफल हो जाती है। बाद में, समर के नेतृत्व में $(n + 4)$ श्रमिकों वाली एक नई टीम, जिनमें से प्रत्येक प्रतिदिन 4 घंटे काम करता है, को परियोजना का शेष भाग सौंपा जाता है और इसे 15 दिनों में पूरा किया जाता है। इस जानकारी के आधार पर, $(n + 13)$ का मान क्या है?

- a) 18
- b) 15
- c) 16
- d) 12
- e) None of these





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73) Priya and Neha agree to complete a task for ₹1,500. Priya can finish the work in 6 days, Neha in 9 days, and with Kajal's help, they complete the work in 3 days. What is Kajal's share?

प्रिया और नेहा ₹1,500 में एक काम पूरा करने के लिए सहमत हैं। प्रिया उस काम को 6 दिन में, नेहा 9 दिन में और काजल की मदद से वे 3 दिन में पूरा कर लेती हैं। काजल का हिस्सा क्या है?

- a) ₹400
- b) ₹500
- c) ₹600
- d) ₹700
- e) None of these

74) Ajay, Bhavesh, and Chirag can collectively complete a task in 5 days. Ajay can finish it alone in 15 days, while Bhavesh is 25% more efficient than Ajay. If the total wages for completing the work amount to Rs. 21,600, what is the difference between the amounts earned by Bhavesh and Chirag?

अजय, भावेश और चिराग मिलकर एक काम 5 दिन में पूरा कर सकते हैं। अजय अकेले इसे 15 दिन में पूरा कर सकता है, जबकि भावेश अजय से 25% अधिक कुशल है। यदि काम पूरा करने के लिए कुल मजदूरी 21,600 रुपये है, तो भावेश और चिराग द्वारा अर्जित राशि के बीच क्या अंतर है?

- a) 4,500
- b) 2,000
- c) 3,000
- d) 3,600
- e) None of these

75) Ravi, Simran, and Radhika can individually complete a project in 8 days, 12 days, and 24 days respectively. They decide to work together on the project and agree to divide the total payment of Rs.2400 in the ratio of the work each person contributes. How much money does Ravi receive?

रवि, सिमरन और राधिका अकेले ही किसी प्रोजेक्ट को क्रमशः 8 दिन, 12 दिन और 24 दिन में पूरा कर सकते हैं। वे प्रोजेक्ट पर साथ मिलकर काम करने का फैसला करते हैं और 2400 रुपये के कुल भुगतान को प्रत्येक व्यक्ति द्वारा किए गए काम के अनुपात में विभाजित करने पर सहमत होते हैं। रवि को कितना पैसा मिलता है?

- a) Rs.1000
- b) Rs.1100





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- c) Rs.1200
- d) Rs.1300
- e) None of these

76) Sanjay, Karan, and Mohit work together on a project that pays a total of Rs.7100. Their individual times to finish the project are in the ratio 3 : 5 : 7. The payment is distributed according to their efficiencies, which are the reciprocals of the times they would take. What is Mohit's share of the total payment?

संजय, करण और मोहित एक प्रोजेक्ट पर मिलकर काम करते हैं, जिसके लिए उन्हें कुल 7100 रुपये मिलते हैं। प्रोजेक्ट को पूरा करने में उन्हें अलग-अलग समय 3 : 5 : 7 के अनुपात में लगता है। भुगतान उनकी कार्यकुशलता के अनुसार वितरित किया जाता है, जो कि उनके द्वारा लिए जाने वाले समय के व्युत्क्रम हैं। कुल भुगतान में मोहित का हिस्सा क्या है?

- a) Rs.1200
- b) Rs.1500
- c) Rs.1600
- d) Rs.1800
- e) None of these

77) Vishal, Jai, and Gopal agreed to complete a typing job for ₹3,080. Vishal and Jai alone can complete the job in 14 days and 11 days, respectively. They both worked together for 4 days, and Gopal completed the remaining work in 6 days. What is Jai's share of the total amount?

विशाल, जय और गोपाल ने ₹3,080 में टाइपिंग का काम पूरा करने का फैसला किया। विशाल और जय अकेले क्रमशः 14 दिन और 11 दिन में काम पूरा कर सकते हैं। वे दोनों 4 दिन साथ काम करते हैं और गोपाल बाकी काम 6 दिन में पूरा करता है। कुल राशि में जय का हिस्सा क्या है?

- a) ₹1,000
- b) ₹1,100
- c) ₹1,120
- d) ₹1,150
- e) None of these





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78) Ravi's efficiency is twice that of Priya, and Seema's efficiency is 150% of Priya's efficiency. Ravi works for 6 days, Priya works for 15 days, and Seema works for 7 days to complete a task. If the total payment for the work is ₹6,000, how much will Seema receive?

रवि की कार्यकुशलता प्रिया की कार्यकुशलता से दोगुनी है, और सीमा की कार्यकुशलता प्रिया की कार्यकुशलता का 150% है। रवि 6 दिन काम करता है, प्रिया 15 दिन काम करती है, और सीमा एक काम पूरा करने के लिए 7 दिन काम करती है। यदि काम के लिए कुल भुगतान ₹6,000 है, तो सीमा को कितना मिलेगा?

- a) ₹1,670
- b) ₹1,680
- c) ₹1,690
- d) ₹1,700
- e) None of these

79) Raj and Simran, working alone, can build a wall in 8 days and 12 days, respectively. They agreed to complete the wall for ₹9,000. Raj and Simran worked together for 2 days and then completed the remaining work with Kiran in another 2 days. How much will Kiran receive?

राज और सिमरन अकेले काम करते हुए क्रमशः 8 दिन और 12 दिन में एक दीवार बना सकते हैं। वे दीवार को ₹9,000 में पूरा करने के लिए सहमत हुए। राज और सिमरन ने 2 दिन साथ काम किया और फिर किरण के साथ मिलकर बाकी काम 2 दिन में पूरा किया। किरण को कितना मिलेगा?

- a) ₹1,000
- b) ₹1,500
- c) ₹2,000
- d) ₹2,500
- e) None of these

80) A group of 46 women earned a total of Rs.82800 by working for 25 days. Now, if each man earns double the daily wage of a woman, how many men are required to work for 24 days in order to earn Rs.110592?

46 महिलाओं के एक समूह ने 25 दिन काम करके कुल 82800 रुपये कमाए। अब, यदि प्रत्येक पुरुष एक महिला की दैनिक मजदूरी का दोगुना कमाता है, तो 110592 रुपये कमाने के लिए कितने पुरुषों को 24 दिन काम करना होगा?

- a) 36
- b) 32





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- c) 30
- d) 28
- e) None of these

81) In a farm, three workers, Ali, Ben, and Charlie, are tasked with harvesting crops. Ali can harvest 50% of the crops in a certain number of days, while Ben can harvest 75% of the crops in the same amount of time. Charlie is twice as efficient as Ali. After working together, they received a total of Rs. 4500 as their payment. How much should Charlie receive as his share?

एक खेत में, तीन मजदूर अली, बेन और चार्ली को फसल काटने का काम सौंपा गया है। अली एक निश्चित समय में 50% फसल काट सकता है, जबकि बेन उसी समय में 75% फसल काट सकता है। चार्ली अली से दोगुना कुशल है। साथ मिलकर काम करने के बाद, उन्हें भुगतान के रूप में कुल 4500 रुपये मिले। चार्ली को अपने हिस्से के रूप में कितना मिलना चाहिए?

- a) Rs.2100
- b) Rs.2000
- c) Rs.2200
- d) Rs.2500
- e) None of these

82) Prisha and Dev can individually complete a project in 15 days and 12 days respectively. Along with another worker, Mehul, they manage to finish the entire work together in just 4 days. If the total payment for the work is Rs. 4500, what is Mehul's fair share of the money based on their individual contributions?

प्रिशा और देव व्यक्तिगत रूप से एक प्रोजेक्ट को क्रमशः 15 दिन और 12 दिन में पूरा कर सकते हैं। एक अन्य कार्यकर्ता मेहुल के साथ मिलकर वे पूरे काम को सिर्फ 4 दिन में पूरा करने में कामयाब हो जाते हैं। यदि काम के लिए कुल भुगतान 4500 रुपये है, तो उनके व्यक्तिगत योगदान के आधार पर मेहुल का उचित हिस्सा कितना है?

- a) Rs.1800
- b) Rs.1600
- c) Rs.2100
- d) Rs.2000
- e) None of these





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- 83) Saanvi and Ishan together can complete a certain project in 12 days. However, when they received help from an additional team member, Veer, they managed to complete the same project in just $6\frac{2}{3}$ days. If the total payment for the project is Rs.18000, what is Veer's share of the earnings?
सानवी और ईशान मिलकर एक निश्चित प्रोजेक्ट को 12 दिनों में पूरा कर सकते हैं। हालाँकि, जब उन्हें टीम के एक अतिरिक्त सदस्य वीर से मदद मिली, तो वे उसी प्रोजेक्ट को सिर्फ $6\frac{2}{3}$ दिनों में पूरा करने में कामयाब रहे। अगर प्रोजेक्ट के लिए कुल भुगतान 18000 रुपये है, तो कमाई में वीर का हिस्सा क्या है?
- a) Rs.6000
b) Rs.7000
c) Rs.8000
d) Rs.9000
e) None of these
- 84) Pranav can finish a job in 24 days, while Iqbal takes 40 days to complete the same work. When Pranav, Iqbal, and Charan work together, they complete the entire task in just 10 days. If they received a total payment of Rs.11760 for their work, how much should Charan receive as his share?
प्रणव एक काम 24 दिन में पूरा कर सकता है, जबकि इकबाल को उसी काम को पूरा करने में 40 दिन लगते हैं। जब प्रणव, इकबाल और चरण एक साथ काम करते हैं, तो वे पूरा काम सिर्फ 10 दिन में पूरा कर लेते हैं। अगर उन्हें अपने काम के लिए कुल 11760 रुपये का भुगतान मिला, तो चरण को अपने हिस्से के रूप में कितना मिलना चाहिए?
- a) Rs.2440
b) Rs.2700
c) Rs.3120
d) Rs.3920
e) None of these
- 85) In a small workshop, Rajesh, Ravi, and Amit (3 men) along with 5 boys can finish a certain task in 15 days. On the other hand, 6 men and 7 boys can complete the same work in just 9 days. Based on this information, what is the relationship between the amount of work done by the boys and the work done by the men?





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एक छोटी सी कार्यशाला में, राजेश, रवि और अमित (3 पुरुष) 5 लड़कों के साथ मिलकर एक निश्चित कार्य को 15 दिनों में पूरा कर सकते हैं। दूसरी ओर, 6 पुरुष और 7 लड़के उसी कार्य को केवल 9 दिनों में पूरा कर सकते हैं। इस जानकारी के आधार पर, लड़कों द्वारा किए गए कार्य की मात्रा और पुरुषों द्वारा किए गए कार्य के बीच क्या संबंध है?

- a) A boy completes half the work of a man per day
- b) A man completes half the work of a boy per day
- c) A boy and a man complete the same amount of work per day
- d) A boy completes double the work of a man per day
- e) None of these

- 86) A gardener can plant all the flowers in 25 days, and the efficiency of a skilled gardener is 25% more than that of an average gardener. The efficiency of a beginner gardener is 20% less than that of an average gardener. 'x' number of average gardeners started the work, after 6 days they left, and then 'x' number of beginner gardeners completed the remaining work in 5 days. If a skilled gardener worked for (y-6) days and left, then an average gardener worked for (y-3) days and left, and then a beginner gardener completed the remaining work in y days. Find the value of (x + y).

एक माली 25 दिनों में सभी फूल लगा सकता है, और एक कुशल माली की दक्षता एक औसत माली की तुलना में 25% अधिक है। एक शुरुआती माली की दक्षता एक औसत माली की तुलना में 20% कम है। औसत माली की 'x' संख्या ने काम शुरू किया, 6 दिनों के बाद वे चले गए, और फिर शुरुआती माली की 'x' संख्या ने शेष काम 5 दिनों में पूरा किया। यदि एक कुशल माली ने (y-6) दिनों तक काम किया और चला गया, तो एक औसत माली ने (y-3) दिनों तक काम किया और चला गया, और फिर एक शुरुआती माली ने शेष काम y दिनों में पूरा किया। (x + y) का मान ज्ञात कीजिए।

- a) 15
- b) 16
- c) 12
- d) 10
- e) None of these





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87) In a publishing house, a female editor is 40% more efficient than a male editor, while a trainee editor is 50% less efficient than a male editor. The female editor alone can finish editing a manuscript in 5 days. In the first scenario, a trainee editor starts editing for $(m-6)$ days and then leaves. After that, a male editor and the female editor work together for the next $(11-m)$ days. Then the male editor leaves, and the female editor completes the remaining work at only 50% of her usual efficiency in $(m-8)$ days. In a second scenario, a trainee editor works for $(n-2)$ days, and then a male editor joins and both work together for the next $(8-n)$ days. Afterward, the female editor, now working with a 42.85% increase in her efficiency, finishes the rest of the work in $(n-5)$ days. Find the value of $(m + n)$.

एक प्रकाशन गृह में, एक महिला संपादक एक पुरुष संपादक की तुलना में 40% अधिक कुशल होती है, जबकि एक प्रशिक्षु संपादक एक पुरुष संपादक की तुलना में 50% कम कुशल होता है। महिला संपादक अकेली एक पांडुलिपि का संपादन 5 दिनों में पूरा कर सकती है। पहले परिदृश्य में, एक प्रशिक्षु संपादक $(m-6)$ दिनों के लिए संपादन शुरू करता है और फिर छोड़ देता है। उसके बाद, एक पुरुष संपादक और महिला संपादक अगले $(11-m)$ दिनों के लिए एक साथ काम करते हैं। फिर पुरुष संपादक छोड़ देता है, और महिला संपादक शेष कार्य को $(m-8)$ दिनों में अपनी सामान्य दक्षता के केवल 50% पर पूरा करती है। दूसरे परिदृश्य में, एक प्रशिक्षु संपादक $(n-2)$ दिनों के लिए काम करता है, और फिर एक पुरुष संपादक शामिल होता है और दोनों अगले $(8-n)$ दिनों के लिए एक साथ काम करते हैं। इसके बाद, महिला संपादक, अब अपनी दक्षता में 42.85% की वृद्धि के साथ काम करते हुए, शेष कार्य को $(n-5)$ दिनों में पूरा करती है।

- a) 13
- b) 14
- c) 15
- d) 16
- e) None of these

88) 22 women can complete _____% of the whole work in 16 days working 15 hours in a day. 20 women working 10 hours in a day can complete 20% of the whole work in _____ days. The values given in which of the following options will fill the blanks in the same order in which is it given to make the above statement true:





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22 महिलाएँ एक दिन में 15 घंटे काम करके 16 दिनों में पूरे काम का _____% पूरा कर सकती हैं। 20 महिलाएँ एक दिन में 10 घंटे काम करके पूरे काम का 20% _____ दिनों में पूरा कर सकती हैं। निम्नलिखित में से किस विकल्प में दिए गए मान रिक्त स्थान को उसी क्रम में भरेंगे जिसमें यह उपरोक्त कथन को सत्य बनाने के लिए दिया गया है:

- A. 33, 16
- B. 60, 16
- C. 36, 10
- a) Only A
- b) Only C
- c) Only B and C
- d) Only C
- e) All A, B and C

89) Vikas was assigned the task of proofreading an entire manuscript. He can complete the task alone in _____ days, while Shruti, a junior editor, would take exactly 90 days to finish the same manuscript on her own. Vikas started editing and continued for 15 days before being reassigned. After that, Shruti took over and worked on the manuscript for 27 days before she also left. The rest of the proofreading was completed by Vikas in _____ days. Which of the following options correctly fills in the blanks in the same sequence?

विकास को एक पूरी पांडुलिपि की प्रूफरीडिंग का काम सौंपा गया था। वह अकेले ही यह काम _____ दिनों में पूरा कर सकता है, जबकि जूनियर एडिटर श्रुति को अकेले ही उसी पांडुलिपि को पूरा करने में ठीक 90 दिन लगेंगे। विकास ने संपादन शुरू किया और 15 दिनों तक काम जारी रखा, फिर उसे फिर से काम सौंपा गया। उसके बाद, श्रुति ने काम संभाला और पांडुलिपि पर 27 दिनों तक काम किया, फिर वह भी चली गई। बाकी प्रूफरीडिंग विकास ने _____ दिनों में पूरी की।

- (i) 42 and 14.4
- (ii) 42 and 16.8
- (iii) 45 and 16.5
- a) Only (i)
- b) Only (ii)
- c) Only (iii)
- d) Both (i) and (iii)
- e) Both (ii) and (iii)





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- 90) At a construction site, 7 trained workers and 8 assistants can complete the building of a shed in 7 days. On the other hand, 5 trained workers and 9 assistants can complete the same shed in 8 days. The daily efficiency difference between a trained worker and an assistant is 7 units. A teenage intern has an efficiency that is 12.5% less than that of a trained worker.

Quantity I: Time required for 8 teenage interns to complete the entire work alone.

Quantity II: Time required for 4 assistants and 4 teenage interns together to finish the same work.

एक निर्माण स्थल पर, 7 प्रशिक्षित श्रमिक और 8 सहायक 7 दिनों में एक शेड का निर्माण पूरा कर सकते हैं। दूसरी ओर, 5 प्रशिक्षित श्रमिक और 9 सहायक उसी शेड को 8 दिनों में पूरा कर सकते हैं। एक प्रशिक्षित श्रमिक और एक सहायक के बीच दैनिक दक्षता का अंतर 7 इकाई है। एक किशोर प्रशिक्षु की दक्षता एक प्रशिक्षित श्रमिक की तुलना में 12.5% कम है।

मात्रा I: 8 किशोर प्रशिक्षुओं को अकेले पूरा काम पूरा करने के लिए आवश्यक समय।

मात्रा II: 4 सहायकों और 4 किशोर प्रशिक्षुओं को एक साथ उसी काम को पूरा करने के लिए आवश्यक समय।

- a) Quantity I > Quantity II
- b) Quantity I $\geq$ Quantity II
- c) Quantity I < Quantity II
- d) Quantity I $\leq$ Quantity II
- e) Quantity I = Quantity II

- 91) Quantity I: Pradeep alone can complete a certain task in 20 days. He works alone for the first 5 days and then gets help from Kunal. Together, they manage to finish the remaining work in 9 more days. What is the efficiency of Pradeep compared to Kunal, in percentage?

Quantity II: A farmer named Harish can plough a field in 15 days, while another farmer, Jitendra, can do the same task in 20 days. They begin the work together, and after a few days, a third farmer, Lakhan, joins them. With Lakhan's help, the entire field is ploughed in 8 days. If all three farmers are paid a total of ₹1500 for the work, how much should Lakhan receive?

मात्रा I: प्रदीप अकेले एक निश्चित कार्य को 20 दिनों में पूरा कर सकता है। वह पहले 5 दिनों तक अकेले काम करता है और फिर कुणाल से मदद लेता है। साथ में, वे शेष कार्य को 9 और दिनों में पूरा करने का प्रबंधन करते हैं। कुणाल की तुलना में प्रदीप की दक्षता प्रतिशत में क्या है?





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मात्रा II: हरीश नाम का एक किसान 15 दिनों में एक खेत जोत सकता है, जबकि एक अन्य किसान, जितेंद्र, उसी कार्य को 20 दिनों में कर सकता है। वे एक साथ काम शुरू करते हैं, और कुछ दिनों के बाद, एक तीसरा किसान, लखन, उनके साथ जुड़ जाता है। लखन की मदद से, पूरा खेत 8 दिनों में जोत लिया जाता है। यदि सभी तीन किसानों को काम के लिए कुल ₹1500 का भुगतान किया जाता है, तो लखन को कितना मिलना चाहिए?

- a) Quantity I > Quantity II
- b) Quantity I $\geq$ Quantity II
- c) Quantity I < Quantity II
- d) Quantity I $\leq$ Quantity II
- e) Quantity I = Quantity II

92) Quantity I: A tailor named Dinesh can stitch 20% of a bulk clothing order in 4 days. After working alone for 5 days, he is joined by another tailor, Manoj. Together, they manage to finish the remaining order in 9 more days. What is the difference between the number of days Manoj would take to complete the entire order alone and the number of days Dinesh would take to do it alone?

Quantity II: A gardener named Surya can plant an entire field in 40 days. Another gardener, Harish, can do the same in 24 days, and a third gardener, Manjeet, can finish it in 30 days. Surya and Manjeet started the planting work and continued for 6 days. Then they handed over the task to Harish, who worked alone for a few days. Finally, Surya and Manjeet returned and completed the remaining work in the last 4 days. For how many days did Harish work alone?

मात्रा I: दिनेश नाम का एक दर्जी 4 दिनों में थोक कपड़ों के ऑर्डर का 20% सिलाई कर सकता है। 5 दिनों तक अकेले काम करने के बाद, उसके साथ एक और दर्जी मनोज जुड़ जाता है। साथ में, वे शेष ऑर्डर को 9 और दिनों में पूरा करने का प्रबंधन करते हैं। मनोज को अकेले पूरा ऑर्डर पूरा करने में लगने वाले दिनों की संख्या और दिनेश को अकेले इसे पूरा करने में लगने वाले दिनों की संख्या के बीच क्या अंतर है?

मात्रा II: सूर्या नाम का एक माली 40 दिनों में एक पूरा खेत रोप सकता है। दूसरा माली हरीश वही काम 24 दिनों में कर सकता है, और तीसरा माली मंजीत इसे 30 दिनों में पूरा कर सकता है। सूर्या और मंजीत ने रोपण का काम शुरू किया और 6 दिनों तक जारी रखा। फिर उन्होंने यह काम हरीश को सौंप दिया, जिसने कुछ दिनों तक अकेले काम किया। अंत में, सूर्या और मंजीत वापस लौटे और शेष काम को अंतिम 4 दिनों में पूरा किया। हरीश ने अकेले कितने दिनों तक काम किया?

- a) Quantity I > Quantity II





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- b) Quantity I $\geq$ Quantity II
- c) Quantity I $<$ Quantity II
- d) Quantity I $\leq$ Quantity II
- e) Quantity I = Quantity II

93) Quantity I: A group of $(x - 5)$ women working 12 hours a day can finish a task in 24 days. In contrast, a group of $(x + 5)$ women working 2.5 hours a day can complete 2.5 times the task in 48 days. Find the value of $(x + 5)$.

Quantity II: 16 men, 20 women, or 24 boys can complete a task in 25 days, working 5 hours per day. How many men must be paired with 10 women and 6 boys to finish the same task in 20 days, working 5 hours a day?

Quantity III: A and C started working, but A left 4 hours and C left 8 hours before the work was finished. B worked for 12 hours. A, B, and C can finish the work individually in 12, 15, and 20 hours, respectively. What is the total time taken to complete the work?

मात्रा I: $(x - 5)$ महिलाओं का एक समूह प्रतिदिन 12 घंटे काम करके 24 दिनों में एक काम पूरा कर सकता है। इसके विपरीत, $(x + 5)$ महिलाओं का एक समूह प्रतिदिन 2.5 घंटे काम करके 48 दिनों में 2.5 गुना काम पूरा कर सकता है। $(x + 5)$ का मान ज्ञात कीजिए।

मात्रा II: 16 पुरुष, 20 महिलाएँ या 24 लड़के प्रतिदिन 5 घंटे काम करके 25 दिनों में एक काम पूरा कर सकते हैं। उसी काम को 20 दिनों में पूरा करने के लिए कितने पुरुषों को 10 महिलाओं और 6 लड़कों के साथ जोड़ा जाना चाहिए, जो प्रतिदिन 5 घंटे काम करें?

मात्रा III: A और C ने काम करना शुरू किया, लेकिन A ने काम खत्म होने से 4 घंटे पहले और C ने 8 घंटे पहले काम छोड़ दिया। B ने 12 घंटे काम किया। A, B और C व्यक्तिगत रूप से क्रमशः 12, 15 और 20 घंटे में काम पूरा कर सकते हैं। काम पूरा करने में लगा कुल समय क्या है?

- a) Quantity I $>$ Quantity II $>$ Quantity III
- b) Quantity I $<$ Quantity II $<$ Quantity III
- c) Quantity II $>$ Quantity I $>$ Quantity III
- d) Quantity III $<$ Quantity II = Quantity I
- e) None of these

94) Quantity I: Pankaj and Rahul are assigned to complete a task. Pankaj can do the work alone in 12 days, while Rahul takes 24 days to finish it alone. If another worker, Shreyas, takes 2.5 times the number of days that Pankaj and Rahul would take together to complete the same work, determine how many days Shreyas alone would need to finish the job.





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Quantity II: Three friends—Chhavi, Deepa, and Eshan—can individually complete a job in 8, 16, and 24 days respectively. All three begin the work together, but Chhavi works for only 2 days. Deepa quits the job 4 days before the work is completed. In how many days is the entire task completed?

Quantity III: Ramesh alone can paint a house in 15 days, while Suresh can do the same job in 12 days. Ramesh begins the work and continues for 5 days before leaving. Afterward, Suresh takes over and completes the remaining task. How many days does Suresh take to finish the leftover work?

मात्रा I: पंकज और राहुल को एक काम पूरा करने के लिए नियुक्त किया गया है। पंकज अकेले काम को 12 दिनों में कर सकता है, जबकि राहुल को इसे अकेले खत्म करने में 24 दिन लगते हैं। यदि एक अन्य कर्मचारी श्रेयस, पंकज और राहुल द्वारा एक साथ उसी काम को पूरा करने में लिए गए दिनों की संख्या का 2.5 गुना समय लेता है, तो निर्धारित करें कि श्रेयस को अकेले काम पूरा करने में कितने दिन लगेंगे।

मात्रा II: तीन दोस्त - छवि, दीपा और ईशान - व्यक्तिगत रूप से क्रमशः 8, 16 और 24 दिनों में एक काम पूरा कर सकते हैं। तीनों एक साथ काम शुरू करते हैं, लेकिन छवि केवल 2 दिन काम करती है। काम पूरा होने से 4 दिन पहले दीपा काम छोड़ देती है। पूरा काम कितने दिनों में पूरा होता है?

मात्रा III: रमेश अकेले एक घर को 15 दिनों में पेंट कर सकता है, जबकि सुरेश उसी काम को 12 दिनों में कर सकता है। रमेश काम शुरू करता है और जाने से पहले 5 दिनों तक जारी रखता है। इसके बाद, सुरेश काम संभालता है और बचा हुआ काम पूरा करता है। सुरेश को बचा हुआ काम पूरा करने में कितने दिन लगते हैं?

- a) Quantity I > Quantity II > Quantity III
- b) Quantity I < Quantity II < Quantity III
- c) Quantity II > Quantity I > Quantity III
- d) Quantity III < Quantity II = Quantity I
- e) None of these

95) Find the time taken by Arun to complete the work alone.

Statement I: Priya and Arun together can complete a task in 12 days, while Mohit can do the same task in 18 days alone.

Statement II: Mohit and Arun worked together for 5 days, then Arun left, and Mohit worked for 4 more days. The remaining task was completed by Priya in 2 days.

अरुण द्वारा अकेले काम पूरा करने में लिया गया समय ज्ञात कीजिए।





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कथन I: प्रिया और अरुण मिलकर एक काम 12 दिनों में पूरा कर सकते हैं, जबकि मोहित अकेले उसी काम को 18 दिनों में पूरा कर सकता है।

कथन II: मोहित और अरुण ने 5 दिनों तक साथ मिलकर काम किया, फिर अरुण चला गया और मोहित ने 4 और दिन काम किया। बचा हुआ काम प्रिया ने 2 दिनों में पूरा किया।

- a) Statement (I) alone is sufficient to answer the question, but statement (II) alone is not sufficient.
- b) Statement (II) alone is sufficient to answer the question, but statement (I) alone is not sufficient.
- c) Both the statements taken together are necessary to answer the question, but neither of the statements alone is sufficient.
- d) Either statement (I) or statement (II) by itself is sufficient to answer the question.
- e) Statements (I) and (II) taken together are not sufficient to answer the question.

96) Find the difference in the number of days taken by Ramesh and Suresh individually to finish a certain task.

Statement I: Ramesh and Suresh together can complete the work in 40 days.

Statement II: Suresh and Mahesh together can complete the same work in 240 days.

Statement III: Ramesh and Mahesh together can finish it in 80 days.

रमेश और सुरेश द्वारा एक निश्चित कार्य को पूरा करने में व्यक्तिगत रूप से लिए गए दिनों की संख्या में अंतर ज्ञात कीजिए।

कथन I: रमेश और सुरेश मिलकर कार्य को 40 दिनों में पूरा कर सकते हैं।

कथन II: सुरेश और महेश मिलकर उसी कार्य को 240 दिनों में पूरा कर सकते हैं।

कथन III: रमेश और महेश मिलकर इसे 80 दिनों में पूरा कर सकते हैं।

- a) Statement I alone is sufficient
- b) All the statements together are sufficient
- c) Statements I and II together are sufficient
- d) Statements II and III together are sufficient
- e) All the statements together are not sufficient

97) In a factory, Worker X worked for 15 hours on a project and then left. How many more hours will Workers Y and Z take to finish the remaining work?

Statement I: The time taken by X alone to complete the project is half the time taken by Y alone.





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Statement II: Workers Y and Z together can finish the entire project in 24 hours.

Statement III: The efficiency of Worker Y is 50% less than that of Worker X. Worker Z works three times faster than Worker Y.

एक फैक्ट्री में, वर्कर X ने एक प्रोजेक्ट पर 15 घंटे काम किया और फिर चला गया। वर्कर Y और Z को बचा हुआ काम पूरा करने में कितने घंटे और लगेंगे?

कथन I: प्रोजेक्ट को पूरा करने में अकेले X द्वारा लिया गया समय अकेले Y द्वारा लिए गए समय का आधा है।

कथन II: वर्कर Y और Z मिलकर पूरे प्रोजेक्ट को 24 घंटे में पूरा कर सकते हैं।

कथन III: वर्कर Y की कार्यदक्षता वर्कर X की कार्यदक्षता से 50% कम है। वर्कर Z, वर्कर Y से तीन गुना तेज काम करता है।

- a) Statements I and III together are sufficient
- b) Statements I, II and III together are sufficient
- c) Statements II and III together are sufficient
- d) Statements II and III together and statements I and III together are sufficient
- e) Any two statements together are sufficient

- 98) Three workers, A, B, and C, are hired to assemble a set of machines. They can finish the assembly in $300/31$ days. A worked for 4 days and then left the task. B and C completed the remaining work. How many days did B and C take to complete the remaining assembly?

Statement I: A can assemble 25% of the machines in 10 days.

Statement II: B and C together can assemble 75% of the machines in $225/16$ days.

Statement III: A and B together can assemble all the machines in 24 days.

तीन कर्मचारी, A, B और C, मशीनों के एक सेट को जोड़ने के लिए काम पर रखे गए हैं। वे $300/31$ दिनों में असेंबली पूरी कर सकते हैं। A ने 4 दिन काम किया और फिर काम छोड़ दिया। B और C ने शेष काम पूरा किया। शेष असेंबली को पूरा करने में B और C को कितने दिन लगे?

कथन I: A 10 दिनों में 25% मशीनें जोड़ सकता है।

कथन II: B और C मिलकर $225/16$ दिनों में 75% मशीनें जोड़ सकते हैं।

कथन III: A और B मिलकर सभी मशीनों को 24 दिनों में जोड़ सकते हैं।

- a) Either Statement I alone or Statement II alone is sufficient, and Statement III alone is not sufficient.





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- b) Statement I and II together are sufficient, and Statement III is not sufficient.
- c) Either Statement III alone is sufficient or Statement I and II together are sufficient.
- d) Any one of the statements is sufficient.
- e) All the statements are necessary.

99) Rahul, Priya, and Sameer are working together to complete a task. Find the time taken by all three to complete the work together.

Statement I: Rahul and Priya together can finish the work in 4 hours. The efficiency of Rahul is twice that of Priya.

Statement II: The ratio of the efficiency of Priya and Sameer is 3:4.

Statement III: Rahul and Sameer together can complete the work in 7 hours.

राहुल, प्रिया और समीर एक साथ मिलकर एक कार्य पूरा कर रहे हैं। तीनों द्वारा एक साथ कार्य पूरा करने में लिया गया समय ज्ञात कीजिए।

कथन I: राहुल और प्रिया मिलकर 4 घंटे में कार्य पूरा कर सकते हैं। राहुल की कार्यकुशलता प्रिया से दोगुनी है।

कथन II: प्रिया और समीर की कार्यकुशलता का अनुपात 3:4 है।

कथन III: राहुल और समीर मिलकर 7 घंटे में कार्य पूरा कर सकते हैं।

- a) Statement II alone is sufficient.
- b) Statement I alone is sufficient, and Statement II alone is not sufficient.
- c) Statement I and II together are sufficient, and Statement III alone is not sufficient.
- d) All the statements are necessary.
- e) Any two of the statements are sufficient.

100) In a factory, Akash worked for 15 days, and then Ravi worked for 7.5 days. After their shifts, how many days will it take for Suresh to finish the remaining work on his own?

Statement I: Akash and Ravi together can complete the work in 27 days. Suresh alone can finish the work in 54 days.

Statement II: Ravi and Suresh together can complete the work in 24 days. Akash alone can finish the work in 72 days.

Statement III: The ratio of the time taken by Akash and Ravi together to the time taken by Ravi and Suresh together is 8:9.





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एक फैक्ट्री में, आकाश ने 15 दिन काम किया और फिर रवि ने 7.5 दिन काम किया। उनकी शिफ्ट के बाद, सुरेश को अकेले शेष काम पूरा करने में कितने दिन लगेंगे?

कथन I: आकाश और रवि मिलकर काम को 27 दिनों में पूरा कर सकते हैं। सुरेश अकेले काम को 54 दिनों में पूरा कर सकता है।

कथन II: रवि और सुरेश मिलकर काम को 24 दिनों में पूरा कर सकते हैं। आकाश अकेले काम को 72 दिनों में पूरा कर सकता है।

कथन III: आकाश और रवि द्वारा मिलकर लिए गए समय का रवि और सुरेश द्वारा मिलकर लिए गए समय से अनुपात 8:9 है।

- a) Statements I and II together and statements I and III together are sufficient.
- b) Any two statements together are sufficient
- c) Statements I and II together and statements II and III together are sufficient.
- d) Statement I alone and statement II and III together are sufficient.
- e) All statements together are sufficient





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Solution

1) Solution

Arjun's rate = $1/40$ per day.

Combined rate = $1/24$ per day.

Priya's rate = $1/24 - 1/40 = 1/60$ per day.

Priya will take 60 days.

2) Solution

Let total work = LCM of 20 and 36 = 180 units

Work per day by Pranav + Qasim = $180 \div 20 = 9$ units/day

Pranav's work per day = $180 \div 36 = 5$ units/day

Qasim's work per day = $9 - 5 = 4$ units/day

Time taken by Qasim alone = $180 \div 4 = 45$ days

3) Solution

Let the efficiency ratio of A to B be 4:3.

A + B together = 20 days

Time taken by A alone = $(20 \times 4) / 3 = 26\frac{2}{3}$ days

4) Solution

Worker A alone can complete 100% of the work in 32 days (since 75% takes 24 days).

Efficiency ratio of A to B = 3:2, and Efficiency ratio of B to C = 5:3.

Thus, the combined efficiency ratio of A:B:C = 15:10:6.

Time taken by A and B to complete the work = $(32 \times 15) / 25 = 96 / 5 = 19\frac{1}{5}$ days.

5) Solution

Efficiency ratio of Sanya and Kabir = 2:3

Combined efficiency = $2 + 3 = 5$ units/day

Total work = $5 \times 20 = 100$ units

Sanya's efficiency = 2 units/day

Time taken by Sanya = $100 \div 2 = 50$ days

6) Solution





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Total work = $14 \times 30 = 420$ units.

To complete double the work, the total work is 840 units.

Since Worker B's efficiency is 5, the time taken by Worker B to finish 840 units of work is $840 / 5 = 168$ days.

7) Solution

Let Raj's work rate be R.

Then, Ananya's work rate = $4R$.

Priya's work rate = 60% of (Ananya's + Raj's work rates) = $0.6 \times (4R + R) = 0.6 \times 5R = 3R$.

Together, their combined work rate is:

$R + 4R + 3R = 8R$.

Since they complete the task in 20 days, their combined 1-day work is $1/20$:

$8R = 1/20$

$R = 1/160$.

Raj's time to complete the task alone = $1 / R = 1 / (1/160) = 160$ days.

8) Solution

Raghav's 5 days = 50% $\rightarrow$ 1 day work = 10% $\rightarrow$ Full work in 10 days

So, Raghav's 1-day work = $1/10$

Deepika's 10 days = 150% $\rightarrow$ 1 day work = 15% $\rightarrow$ Full work in 6.67 days

So, Deepika's 1-day work = $1.5 / 10 = 0.15 = 3/20$

Together 1-day work = $1/10 + 3/20 = (2 + 3)/20 = 5/20 = 1/4$

To finish 50% $\rightarrow 0.5 / (1/4) = 2$ days

9) Solution

Linda's 1-day work = $30\% / 5 = 6\%$ per day.

Mark's 1-day work = $20\% / 5 = 4\%$ per day.

Together, their combined 1-day work = $6\% + 4\% = 10\%$ per day.

To finish the entire project (100% work), they will take:

$100\% / 10\% = 10$ days.

10) Solution

Aryan's 1-day work = $1/16$

Bhaves's 1-day work = $1/12$

Charan's 1-day work = $1/15$





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Let's calculate 2-day work:

Day 1: Aryan + Charan = $\frac{1}{16} + \frac{1}{15} = \frac{(15 + 16)}{240} = \frac{31}{240}$

Day 2: Bhavesh + Charan = $\frac{1}{12} + \frac{1}{15} = \frac{(5 + 4)}{60} = \frac{9}{60} = \frac{36}{240}$

In 2 days: $\frac{31}{240} + \frac{36}{240} = \frac{67}{240}$

In 6 such days (3 cycles): $3 \times \frac{67}{240} = \frac{201}{240}$

Remaining work = $1 - \frac{201}{240} = \frac{39}{240}$

Next day: Aryan + Charan = $\frac{31}{240}$

Remaining = $39 - 31 = \frac{8}{240}$

Next day: Bhavesh + Charan = $\frac{36}{240}$

They can finish remaining $\frac{8}{240}$ in $(\frac{8}{240}) \div (\frac{36}{240}) = \frac{8}{36} = \frac{2}{9}$ day

Total = 6 days + 1 day + $\frac{2}{9}$ day = $7 + \frac{2}{9} = 7.22$ days

11) Solution

Together in 1 day = $\frac{3}{40}$.

Nakul alone in 1 day = $\frac{1}{24} \rightarrow$ So, Rafi in 1 day = $\frac{3}{40} - \frac{1}{24} = \frac{1}{30}$.

In 2 days: $(\frac{1}{24} + \frac{1}{30}) = \frac{9}{120} = \frac{3}{40}$.

Every 2 days they do $\frac{3}{40}$ work.

In 26 days (13 such pairs) they do $\frac{39}{40}$ work.

Work left = $\frac{1}{40}$, Nakul's next turn, he does $\frac{1}{24}$ per day.

Time to finish = $(\frac{1}{40}) \div (\frac{1}{24}) = \frac{24}{40} = \frac{3}{5}$ day.

Total = $26 + \frac{3}{5} = 26.6$ days

12) Solution

One day work of Asha and Bharat = $\frac{1}{12} + \frac{1}{18} = \frac{30}{(12 * 18)}$

Three-day work of Asha and Bharat = $3 * \frac{30}{(12 * 18)} = \frac{15}{36}$

Total work done in three days = $\frac{15}{36} + \frac{1}{24} = \frac{(30 + 3)}{72} = \frac{33}{72}$

Six days work = $(\frac{33}{72}) * 2 = \frac{66}{72} = \frac{11}{12}$

Remaining work = $\frac{1}{12}$

On the 7th day, Asha and Bharat are working together.

One day work of Asha and Bharat $\rightarrow \frac{5}{36}$

Time required for Asha and Bharat to complete $\frac{1}{12}$ th of the work = $\frac{36}{(12 * 5)} = \frac{3}{5}$

Total time required = $6 + \frac{3}{5} = (6 \frac{3}{5})$ days

13) Solution

Let's add the given equations:





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John and Liam: $J + L = 1/30$

Liam and Sarah: $L + S = 1/15$

John and Sarah: $J + S = 1/40$

Adding all three equations gives:

$$2(J + L + S) = 1/30 + 1/15 + 1/40$$

Simplifying the right side:

$$2(J + L + S) = 1/8$$

$$\text{So, } J + L + S = 1/16.$$

Thus, they will complete the work in $1 / (1/16) = 16$ days.

14) Solution

Work done by Arjun in 1 day = $1/55$

Work done by Bhavesh in 1 day = $1/66$

Work done by all three in 1 day = $1/12$

$$1/55 + 1/66 + 1/x = 1/12$$

Solving for x: $x = 20$

So, Charan can paint the house in 20 days.

15) Solution

$$1/A + 1/B = 1/10$$

$$1/B + 1/C = 1/15$$

$$1/C + 1/D = 1/20$$

$$1/D + 1/A = 1/12$$

Adding all the equations:

$$2 * (1/A + 1/B + 1/C + 1/D) = 1/10 + 1/15 + 1/20 + 1/12$$

$$2 * (1/A + 1/B + 1/C + 1/D) = (6 + 4 + 3 + 5)/60$$

$$2 * (1/A + 1/B + 1/C + 1/D) = 18/60$$

$$2 * (1/A + 1/B + 1/C + 1/D) = 3/10$$

$$(1/A + 1/B + 1/C + 1/D) = 3/20$$

Work done in 2.4 hours:

$$(1/A + 1/B + 1/C + 1/D) * 2.4 = 9/25$$

Asha and Bharat's one-hour work: $1/10$

Remaining work = $16/25$

Time required to complete the remaining work = $(16/25) * 10 = 6.4$ hours

16) Solution





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Let the least common multiple (LCM) be 150.

Efficiency of A + B = 6, and Efficiency of B + C = 5.

Efficiency of A - C = 6 - 5 = 1.

Since A is 66.67% more efficient than C,

Efficiency ratio of A and C = 5x:3x, and solving for x gives x = 0.5.

Thus,

Efficiency of A = 2.5, Efficiency of C = 1.5, Efficiency of B = 3.5.

Required days = $150 / (2.5 + 3.5 + 1.5) = 150 / 7.5 = 20$ days.

17) Solution

Let Aditya's work rate be x.

Then Ishita's work rate = 4x.

Combined rate of Ishita and Aditya = $x + 4x = 5x$.

Tanvi's work rate = 60% of $5x = 3x$.

Total combined rate = $x + 4x + 3x = 8x$.

They finish the task in 20 days → Total work = $20 \times 8x = 160x$.

Aditya alone would take $160x / x = 160$ days.

18) Solution

Aarav's daily efficiency = $50\% \div 20 = 2.5\%$ per day

Aarav and Riya's combined daily efficiency = $100\% \div 24 = 4.17\%$ per day

Riya's daily efficiency = $4.17\% - 2.5\% = 1.67\%$ per day

Work done in 12 days = $12 \times 4.17\% = 50\%$

Remaining work = $100\% - 50\% = 50\%$

Time taken by Riya to complete the remaining work = $50\% \div 1.67\% = 30$ days

19) Solution

Anjali's 1-day work = $1/8$

Bhavna's 1-day work = $1/12$

Together with Charu:

$1/8 + 1/12 + C = 1/4$

$\Rightarrow (3 + 2)/24 + C = 1/4$

$\Rightarrow 5/24 + C = 1/4$

$\Rightarrow C = 1/4 - 5/24 = (6 - 5)/24 = 1/24$

Charu's 1-day work = $1/24$





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$$\text{Charu} + \text{Diya} = 1/15$$

$$\Rightarrow 1/24 + D = 1/15$$

$$\Rightarrow D = 1/15 - 1/24 = (8 - 5)/120 = 3/120 = 1/40$$

20) Solution

Let the total work be 100 units.

$$\text{Efficiency of A} = 100 / 50 = 2$$

$$\text{Efficiency of B} = 100 / 20 = 5$$

$$\text{Efficiency of A, B, and C together} = 100 / 10 = 10$$

$$\text{Efficiency of C} = 10 - (2 + 5) = 3$$

$$\text{Time taken by C alone} = 100 / 3 = 33(1/3) \text{ days}$$

21) Solution

A does $2/5$ work in 30 days $\rightarrow$ total work = 1

$$\text{A's rate} = (2/5)/30 = 1/75$$

B does 83.33% = $5/6$ work in 30 days $\rightarrow$ B's rate = $(5/6)/30 = 1/36$

Let C's rate = $1/x$

$$\text{A} + \text{B} + \text{C together} = 1/15$$

$$\Rightarrow 1/75 + 1/36 + 1/x = 1/15$$

Take LCM of 75 and 36 = 900

$$(12 + 25)/900 + 1/x = 1/15 \rightarrow 37/900 + 1/x = 1/15$$

$$1/x = 1/15 - 37/900 = (60 - 37)/900 = 23/900$$

So, C = $900/23$ days

Now,

$$\text{A} + \text{C worked for } (n + 7) \text{ days} \rightarrow \text{work} = (1/75 + 23/900) \times (n + 7)$$

$$= (12 + 23)/900 = 35/900 = 7/180$$

$$\text{Work} = (7/180) \times (n + 7)$$

$$\text{B worked alone for } (n + 19) \text{ days} \rightarrow \text{work} = 1/36 \times (n + 19)$$

Total work:

$$(7/180)(n + 7) + (1/36)(n + 19) = 1$$

Multiply all terms by 180 to eliminate denominator:

$$7(n + 7) + 5(n + 19) = 180$$

$$7n + 49 + 5n + 95 = 180 \rightarrow 12n = 36 \rightarrow n = 3$$

22) Solution

$$\text{M's rate} = 1 / (x + 5), \text{J's rate} = 1 / (2x)$$





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$$M + J \text{ together} = 1 / (22 + 2/19) = 1 / (420/19) = 19/420$$

So,

$$1/(x + 5) + 1/(2x) = 19/420$$

Take LCM: $(2x(x + 5))$

$$(2x + x + 5) / (2x(x + 5)) = 19/420$$

$$(3x + 5) / (2x(x + 5)) = 19/420$$

Cross-multiply:

$$(3x + 5) \times 420 = 19 \times 2x(x + 5)$$

$$1260x + 2100 = 38x^2 + 380x$$

Bring all terms to one side:

$$38x^2 + 380x - 1260x - 2100 = 0$$

$$38x^2 - 880x - 2100 = 0$$

Solve using quadratic formula:

$$x = [880 \pm \sqrt{(880^2 + 4 \times 38 \times 2100)}] / (2 \times 38)$$

$$x = 30 \text{ (positive root)}$$

$$\text{Now, } M = x + 5 = 35 \text{ days}$$

$$J = 2x = 60 \text{ days}$$

$$K = x - 10 = 20 \text{ days}$$

$$1/M + 1/J + 1/K = 1/35 + 1/60 + 1/20$$

$$\text{LCM} = 420$$

$$= (12 + 7 + 21)/420 = 40/420 = 2/21$$

$$\text{So, time} = 21/2 = 10.5 \text{ days}$$

23) Solution

$$\text{Simran's 1 day work} = 1/50$$

$$\text{Simran + Karthik's 1 day work} = 1/30$$

$$\text{Karthik's 1 day work} = 1/30 - 1/50 = (5 - 3)/150 = 2/150 = 1/75$$

$$\text{Work done in first 10 days} = 10 \times (1/30) = 1/3$$

$$\text{Remaining work} = 1 - 1/3 = 2/3$$

$$\text{Time Karthik needs} = (2/3) \div (1/75) = 2/3 \times 75 = 50 \text{ days}$$

24) Solution

$$\text{Vikram's 1-day work} = 1/20$$

$$\text{Karan's 1-day work} = 1/30$$

Let the total time taken be x days.

$$\text{Vikram works for } (x - 5) \text{ days.}$$





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Karan works for x days.

$$\text{Total work} = (x - 5)(1/20) + x(1/30) = 1$$

Taking LCM of 20 and 30 = 60,

$$(3(x - 5) + 2x)/60 = 1$$

$$(3x - 15 + 2x)/60 = 1$$

$$(5x - 15)/60 = 1$$

$$5x - 15 = 60$$

$$5x = 75$$

$$x = 15$$

25) Solution

Nina's work rate = $1/21$, Vikram's rate = $1/12 - 1/21 = 1/28$.

Let x be the number of days Nina worked. Then:

$$x/21 + 16 \times 1/28 = 1.$$

Solving:

$$x = 9.$$

So, Nina left 9 days before the task was completed.

26) Solution

Manish's 1-day work = $1/20$

Tarun's 1-day work = $1/30$

Sneha's 1-day work = $1/60$

Let total days be x .

Manish worked for $(x - 8)$ days.

Tarun worked for $(x - 3)$ days (5 more than Manish).

Sneha worked for x days.

$$\text{Total work} = (x - 8)(1/20) + (x - 3)(1/30) + x(1/60) = 1$$

Take LCM of 20, 30, and 60 = 60

$$(1/20)(x - 8) = (x - 8)/20 = 3(x - 8)/60$$

$$(1/30)(x - 3) = 2(x - 3)/60$$

$$(1/60)(x) = x/60$$

Now,

$$[3(x - 8) + 2(x - 3) + x]/60 = 1$$

$$[3x - 24 + 2x - 6 + x]/60 = 1$$

$$(6x - 30)/60 = 1$$

$$6x - 30 = 60$$



$$6x = 90$$

$$x = 15$$

27) Solution

Let total work = LCM of 60, 100, and 120 = 600 units

Aditya's 1-day work = 10 units

Bharat's = 6 units

Chandan's = 5 units

Phase 1 (Days 1–6): All three worked $\rightarrow (10 + 6 + 5) \times 6 = 21 \times 6 = 126$ units

Phase 2 (Days 7–12): Aditya + Bharat $\rightarrow (10 + 6) \times 6 = 16 \times 6 = 96$ units

Total completed in 12 days = $126 + 96 = 222$ units

Remaining = $600 - 222 = 378$ units

Phase 3: Done by Aditya alone $\rightarrow 378 \div 10 = 37.8$ days

Work done by Aditya =

In Phase 1: $10 \times 6 = 60$

In Phase 2: $10 \times 6 = 60$

In Phase 3: 378

Total by Aditya = $60 + 60 + 378 = 498$ units

Percentage = $(498 / 600) \times 100 = 83\%$

28) Solution

Naman completes $4/5$ of the work in 20 days, so his 1-day work is $1/25$.

Assume $x = 10$.

Raghav completes $4/15$ of the work in 10 days, so his 1-day work is $2/75$.

Chetan completes $2/5$ of the work in 8 days, so his 1-day work is $1/20$.

Together, their 1-day work is $1/25 + 2/75 + 1/20 = 7/60$.

Naman and Raghav together:

$1/25 + 2/75 = 1/15$.

Thus, Naman and Raghav together will take 15 days to complete the work.

29) Solution

Let total work = LCM of 24, 30, and 40 = 120 units

Rahul's 1-day work = 5, Suresh's = 4, Tapan's = 3

Let total time = x days

Rahul worked for $(x - 8)$ days



Suresh worked for 8 days

Tapan worked for x days

Total work done:

Rahul: $5(x - 8)$

Suresh: $4(8) = 32$

Tapan: $3x$

Total = $5(x - 8) + 32 + 3x = 120$

$\rightarrow 5x - 40 + 32 + 3x = 120$

$\rightarrow 8x - 8 = 120$

$\rightarrow 8x = 128$

$\rightarrow x = 16$

30) Solution

Let total work = LCM of 15, 12, and 14 = 420 units

Neeraj's 1-day work = 28, Om's = 35, Prateek's = 30

Total time = $3x - 2$ days

Neeraj worked for $x + 1$ days $\rightarrow$ did $(x + 1) \times 28$

Om worked for $(3x - 2 - 4x/3) = (9x - 6 - 4x)/3 = (5x - 6)/3$ days $\rightarrow$ did $(5x - 6)/3 \times 35$

Prateek worked for all $3x - 2$ days $\rightarrow$ did $(3x - 2) \times 30$

Add all work and equate to 420:

$\rightarrow 28(x + 1) + 35 \times (5x - 6)/3 + 30(3x - 2) = 420$

$\rightarrow 28x + 28 + (175x - 210)/3 + 90x - 60 = 420$

$\rightarrow$ Combine:

$(28x + 28) + (175x/3 - 70) + 90x - 60 = 420$

Group terms:

$(28x + 90x + 175x/3) + (28 - 70 - 60) = 420$

$(118x + 175x/3) - 102 = 420$

Take LCM: $(354x + 175x)/3 = 529x/3$

$\rightarrow 529x/3 - 102 = 420$

$\rightarrow 529x/3 = 522$

$\rightarrow x = (522 \times 3) / 529 = 1566 / 529 = 3$

Now find $x^2 + 25 = 9 + 25 = 34$

31) Solution





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Time for Anaya and Kabir together = 7.5 days = $15/2$ days

Kabir does half work in 10 days $\rightarrow$ full work = 20 days

Q's 1 day work = $1/20$

(P + Q)'s 1 day work = $2/15$

So, Anaya's 1 day work = $2/15 - 1/20 = (8 - 3)/60 = 5/60 = 1/12$

So Anaya alone finishes in 12 days

Now, 5 days of joint work:

Work done in 5 days = $5 \times (2/15) = 10/15 = 2/3$

Remaining work = $1 - 2/3 = 1/3$

Let x be time Anaya takes to do $1/3$ work:

$x \times (1/12) = 1/3 \rightarrow x = 4$ days

32) Solution

Let Ritika's time alone = x days

Ritika and Neha together = 20 days

Work done in 4 days together = $4/20 = 1/5$

Remaining work = $1 - 1/5 = 4/5$

Ritika did remaining $4/5$ work in 24 days

So, Ritika alone:

$24/x = 4/5 \rightarrow$ Multiply both sides by x:

$24 = (4x)/5 \rightarrow x = 30$ days

Now,

$1/30 + 1/y = 1/20 \rightarrow y$ is Neha's time alone

$1/y = 1/20 - 1/30 = (3 - 2)/60 = 1/60$

So, Neha alone takes 60 days

33) Solution

Let the total work be 1 unit.

Amit's one day work = $1/10$

Karan's one day work = $1/15$

Work done in the first 4 days by both Amit and Karan together:

$(1/10 + 1/15) \times 4 = (3/30 + 2/30) \times 4 = 5/30 \times 4 = 2/3$ of the work.

Remaining work after 4 days:

Remaining work = $1 - 2/3 = 1/3$ of the work.

Karan's work rate:

Karan's work per day = $1/15$.





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Time taken by Karan to finish the remaining work:

$$\text{Time} = (\text{Remaining work}) / (\text{Karan's work per day}) = (1/3) / (1/15) = 5 \text{ days.}$$

Total time taken to complete the work:

$$\text{Time} = 4 \text{ days (both working)} + 5 \text{ days (Karan alone)} = 9 \text{ days.}$$

34) Solution

Let the total work be represented by $20x$ units.

Raj's work per day = 4 units

Priya's work per day = 5 units

Simran's work per day = 10 units

Raj worked alone for 5 days:

$$\text{Work done by Raj} = 4 * 5 = 20 \text{ units}$$

Next, Raj and Priya worked together for 4 days:

$$\text{Work done by Raj and Priya} = (4 + 5) * 4 = 36 \text{ units}$$

Then, Raj and Simran worked together for $3 \frac{1}{7}$ days:

$$\text{Work done by Raj and Simran} = (4 + 10) * (22/7) = 44 \text{ units}$$

$$\text{Total work completed} = 20 + 36 + 44 = 100 \text{ units}$$

Therefore, the time Simran would take to complete the work on her own = $2 * 5 = 10$ days

35) Solution

Vikram's one day work = $1/24$

Meera's one day work = $1/32$

Priya's efficiency is twice that of Meera, so Priya's one day work = $1/16$

$$\text{Vikram's work in 6 days} = (1/24) * 6 = 1/4$$

$$\text{Vikram and Meera's combined work in 8 days} = (1/24 + 1/32) * 8 = 1/3 + 1/4 = 7/12$$

$$\text{Total work done} = 1/4 + 7/12 = (3 + 7)/12 = 10/12 = 5/6$$

Remaining work = $1/6$

$$\text{Vikram and Priya's one day work} = (1/24 + 1/16) = 5/48$$

$$\text{Time required to complete the remaining } 1/6 \text{ of the work} = (1/6) * (48/5) = 8/5 \text{ days}$$

$$\text{Total time required} = 6 + 8 + 8/5 = 14 + 8/5 = 78/5 \text{ days}$$

36) Solution

Let the total time to complete the work be ' x ' days.





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Ankur and Sunita worked for 'x' days, while Meera worked for '(x - 2)' days.

The equation becomes:

$$(1/40 + 1/60) * x + (1/20) * (x - 2) = 1$$

Simplifying:

$$(100/2400) * x + (x - 2)/20 = 1$$

$$x/24 + (x - 2)/20 = 1$$

Multiply both sides by 240:

$$20x + 24 * (x - 2) = 480$$

$$20x + 24x - 48 = 480$$

$$44x = 528$$

$$x = 528/44$$

$$x = 12 \text{ days}$$

Thus, the total time taken to complete the work is 12 days.

37) Solution

Let the total time to complete the work be 'x' days.

Rajesh and Priya worked for 'x' days, while Anil worked for '(x - 9)' days.

The equation becomes:

$$(1/30 + 1/60) * x + (1/90) * (x - 9) = 1$$

Simplifying:

$$(90/1800) * x + (x - 9)/90 = 1$$

$$x/20 + (x - 9)/90 = 1$$

Multiply both sides by 90:

$$90x + 20 * (x - 9) = 1800$$

$$90x + 20x - 180 = 1800$$

$$110x = 1980$$

$$x = 1980/110$$

$$x = 18 \text{ days}$$

Thus, Rajesh, Priya, and Anil worked together for (x - 9) days:

$$18 - 9 = 9 \text{ days}$$

38) Solution

$$\text{Arun and Bina's combined one-day work} = 1/24 + 1/48 = 3/48 = 1/16$$

$$\text{Work done by Arun and Bina in 10 days} = (1/16) * 10 = 5/8$$

$$\text{Remaining work} = 3/8$$





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Now, after 10 days, the remaining work is completed by Arun, Bina, and Chitra together in 3 days:

$$(1/24 + 1/48 + 1/C) * 3 = 3/8$$

$$(1/16 + 1/C) = 1/8$$

Solving for $1/C$:

$$1/C = 1/8 - 1/16$$

$$1/C = 8/(8 * 16)$$

$$1/C = 1/16$$

So, Chitra can complete the work in 16 days.

39) Solution

$$B : C = 5 : 6, C : D = 1 : 2 \rightarrow B : C : D = 5 : 6 : 12$$

$$A's \text{ 1-day work} = 1/20$$

$$B's = 1/5x, C's = 1/6x, D's = 1/12x$$

$$(1/20 + 1/5x + 1/6x + 1/12x) \times 5 + (1/5x + 1/6x + 1/12x) \times 3 = 1$$

$$5/20 + (8/5x + 8/6x + 8/12x) = 1$$

$$1/4 + 8(1/5x + 1/6x + 1/12x) = 1$$

$$8 \times (12 + 10 + 5)/60x = 3/4$$

$$8 \times 27/60x = 3/4 \rightarrow 216/60x = 3/4$$

$$9/20x = 3/32 \rightarrow x = (3 \times 32)/9 \times 20 = 4.8$$

$$\text{Time by B \& C} = 1/5x + 1/6x = 11/30x \rightarrow 30x/11 = 30 \times 4.8/11 = 144/11 = 13 \frac{1}{11} \text{ days}$$

40) Solution

Kavya is 3 times as efficient as Harsh (since twice more = $3x$).

$$A : B \text{ efficiency} = 3 : 1 \rightarrow \text{time} = x, 3x$$

$$C's \text{ time} = \text{time taken by A and B together} = \text{LCM of } x \text{ and } 3x = 3x$$

$$1 \text{ day work of C} = 1/x + 1/3x = 4/3x$$

$$C \text{ alone takes} = 3x/4 \text{ days}$$

$$(1/3x + 4/3x) \times 3 + (1/3x + 1/x) \times 15/4 = 1$$

$$1/x + 4/x + 5/4x + 15/4x = 1$$

$$(4 + 16 + 5 + 15)/4x = 40/4x = 1 \rightarrow x = 10$$

$$C's \text{ time} = 3x/4 = 3 \times 10 / 4 = 7.5 \text{ days}$$

41) Solution

$$\text{Nisha's 1-day work} = 1/20$$





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Mohan : Priya efficiency = 2 : 5 $\rightarrow$ Time = $5x : 2x$

Mohan's 1-day work = $1/5x$, Priya's = $1/2x$

$$(1/20 + 1/5x + 1/2x) \times 4 + (1/20 + 1/5x) \times 8/3 = 1$$

$$1/5 + 4/5x + 2/x + 2/15 + 8/15x = 1$$

$$(12 + 30 + 8)/15x = (15 - 3 - 2)/15$$

$$50/15x = 10/15 \rightarrow 50 = 10x \rightarrow x = 5$$

Mohan's time = $5x = 25$ days

42) Solution

Time taken ratio = 10:6:5.

Efficiency ratio = $(1/10):(1/6):(1/5) = 3:5:6$.

Required percentage = $[(6 - 3) / 6] * 100 = 50\%$ less efficient.

43) Solution

Time taken ratio = 10:6:5.

B alone finishes the work in 30 days.

A alone takes 50 days, and C alone takes 25 days.

LCM = 150.

Efficiency of A = 3, B = 5, and C = 6.

Required days = $150 / 14 = 75 / 7 = 10 (5/7)$ days.

44) Solution

Let the number of days Aarti takes to finish the work be x .

Then Simran takes $x + 63$ days to complete the same task.

Since Aarti is 45% more efficient, this means:

Aarti's work rate = $1/x$, and Simran's work rate = $1/(x + 63)$.

Given that Aarti's efficiency is 45% higher, we have the relation:

$$1/x = 1.45 \times 1/(x + 63)$$

Now, solving this equation:

$$(x + 63) = 1.45x$$

$$x + 63 = 1.45x$$

$$63 = 0.45x$$

$$x = 63 \div 0.45 = 140 \text{ days}$$

45) Solution

Let the efficiency of Anil be 7.





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Efficiency of Bharat = $(8/7) \times 7 = 8$.

Total work = $7 \times 20 = 140$ units.

Time taken by both together = $140 / (7 + 8) = 140 / 15 = 28/3 = 9(1/3)$ days.

46) Solution

Efficiency ratio of John, Paul, and George = 2:3:4

Time taken ratio of John, Paul, and George = $1/2 : 1/3 : 1/4 = 6:4:3$

LCM = 12

John and Paul together will take = $12 / (2 + 3) = 12 / 5 = 2.4$ days

47) Solution

Tarun's $1/7$ work takes 5 days $\Rightarrow$ full work = $5 \times 7 = 35$ days

So Tarun's efficiency = $1/35$ per day

Charu's efficiency = $1/21$ per day

At 80%, Charu's efficiency = $(1/21) \times 0.8 = 4/105$

Tarun's efficiency = $1/35 = 3/105$

Combined efficiency = $3/105 + 4/105 = 7/105 = 1/15$

So, time to complete the work = 15 days

48) Solution

Let work = LCM of ratio = 70 units

Arjun's rate = 7 units/day $\rightarrow$ Time = $70/7 = 10$ days

But it is given Arjun takes 84 days, so 7 units/day = $1/84$ work/day

$\Rightarrow 1$ unit = $1/588$ work

$\Rightarrow$ Total work = 70 units = $70 \times (1/588) = 70/588 = 35/294 = 5/42$ (not helpful here)

Better to use ratio approach directly:

If Arjun's rate = 7 units/day, and he takes 84 days $\rightarrow$ Total work = $7 \times 84 = 588$ units

Then Karthik's rate = $(10/7) \times$ Arjun's rate = 10 units/day

So Karthik's rate = $(10/7) \times (1/84) = 10 / 588 = 5 / 294$ per day

Karthik to do $5/7$ of work = $5/7 \times 1 = 5/7$

Time = Work / Rate = $(5/7) \div (5/294) = (5/7) \times (294/5) = 294 / 7 = 42$ days

49) Solution





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Prateek's 1-day work = $1/30$

Karan's 1-day work = $1/6$

Combined 1-day work = $1/5$ (since together they complete in 5 days)

Check:

$$1/30 + 1/6 = (1 + 5)/30 = 6/30 = 1/5 \quad \checkmark$$

Now, Suraj is 20% more efficient than Prateek.

$$\text{So, Suraj's efficiency} = 1/30 \times 120\% = (1/30) \times (120/100) = 1/25$$

Therefore, Suraj alone will complete the work in 25 days.

50) Solution

Let 1 woman's 1-day work = 1 unit

Then, 1 man's 1-day work = 2 units

Work done per day by 3 men and 2 women = $(3 \times 2 + 2 \times 1) = 6 + 2 = 8$ units/day

Total work in 2 days = $8 \times 2 = 16$ units

1 woman does 1 unit/day $\rightarrow$ Time to do 16 units = $16 \div 1 = 16$ days

51) Solution

Let total work = LCM = 120 units (to keep all values clean).

Ravindra does it in 20 days $\rightarrow$ Efficiency = $120 / 20 = 6$ units/day

Shahid is 20% less efficient $\Rightarrow$ Efficiency = 80% of 6 = 4.8 units/day

Then, Shahid's time = $120 / 4.8 = 25$ days $\Rightarrow$ So, $Y + 5 = 25 \Rightarrow Y = 20$

Thus, Neeraj does in 20 days $\Rightarrow$ Efficiency = $120 / 20 = 6$ units/day

Now total efficiency of all three:

Neeraj = 6, Shahid = 4.8, Ravindra = 6 $\Rightarrow$ Total = 16.8 units/day

We need time to do 60% of 120 = 72 units

Time = $72 / 16.8 = 4.2857$ days = $30/7$ days

52) Solution

Arjun + Kabir together finish in $3(1/13) = 40/13$ days $\rightarrow$ efficiency = $13/40$

Arjun + Kabir + Ritesh together finish in $2(6/17) = 40/17$ days $\rightarrow$ efficiency = $17/40$

Ritesh's efficiency = $(17/40) - (13/40) = 4/40 = 1/10$

So Ritesh alone does $1/10$ of work per day $\Rightarrow$ Time = 10 days

Now, Ritesh becomes 50% more efficient:

New efficiency = $1.5 \times (1/10) = 3/20$





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$$\text{New time} = 1 \div (3/20) = 20/3 = 6 \frac{2}{3} \text{ days}$$

53) Solution

Harshil takes 25 days to complete $5/6$ of the work, so for full work he would take $25 \times 6/5 = 30$ days.

Efficiency ratio of Harshil to Rudra is 3:10, so time ratio is 10:3.

Rudra's time to complete the full work = $30 \times 3/10 = 9$ days.

Nakul completes 80% of the work in 14.4 days, so full work = $14.4 \times 100/80 = 18$ days.

Now, Rudra and Nakul together:

$$1/9 + 1/18 = (2 + 1)/18 = 3/18 = 1/6.$$

So, together they will complete the work in 6 days.

Answer: d) 6 days

54) Solution

Anish completes 75% of the work in 15 days.

Thus, the time taken by Anish to complete the entire task is:

$$15 \times 100 / 75 = 20 \text{ days.}$$

The efficiency ratio of Anish and Bharat is 3:2.

Thus, the time ratio of Anish and Bharat is 2:3.

The time taken by Bharat to complete the work is:

$$(20 / 2) \times 3 = 30 \text{ days.}$$

Now, the combined time taken by Anish and Bharat to complete the work is:

$$1/20 + 1/30 = (3 + 2) / 60 = 5 / 60 = 1/12.$$

Thus, the time taken by Anish and Bharat together to complete the entire work is:

12 days.

The time taken by Anish and Bharat together to complete 25% of the work is:

$$12 \times 25 / 100 = 3 \text{ days.}$$

55) Solution

The efficiency ratio of Priya and Ravi is 100:75, which simplifies to 4:3.

Thus, the time ratio of Priya and Ravi is 3:4.





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The efficiency ratio of Priya and Karan is 2:1, so the time ratio of Priya and Karan is 1:2, which simplifies to 3:6.

Thus, the time ratio of Priya, Ravi, and Karan together is 3:4:6.

Let the time taken by Priya, Ravi, and Karan to complete the work individually be $3x$, $4x$, and $6x$ days, respectively.

Now, using the equation for their combined work rate:

$$1/(3x) + 1/(4x) + 1/(6x) = 1/8$$

Simplifying the equation:

$$(8 + 6 + 4) / 24x = 1/8$$

$$18 / 24x = 1/8$$

$$18 = 3x$$

$$x = 6$$

Thus, the time taken by Priya alone to complete the work is:

$$3 * 6 = 18 \text{ days.}$$

56) Solution

Ajay completes $1/4$ of the project in 6 days.

Thus, the time taken by Ajay to finish the entire project is:

$$6 * 4 = 24 \text{ days.}$$

Ajay is 50% more efficient than Bharat, so the efficiency ratio of Ajay to Bharat is 3:2.

The time ratio of Ajay to Bharat is 2:3.

Thus, the time taken by Bharat to complete the project is:

$$(24/2) * 3 = 36 \text{ days.}$$

Chitra is 20% more efficient than Bharat, so the efficiency ratio of Bharat to Chitra is 5:6.

The time ratio of Bharat to Chitra is 6:5.

Thus, the time taken by Chitra to complete the project is:

$$(36/6) * 5 = 30 \text{ days.}$$

Now, the combined time taken by Ajay, Bharat, and Chitra to complete the project:

$$1/24 + 1/36 + 1/30 = (15 + 10 + 12) / 360 = 37/360.$$

Thus, the time taken for Ajay, Bharat, and Chitra together to complete the project is:

$$360/37 \text{ days.}$$





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57) Solution

Total work = 180 units (LCM method)

1 developer's efficiency = 1 unit/day

1 tester's efficiency = 0.75 units/day

Work per day = $(6 \times 1) + (8 \times 0.75) = 6 + 6 = 12$ units/day

Time = $180 \div 12 = 15$ days

58) Solution

The total work done is represented in researcher-hours. First, calculate the total researcher-hours for the experiment when 15 researchers work 4 hours a day for 12 days:

$15 \text{ researchers} \times 4 \text{ hours/day} \times 12 \text{ days} = 720 \text{ researcher-hours}$

Now, the team consists of $15 + 5 = 20$ researchers, and they will work 3 hours a day.

The total number of researcher-hours per day will be:

$20 \text{ researchers} \times 3 \text{ hours/day} = 60 \text{ researcher-hours/day}$

To find how many days it will take to complete the same 720 researcher-hours of work, divide the total work by the daily researcher-hours:

$720 \text{ researcher-hours} \div 60 \text{ researcher-hours/day} = 12 \text{ days}$

59) Solution

The total number of hours worked by 12 gardeners in 18 days is:

$12 \text{ gardeners} \times 8 \text{ hours/day} \times 18 \text{ days} = 1728 \text{ hours}$

In 1728 hours, 46 trees are planted.

So, the number of hours required to plant 1 tree = $1728 / 46 = 37.56$ hours

Now, 9 gardeners are working 12 hours a day for 8 days. The total hours worked by 9 gardeners in 8 days is:

$9 \text{ gardeners} \times 12 \text{ hours/day} \times 8 \text{ days} = 864 \text{ hours}$

The number of trees that can be planted in 864 hours is:

$864 \text{ hours} / 37.56 \text{ hours/tree} = 23 \text{ trees}$

60) Solution

Total work = $18 \text{ workers} \times 48 \text{ days} \times 8 \text{ hours} = 6912 \text{ worker-hours}$.

Work for half the project = $6912 / 2 = 3456 \text{ worker-hours}$.

Now, for the new scenario:

The new time is 24 days and each worker works 12 hours per day, so:





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Required work = Number of workers $\times$ 24 days $\times$ 12 hours = 3456 worker-hours.

Let the number of workers be x .

$$x \times 24 \times 12 = 3456$$

$$288x = 3456$$

$$x = 3456 / 288 = 12 \text{ workers.}$$

61) Solution

Let the work done by one man in a day be x , and the work done by one woman in a day be y .

From the given information:

$$6/x + 9/y = 1/2 \text{ ----} > (1)$$

$$3/x + 3/y = 5/24 \text{ ----} > (2)$$

Multiply equation (2) by 2:

$$6/x + 6/y = 5/12$$

Now subtract equation (1) from the modified equation (2):

$$(6/x + 6/y) - (6/x + 9/y) = 5/12 - 1/2$$

This simplifies to:

$$-3/y = -5/12$$

Solving for y :

$$y = 36 \text{ days}$$

Now substitute $y = 36$ into equation (1):

$$6/x + 9/36 = 1/2$$

$$6/x = 1/2 - 1/4$$

$$6/x = 1/4$$

Solving for x :

$$x = 24 \text{ days}$$

The time taken for 4 men to complete the task is:

$$24/4 = 6 \text{ days}$$

62) Solution

$$\text{Male workers' 1-day work} = 1 / (10 \times 20) = 1/200$$

$$\text{Female workers' 1-day work} = 1 / (10 \times 40) = 1/400$$

$$\text{Total work done by 20 men} = 20 \times (1/200) = 1/10$$

$$\text{Total work done by 40 women} = 40 \times (1/400) = 1/10$$

$$\text{Together, they complete } 1/10 + 1/10 = 2/10 = 1/5 \text{ of the work per day.}$$





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Total days to complete the project = $1 / (1/5) = 5$ days.

63) Solution

Let the number of days taken by 14 workers to finish the task at half their efficiency be x .

Using the work formula:

$$(12 * 7 * 5) = (14 * 6 * x * 1/2)$$

Solving for x :

$$x = (24 * 35) / (14 * 6)$$

$$x = 2 * 5$$

$$x = 10 \text{ days}$$

64) Solution

Let the efficiency of a man be M and that of a boy be B .

From the first condition:

$$3M + 4B = 1/6 \text{ (since they complete the work in 6 days)}$$

From the second condition:

$$6M + 3B = 1/4 \text{ (since they complete the work in 4 days)}$$

Now, solving these two equations:

$$(3M + 4B) = 1/6 \text{ and } (6M + 3B) = 1/4.$$

Multiply the first equation by 2 and subtract it from the second equation to eliminate B :

$$(6M + 8B) = 1/3$$

$$(6M + 3B) = 1/4$$

Now subtract:

$$(6M + 8B) - (6M + 3B) = 1/3 - 1/4,$$

$$5B = 1/12,$$

$$B = 1/60.$$

Substitute $B = 1/60$ into the first equation:

$$3M + 4(1/60) = 1/6,$$

$$3M + 1/15 = 1/6,$$

$$3M = 1/6 - 1/15,$$

$$3M = (5 - 2)/30,$$

$$3M = 3/30,$$

$$M = 1/30.$$

Now, comparing the efficiency of a man to that of a boy:





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Efficiency ratio (M:B) = $(1/30) : (1/60) = 2:1$.

So, the efficiency of a man is twice that of a boy.

65) Solution

Total food = 75 people $\times$ 35 days = 2625 person-days

New total people = 75 + 15 = 90

New number of days = $2625 \div 90 = 29.16 \approx 29$ days

Answer: e) None of these

66) Solution

Total work = 1 (in terms of full job)

Work completed in 24 days = $1/3 \rightarrow$ Work left = $2/3$

Time left = 48 - 24 = 24 days

20 workers did $1/3$ of work in 24 days

So, 20 workers can do $1/3$ work in 24 days $\Rightarrow$ Their rate = $(1/3)/24 = 1/72$ per day

So, 20 workers' efficiency = $1/72$ per day $\Rightarrow$ 1 worker = $1/(72 \times 20) = 1/1440$

Let x be the total workers needed to finish remaining $2/3$ work in 24 days

So, $x \times 1/1440 \times 24 = 2/3$

$\Rightarrow x \times 24 = (2/3) \times 1440$

$\Rightarrow x \times 24 = 960$

$\Rightarrow x = 40$

So, required number of workers = 40

Already have 20 workers $\Rightarrow$ Extra workers = 40 - 20 = 20

67) Solution

20 men finish $1/2$ work in 8 days

So, 20 men's 1 day work = $(1/2) / 8 = 1/16$

$\rightarrow$ 1 man's 1 day work = $1/320$

Efficiency: 1 woman = 2 men

8 men = 8 units

12 women = $12 \times 2 = 24$ units

Total = 32 men-units

Team's 1 day work = $32 \times (1/320) = 1/10$

Time to complete full work = $1 / (1/10) = 10$ days



68) Solution

$$\text{Total work} = 105 \times 8 \times 50 = 42000 \text{ man-hours}$$

$$\text{Work done in 25 days} = 105 \times 8 \times 25 = 21000 \text{ man-hours}$$

Since this is $\frac{2}{5}$ of total work:

$$\Rightarrow \text{Total work} = 21000 \times \left(\frac{5}{2}\right) = 52500 \text{ man-hours}$$

$$\text{Remaining work} = 52500 - 21000 = 31500 \text{ man-hours}$$

Let additional workers = x

$$\text{New total} = (105 + x) \times 9 \times 25 = 31500$$

$$\Rightarrow (105 + x) \times 225 = 31500$$

$$\Rightarrow 105 + x = 140$$

$$\Rightarrow x = 35$$

69) Solution

Let original number of persons = x

$$\text{First part of work} = \left(\frac{2}{3}\right)W = x \times 15 \times 6$$

$$\text{Remaining } \left(\frac{1}{3}\right)W = (x-8) \times 7.5 \times 8$$

Set both expressions equal to respective portions of work:

$$x \times 15 \times 6 = \left(\frac{2}{3}\right)W \rightarrow 90x = \left(\frac{2}{3}\right)W$$

$$(x-8) \times 7.5 \times 8 = \left(\frac{1}{3}\right)W \rightarrow 60(x-8) = \left(\frac{1}{3}\right)W$$

Now add both:

$$90x + 60(x-8) = W$$

$$90x + 60x - 480 = W$$

$$150x - 480 = W$$

Also from above:

$$\left(\frac{2}{3}\right)W = 90x \Rightarrow W = \left(\frac{3}{2}\right) \times 90x = 135x$$

Now set both values of W equal:

$$150x - 480 = 135x$$

$$15x = 480 \Rightarrow x = 32$$

70) Solution

Let original number of men = x

$$\text{Work} = \text{men} \times \text{days} \times \text{hours} = x \times 90 \times 8$$

New scenario: $(x+8)$ men $\times$ 45 days $\times$ 12 hours = same work

$$x \times 90 \times 8 = (x+8) \times 45 \times 12$$

$$\Rightarrow 720x = 540x + 4320$$

$$\Rightarrow 180x = 4320 \Rightarrow x = 24$$





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Now, find men for 72 days $\times$ 6 hours:

$$\text{Work} = 24 \times 90 \times 8 = 17280$$

$$\text{Required men} = 17280 / (72 \times 6) = 40$$

71) Solution

Let the total work be represented in gardener-hours. First, calculate the total work done by $(a + 5)$ gardeners working 9 hours/day for 24 days to complete 75% of the work:

$$(a + 5) \text{ gardeners} \times 9 \text{ hours/day} \times 24 \text{ days} = 75\% \text{ of the total work}$$

This is equivalent to:

$$(a + 5) \times 9 \times 24 = 0.75 \times \text{Total Work}$$

$$\Rightarrow (a + 5) \times 216 = 0.75 \times \text{Total Work}$$

Next, calculate the total work done by $(a - 4)$ gardeners working 2.5 hours/day for 36 days to complete the remaining 25% of the work:

$$(a - 4) \text{ gardeners} \times 2.5 \text{ hours/day} \times 36 \text{ days} = 25\% \text{ of the total work}$$

This is equivalent to:

$$(a - 4) \times 2.5 \times 36 = 0.25 \times \text{Total Work}$$

$$(a - 4) \times 90 = 0.25 \times \text{Total Work}$$

$$(a + 5) \times 216 / (a - 4) \times 90 = 0.75 / 0.25$$

$$(a + 5) \times 216 / (a - 4) \times 90 = 3$$

$$(a + 5) \times 216 = 3 \times (a - 4) \times 90$$

$$(a + 5) \times 216 = 270 \times (a - 4)$$

$$216a + 1080 = 270a - 1080$$

$$270a - 216a = 1080 + 1080$$

$$54a = 2160$$

$$a = 40$$

So, the number of gardeners who completed 75% of the work is:

$$a + 5 = 40 + 5 = 45$$

72) Solution

Assume total work = 600 units (to keep values integer).

First team: $(n + 2)$ men, 6 hrs/day for 6 days = $36(n + 2)$ hours.

Given they complete 30% = 180 units:

$$\Rightarrow 36(n + 2) = 180 \rightarrow n + 2 = 5 \rightarrow n = 3$$

Second team: $(n + 4)$ men, 4 hrs/day for 15 days = $60(n + 4)$ hours.

They do 70% = 420 units:





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$$\Rightarrow 60(n + 4) = 420 \rightarrow n + 4 = 7 \rightarrow n = 3 \quad \checkmark$$

$$\text{So, } (n + 13) = 3 + 13 = 16$$

73) Solution

Total work = 1 unit.

Priya's rate = $1/6$ per day, Neha's rate = $1/9$ per day.

Priya and Neha together complete $5/18$ of the work per day.

In 3 days, they complete $5/6$ of the work.

Remaining work = $1/6$.

Priya, Neha, and Kajal together complete $1/6$ in 1 day.

Kajal's rate = $1/9$ per day.

Kajal's share of the work = $1/3$.

Kajal's payment = $\text{₹}1,500 \times 1/3 = \text{₹}500$.

74) Solution

Let the amount of work Ajay, Bhavesh, and Chirag can do per day be represented by their rates.

Ajay's rate = $1/15$ (since Ajay can complete the work in 15 days).

Bhavesh is 25% more efficient, so Bhavesh's rate = $1.25 \times (1/15) = 1/12$.

Ajay, Bhavesh, and Chirag together complete the work in 5 days, so their combined rate is $1/5$.

Therefore, the rate of Chirag = $1/5 - (1/15 + 1/12)$.

To calculate the value of Chirag's rate:

$$1/5 - (1/15 + 1/12) = 1/5 - (4/60 + 5/60) = 1/5 - 9/60 = 12/60 - 9/60 = 3/60 = 1/20.$$

So, Chirag's rate is $1/20$.

Now, we calculate their respective shares of the total wages based on their individual rates:

$$\text{Total rate} = 1/15 + 1/12 + 1/20 = 4/60 + 5/60 + 3/60 = 12/60 = 1/5.$$

$$\text{Ajay's share} = (1/15) / (1/5) \times 21,600 = (1/15) \times 5 \times 21,600 = 7,200.$$

$$\text{Bhavesh's share} = (1/12) / (1/5) \times 21,600 = (1/12) \times 5 \times 21,600 = 9,000.$$

$$\text{Chirag's share} = (1/20) / (1/5) \times 21,600 = (1/20) \times 5 \times 21,600 = 5,400.$$

$$\text{The difference between Bhavesh's and Chirag's share} = 9,000 - 5,400 = 3,600.$$

75) Solution





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Ravi's 1 day work = $1/8$

Simran's 1 day work = $1/12$

Radhika's 1 day work = $1/24$

LCM of 8, 12, and 24 = 24. Let total work = 24 units

Ravi does 3 units/day, Simran 2 units/day, Radhika 1 unit/day

Work ratio = 3 : 2 : 1 $\rightarrow$ Total = 6 parts

Ravi's share = $(3/6) \times 2400 = \text{Rs.}1200$

76) Solution

Time ratio = 3 : 5 : 7

Efficiency ratio = $1/3 : 1/5 : 1/7$

LCM of 3, 5, and 7 = 105

Efficiency = 35 : 21 : 15

Total efficiency = 35 + 21 + 15 = 71

Mohit's share = $(15/71) \times 7100 = \text{Rs.}1500$

77) Solution

Let the total work be 1 unit.

Vishal's rate = $1/14$ per day, Jai's rate = $1/11$ per day.

Vishal and Jai together complete:

$1/14 + 1/11 = 11/154 + 14/154 = 25/154$ of the work per day.

In 4 days, Vishal and Jai together will complete:

$(25/154) \times 4 = 100/154 = 50/77$ of the work.

Thus, the remaining work = $1 - 50/77 = 27/77$.

Gopal completes the remaining $27/77$ of the work in 6 days.

So, Gopal's rate is:

$(27/77) / 6 = 27/462 = 1/17$ per day.

Now, let's calculate the work done by each person:

Vishal's share of work = $4 \times (1/14) = 4/14 = 2/7$.

Jai's share of work = $4 \times (1/11) = 4/11$.

Now, the payment is split according to the work done.

Total amount = ₹3,080.

Jai's share = ₹3,080 $\times (4/11) = ₹1,120$.

78) Solution





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Let the efficiencies of Ravi, Priya, and Seema be represented as Ravi, Priya, and Seema units of work per day, respectively.

From the given information:

$$\text{Ravi} = 2 * \text{Priya}$$

$$\text{Seema} = 1.5 * \text{Priya} = (150/100) * \text{Priya} = 3 * \text{Priya} / 2.$$

Thus, the ratio of their efficiencies Ravi : Priya : Seema is:

$$\text{Ravi} : \text{Priya} : \text{Seema} = 4 : 2 : 3.$$

Now, calculate the amount of work done by Ravi, Priya, and Seema in terms of their efficiencies and the number of days worked:

$$\text{Work done by Ravi} = 4 * 6 = 24 \text{ units.}$$

$$\text{Work done by Priya} = 2 * 15 = 30 \text{ units.}$$

$$\text{Work done by Seema} = 3 * 7 = 21 \text{ units.}$$

So, the total work ratio is:

$$24 : 30 : 21 = 8 : 10 : 7.$$

Now, to calculate Seema's share of the total payment:

$$\text{Seema's share} = ₹6,000 * (7 / 25) = ₹1,680.$$

79) Solution

$$\text{Total work} = 1 \text{ unit.}$$

$$\text{Raj's rate} = 1/8 \text{ per day, Simran's rate} = 1/12 \text{ per day.}$$

Raj and Simran together complete $5/24$ work per day. In 2 days, they complete $5/12$ work.

$$\text{Remaining work} = 7/12.$$

In the next 2 days, Raj, Simran, and Kiran complete $7/12$ work.

$$\text{Let Kiran's rate} = 1/12.$$

$$\text{Kiran's share of the total work} = 2 * (1/12) = 1/6.$$

$$\text{Kiran's payment} = ₹9,000 * (1/6) = ₹1,500.$$

80) Solution

$$\text{Total wage per day of 46 women} = 82800 / 25 = 3312$$

$$\text{Wage of 1 woman per day} = 3312 / 46 = \text{Rs.}72$$

$$\text{Wage of 1 man per day} = 72 * 2 = \text{Rs.}144$$

$$\text{Let number of men} = x$$

$$\text{Total wage of } x \text{ men in 24 days} = 144 * x * 24 = 110592$$

$$\Rightarrow x = 110592 / (144 * 24) = 32$$





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81) Solution

Let the time taken by Ali to finish 50% of the work be $30x$ days.

Thus, Ali takes $60x$ days to complete the full work.

Ben takes $30x$ days to finish 75% of the work, so he takes $40x$ days to finish 100% of the work.

Charlie is twice as efficient as Ali, so Charlie's time to finish the work is half of Ali's, i.e., $30x$ days.

The least common multiple (LCM) of the times for Ali, Ben, and Charlie = $120x$ days.

Efficiency of Ali = 2, Ben = 3, and Charlie = 4.

Total efficiency = $2 + 3 + 4 = 9$

Charlie's share = $(4/9) \times 4500 = \text{Rs.}2000$

82) Solution

Prisha's 1 day work = $1/15$

Dev's 1 day work = $1/12$

Together with Mehul, they finish in 4 days $\rightarrow$ 1 day work = $1/4$

So,

$$1/15 + 1/12 + 1/Z = 1/4$$

Find $1/Z$:

LCM of 15, 12, 4 = 60

$$(4 + 5 + ?)/60 = 15/60$$

$$\text{So, } 1/Z = 1/4 - 1/15 - 1/12 = (15 - 4 - 5)/60 = 6/60 = 1/10$$

So Mehul can do $1/10$ work per day.

Efficiencies:

$$\text{Prisha : Dev : Mehul} = 1/15 : 1/12 : 1/10 = 4:5:6$$

Total ratio = $4 + 5 + 6 = 15$ parts

$$\text{Mehul's share} = (6/15) \times 4500 = \text{Rs.}1800$$

Answer: a) Rs.1800

83) Solution

Work done by Saanvi and Ishan together per day = $1/12$

Time taken with Veer = $6\frac{2}{3} = 20/3$ days

Combined work per day with Veer = $1 / (20/3) = 3/20$

So,

$$1/12 + 1/x = 3/20$$





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Let x be Veer's time to finish alone

$$1/x = 3/20 - 1/12 = (9 - 5)/60 = 4/60 = 1/15$$

So Veer alone takes 15 days

Efficiency:

$$\text{Saanvi and Ishan together:Veer} = 1/12 : 1/15 = 5:4$$

$$\text{Total ratio} = 5 + 4 = 9 \text{ parts}$$

$$\text{Veer's share} = (4/9) \times 18000 = \text{Rs.}8000$$

84) Solution

Work done in 1 day:

$$A = 1/24, B = 1/40$$

$$A + B + C = 1/10$$

$$C = 1/10 - (1/24 + 1/40) = 1/10 - (1/15) = (3-2)/30 = 1/30$$

So, C does $1/30$ of work per day

$$\text{In 10 days, C's work} = 10 \times 1/30 = 1/3 \text{ of total work}$$

$$\text{C's share} = (1/3) \times 11760 = \text{Rs.}3920$$

85) Solution

Let the amount of work a man can do in 1 day be m and the amount of work a boy can do in 1 day be b .

From the first scenario:

$$3 \text{ men} + 5 \text{ boys} = \text{total work done in 1 day (work per day)}$$

$$(3m + 5b) = 1/15 \text{ (since the task is completed in 15 days)}$$

From the second scenario:

$$6 \text{ men} + 7 \text{ boys} = \text{total work done in 1 day (work per day)}$$

$$(6m + 7b) = 1/9 \text{ (since the task is completed in 9 days)}$$

Now, we have the system of equations:

$$3m + 5b = 1/15$$

$$6m + 7b = 1/9$$

To solve this system, we first multiply the first equation by 2:

$$2(3m + 5b) = 2 \times 1/15$$

$$\Rightarrow 6m + 10b = 2/15$$

Now subtract equation 2 from the new equation:

$$(6m + 10b) - (6m + 7b) = 2/15 - 1/9$$

$$\Rightarrow 3b = (6 - 5)/45$$

$$\Rightarrow 3b = 1/45$$





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$$\Rightarrow b = 1/135$$

Now, substitute $b = 1/135$ into equation 1:

$$3m + 5(1/135) = 1/15$$

$$\Rightarrow 3m + 5/135 = 1/15$$

$$\Rightarrow 3m = 1/15 - 5/135$$

$$\Rightarrow 3m = (9 - 5)/135$$

$$\Rightarrow 3m = 4/135$$

$$\Rightarrow m = 4/405 = 1/135$$

Thus, both the work rate of a man and a boy are the same — each completes $1/135$ of the task per day.

Answer: The work done by a man and a boy is the same.

86) Solution

Efficiency of skilled gardener (S) = 125

Efficiency of average gardener (A) = 100

Efficiency of beginner gardener (B) = 80

Total work = $80 * 25 = 400$ units

From the first part of the problem:

$$(100 * 6) + (80 * 5) = 400$$

$$600 + 400 = 400$$

Thus, $x = 2$

For the second part:

$$125(y - 6) + 100(y - 3) + 80y = 400$$

$$125y - 750 + 100y - 300 + 80y = 400$$

$$305y - 1050 = 400$$

$$305y = 1450$$

$$y = 10$$

$$\text{Thus, } x + y = 2 + 10 = 12.$$

87) Solution

Let man's efficiency = 10 units

Then woman = 14 units, boy = 5 units

Work = $14 \times 5 = 70$ units

First case:

Boy works $(m-6)$ days $\rightarrow 5(m-6)$

Man and woman: $(11-m)$ days $\rightarrow (10 + 14)(11-m) = 24(11-m)$





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Then woman at 50%: $(m-8)$ days $\rightarrow 7(m-8)$

Equation:

$$5(m-6) + 24(11-m) + 7(m-8) = 70$$

$$\Rightarrow 5m - 30 + 264 - 24m + 7m - 56 = 70$$

$$\Rightarrow -12m + 178 = 70$$

$$\Rightarrow m = 9$$

Second case:

$$\text{Woman's increased efficiency} = 14 \times 10/7 = 20$$

Boy works $(n-2)$ days $\rightarrow 5(n-2)$

$$\text{Boy + man: } (8-n) \text{ days} \rightarrow (5 + 10)(8-n) = 15(8-n)$$

Woman works $(n-5)$ days $\rightarrow 20(n-5)$

Equation:

$$5(n-2) + 15(8-n) + 20(n-5) = 70$$

$$\Rightarrow 5n - 10 + 120 - 15n + 20n - 100 = 70$$

$$\Rightarrow 10n + 10 = 70$$

$$\Rightarrow n = 6$$

$$\text{So, } m + n = 9 + 6 = 15$$

88) Solution

Let the total work be W .

First part:

Total hours worked by 22 women in 16 days, working 15 hours per day:

$$22 \times 15 \times 16 = 5280 \text{ hours.}$$

$$\text{Percentage of work completed} = (5280 / \text{total work hours}) \times 100.$$

If $W = 5280$, the percentage of work completed is 33%.

Second part:

Work done by 20 women working 10 hours a day for ' d ' days to complete 20% of the total work:

$$20 \times 10 \times d = 200d \text{ hours.}$$

Since this is 20% of the total work, we have:

$$200d = 0.2 \times W.$$

We know $W = 5280$ from the first part.

$$200d = 0.2 \times 5280 \rightarrow d = 16 \text{ days.}$$

Thus, the values for the blanks are:

33% for the first blank.

16 days for the second blank.





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The correct option is A.

89) Solution

Try (i):

Vikas = $1/42$ per day, Shruti = $1/90$ per day

Work done in 15 days by Vikas = $15/42 = 5/14$

Work done in 27 days by Shruti = $27/90 = 3/10$

Combined = $5/14 + 3/10 = (50 + 42)/140 = 92/140 = 46/70$

Remaining work = $1 - 46/70 = 24/70 = 12/35$

Time by Vikas = $(12/35) \div (1/42) = 12/35 \times 42 = 504/35 = 14.4$ days $\rightarrow$

Valid

Try (iii):

Vikas = $1/45$ per day

Work by Vikas in 15 days = $15/45 = 1/3$

Shruti = $1/90 \rightarrow 27/90 = 3/10$

Combined = $1/3 + 3/10 = 19/30$

Remaining = $11/30$

Time by Vikas = $(11/30) \div (1/45) = 11/30 \times 45 = 495/30 = 16.5$ days $\rightarrow$

Valid

Try (ii):

If Vikas takes 42 days, the remaining time must be 14.4 (already calculated), not 16.8 $\rightarrow$ Invalid

Correct Answer: d) Both (i) and (iii)

90) Solution

Let trained worker = m , assistant = w

Given: $(7m + 8w) * 7 = (5m + 9w) * 8$

$\Rightarrow 49m + 56w = 40m + 72w$

$\Rightarrow 9m = 16w$

$\Rightarrow m/w = 16/9$, so $m = 16$ units, $w = 9$ units

Difference = $16 - 9 = 7$ (satisfies)

Efficiency of a child (teenage intern) = $16 * (7/8) = 14$ units

Total work = $(16 \times 7 + 9 \times 8) \times 7 = (112 + 72) \times 7 = 1288$ units

Quantity I:

8 childre



$$n = 14 \times 8 = 112 \text{ units/day}$$

$$\text{Time} = 1288/112 = 11.5 \text{ days}$$

Quantity II:

$$4 \text{ assistants} + 4 \text{ children} = 4 \times 9 + 4 \times 14 = 36 + 56 = 92 \text{ units/day}$$

$$\text{Time} = 1288/92 = 14 \text{ days}$$

So, Quantity I < Quantity II

Correct option: c) Quantity I < Quantity II

91) Solution

Quantity I:

$$\text{Pradeep's one day work} = 1/20$$

$$\text{Work done by Pradeep in 5 days} = 5 \times (1/20) = 1/4$$

$$\text{Remaining work} = 1 - 1/4 = 3/4$$

Let Kunal's one day work be x .

$$\text{Together in one day, they do: } 1/20 + x$$

$$\text{In 9 days, total work} = 9 \times (1/20 + x) = 3/4$$

$$\Rightarrow (1/20 + x) = (3/4) \div 9 = 1/12$$

$$\Rightarrow x = 1/12 - 1/20 = (5 - 3)/60 = 2/60 = 1/30$$

$$\text{So, Pradeep's efficiency : Kunal's efficiency} = (1/20) : (1/30) = 3 : 2$$

$$\Rightarrow \text{Efficiency of Pradeep is } (3/2) \times 100 = 150\%$$

Quantity II:

$$\text{Total work} = 60 \text{ units}$$

$$\text{Harish + Jitendra's 1-day work} = 4 + 3 = 7 \text{ units}$$

$$\text{Let Lakhan joined after 4 days} \rightarrow \text{work done in 4 days} = 4 \times 7 = 28 \text{ units}$$

$$\text{Remaining} = 60 - 28 = 32 \text{ units}$$

$$\text{Lakhan worked for last 4 days} \rightarrow \text{total work in 4 days} = 32$$

$$\Rightarrow (7 + \text{Lakhan}) \times 4 = 32 \rightarrow \text{Lakhan's 1-day work} = 1 \text{ unit}$$

$$\text{Lakhan's total work} = 4 \text{ units}$$

$$\text{His share} = (4/60) \times 1500 = ₹100$$

Answer: b) Quantity I < Quantity II

92) Solution

Quantity I:

$$\text{Dinesh does 20\% in 4 days} \rightarrow 100\% \text{ in 20 days}$$

$$\text{Dinesh's 1-day work} = 1/20$$





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Work done in 5 days = $5 \times \frac{1}{20} = \frac{1}{4}$

Remaining = $1 - \frac{1}{4} = \frac{3}{4}$

Let Manoj's 1-day work = x

(Dinesh + Manoj)'s 1-day work = $\frac{1}{20} + x$

Together in 9 days they finish $\frac{3}{4} \rightarrow$

$(\frac{1}{20} + x) \times 9 = \frac{3}{4}$

$\Rightarrow \frac{9}{20} + 9x = \frac{3}{4}$

$\Rightarrow 9x = \frac{3}{4} - \frac{9}{20} = \frac{(15 - 9)}{20} = \frac{6}{20} = \frac{3}{10}$

$\Rightarrow x = \frac{3}{90} = \frac{1}{30}$

So Manoj alone can do the work in 30 days

Dinesh alone can do it in 20 days

Difference = $30 - 20 = 10$ days

Quantity II:

Surya's 1-day work = $\frac{1}{40}$, Manjeet's = $\frac{1}{30}$

Combined 1-day work = $\frac{1}{40} + \frac{1}{30} = \frac{(3 + 4)}{120} = \frac{7}{120}$

Work done in first 6 days = $6 \times \frac{7}{120} = \frac{42}{120}$

Last 4 days again by Surya and Manjeet = $4 \times \frac{7}{120} = \frac{28}{120}$

Total work done by Surya and Manjeet = $(42 + 28)/120 = \frac{70}{120}$

Remaining work = $1 - \frac{70}{120} = \frac{50}{120}$

Harish's 1-day work = $\frac{1}{24}$

Let Harish worked for x days $\rightarrow$

$x \times \frac{1}{24} = \frac{50}{120}$

$\Rightarrow x = (\frac{50}{120}) \times 24 = 10$ days

Answer: b) Quantity I = Quantity II

93) Solution

Quantity I

Work = $(x - 5) \cdot 12 \cdot 24$

Also $2.5 \cdot \text{Work} = (x + 5) \cdot 2.5 \cdot 48$

$\Rightarrow (x + 5) \cdot 48 = (x - 5) \cdot 288$

$\Rightarrow 48x + 240 = 288x - 1440$

$\Rightarrow 240 + 1440 = 240x$

$\Rightarrow x = 7 \Rightarrow x + 5 = 12$

Quantity II

Total work = $16 \cdot 25 \cdot 5 = 2000$ man-hours

A woman = 0.8 man-rate, a boy = $\frac{2}{3}$ man-rate





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10 women + 6 boys in 20.5 = 100 h do

$$10 \cdot 100 \cdot 0.8 + 6 \cdot 100 \cdot (2/3) = 800 + 400 = 1200 \text{ man-h}$$

$$\text{Remaining} = 2000 - 1200 = 800 \text{ man-h} \Rightarrow \text{men needed} = 800/100 = 8$$

Quantity III

$$A \text{ rate} = 1/12, B = 1/15, C = 1/20$$

Let total time = T

$$A \text{ works } T-4, B=12, C=T-8:$$

$$(T-4)/12 + 12/15 + (T-8)/20 = 1 \Rightarrow 2T/15 + 1/15 = 1 \Rightarrow T = 7 \text{ h}$$

Quantity I > Quantity II > Quantity III

94) Solution

Quantity I:

$$\text{Pankaj} = 1/12 \text{ work per day, Rahul} = 1/24 \text{ work per day}$$

$$\text{Together: } 1/12 + 1/24 = (2 + 1)/24 = 1/8 \rightarrow \text{Pankaj and Rahul together take 8 days}$$

$$\text{Shreyas takes } 2.5 \times 8 = 20 \text{ days}$$

$$\text{Quantity I} = 20 \text{ days}$$

Quantity II:

$$\text{Chhavi} = 1/8, \text{Deepa} = 1/16, \text{Eshan} = 1/24 \text{ work per day}$$

$$\text{For 2 days: } (1/8 + 1/16 + 1/24) \times 2 = (6 + 3 + 2)/48 \times 2 = 11/24$$

$$\text{Remaining work} = 1 - 11/24 = 13/24$$

$$\text{Let total days} = x$$

$$\text{Deepa works } (x - 6), \text{Eshan works } (x - 2)$$

$$(1/16)(x - 6) + (1/24)(x - 2) = 13/24$$

Multiply both sides by 48:

$$3(x - 6) + 2(x - 2) = 26 \rightarrow 3x - 18 + 2x - 4 = 26 \rightarrow 5x = 48 \rightarrow x = 9.6$$

$$\text{Quantity II} = 9.6 \text{ days}$$

Quantity III:

$$\text{Ramesh} = 1/15 \text{ work per day} \rightarrow 5 \text{ days} = 5/15 = 1/3 \text{ work done}$$

$$\text{Remaining work} = 1 - 1/3 = 2/3$$

$$\text{Suresh} = 1/12 \text{ work per day}$$

$$\text{Time} = (2/3) \div (1/12) = (2/3) \times 12 = 8 \text{ days}$$

$$\text{Quantity III} = 8 \text{ days}$$

Quantity I > Quantity II > Quantity III

95) Solution





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Let the work done by Arun, Priya, and Mohit per day be denoted by A, P, and M, respectively.

From Statement I:

Priya and Arun together can complete the task in 12 days, so:

$$A + P = 1/12$$

Mohit can complete the task in 18 days, so:

$$M = 1/18$$

This gives us two equations, but there are still two unknowns, A and P, so Statement I alone does not provide enough information to determine Arun's individual time.

From Statement II:

Mohit and Arun worked together for 5 days. The work done by them in 5 days is:

$$5(A + M) = 5(A + 1/18) = 5A + 5/18$$

Mohit then worked for 4 more days. The work done by Mohit in 4 days is:

$$4M = 4 \times 1/18 = 2/9$$

Finally, Priya completed the remaining task in 2 days. This means the remaining work is:

$$1 - (5A + 5/18 + 2/9) = 1 - 5A - 5/18 - 2/9 = 1 - 5A - 19/18$$

The remaining work completed by Priya in 2 days is:

$$2P = 1 - 5A - 19/18 \Rightarrow 2P = 18/18 - 5A - 19/18 = -5A - 1/18$$

Now, solving this for A, we can find the time taken by Arun.

Conclusion:

Both Statement I and Statement II are necessary to determine Arun's individual time to complete the task.

96) Solution

Let total work = LCM of 40, 240, and 80 = 240 units

Ramesh + Suresh $\rightarrow$ 40 days $\rightarrow 240 \div 40 = 6$ units/day

Suresh + Mahesh $\rightarrow$ 240 days $\rightarrow 240 \div 240 = 1$ unit/day

Ramesh + Mahesh $\rightarrow$ 80 days $\rightarrow 240 \div 80 = 3$ units/day

Add all:

$$(R + S) + (S + M) + (R + M) = 6 + 1 + 3 = 10$$

$$\text{So, } 2R + 2S + 2M = 10 \rightarrow R + S + M = 5 \text{ units/day}$$

Now,

$$M = 5 - (R + S) = 5 - 6 = -1$$





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$$R = 5 - (S + M) = 5 - 1 = 4$$

$$S = 5 - (R + M) = 5 - 3 = 2$$

Time:

$$\text{Ramesh} = 240/4 = 60 \text{ days}$$

$$\text{Suresh} = 240/2 = 120 \text{ days}$$

$$\text{Difference} = 60 \text{ days}$$

Answer: b) All the statements together are sufficient

97) Solution

Analyze Statement I

Let the time taken by Machine X to complete the work be x hours.

From Statement I, Machine Y takes $2x$ hours to complete the work.

The efficiency of Machine X = $1/x$, and the efficiency of Machine Y = $1/2x$.

Analyze Statement II

Machine Y and Machine Z together complete the work in 24 hours, so their combined efficiency is $1/24$.

Analyze Statement III

Machine Y's efficiency is half of Machine X's, i.e., Machine Y's efficiency = $1/2x$.

Machine Z's efficiency is three times Machine Y's, i.e., Machine Z's efficiency = $3 * (1/2x) = 3/2x$.

Now, we can set up the equation:

$$\text{Combined efficiency of Machine Y and Z} = 1/2x + 3/2x = 1/24$$

$$\text{This simplifies to } 4/2x = 1/24$$

$$\text{Hence, } 2/x = 1/24$$

$$\text{Solving for } x, \text{ we get: } x = 48.$$

Calculate Remaining Work After Machine X Worked

$$\text{Efficiency of Machine X} = 1/48.$$

$$\text{Work done by Machine X in 15 hours} = 15 * 1/48 = 15/48 = 5/16 \text{ of the total work.}$$

$$\text{Remaining work} = 1 - 5/16 = 11/16 \text{ of the total work.}$$

Time Taken by Machines Y and Z to Complete the Remaining Work

$$\text{Combined efficiency of Machines Y and Z} = 1/24.$$

$$\text{Time taken to complete the remaining work} = (11/16) / (1/24) = (11/16) * (24) = 16.5 \text{ hours.}$$

Statements II and III together are sufficient.





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98) Solution

The total time taken by A, B, and C together to assemble the machines is $300/31$ days.

Statement I:

A can assemble 25% of the machines in 10 days, meaning A's rate of work is $1/40$ per day.

The combined rate of A, B, and C is $31/300$ per day.

Thus, B and C together work at a rate of $(31/300 - 1/40) = 4/75$ per day.

A worked for 4 days, completing $4 * 1/40 = 1/10$ of the work. The remaining work is $9/10$ of the machines.

B and C can complete the remaining work in:

$$(9/10) \div (4/75) = 15 \text{ days.}$$

Statement II:

B and C together can assemble 75% of the machines in $225/16$ days.

This implies B and C can finish the entire assembly in $75/4$ days = 18.75 days.

Thus, the remaining work (25%) will take B and C 15 days to complete.

Statement III:

This statement alone does not give enough information to calculate how long it would take B and C to complete the remaining work, as it only tells us about the rate at which A and B work together.

Conclusion:

Either Statement I or Statement II alone is sufficient to determine how long B and C will take to complete the remaining assembly. Statement III is not sufficient on its own.

Answer: Option A

99) Solution

Let the efficiency of Rahul be R, Priya be P, and Sameer be S.

From Statement I:

Rahul and Priya together can finish the work in 4 hours, so their combined efficiency is:

$$R + P = 1/4$$

Also, we know that $R = 2P$.





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Substitute $R = 2P$ into the equation:

$$2P + P = 1/4$$

$$3P = 1/4$$

$$P = 1/12$$

Thus, Priya's efficiency is $1/12$, and Rahul's efficiency is $R = 2P = 1/6$.

From Statement II:

The ratio of Priya's efficiency to Sameer's efficiency is $3:4$, so:

$$P/S = 3/4$$

$$S = 4/3 \times P = 4/3 \times 1/12 = 1/9$$

Thus, Sameer's efficiency is $1/9$.

From Statement III:

Rahul and Sameer together can complete the work in 7 hours, so their combined efficiency is:

$$R + S = 1/7$$

Substitute $R = 1/6$ and $S = 1/9$:

$$1/6 + 1/9 = 1/7$$

Simplifying:

$$3/18 + 2/18 = 1/7$$

$$5/18 \neq 1/7, \text{ which is incorrect.}$$

Since Statement III is inconsistent, it is not sufficient.

Conclusion: Statement I and Statement II together are sufficient to calculate the time taken by all three to complete the work, but Statement III is not.

Answer: c) Statement I and II together are sufficient, and Statement III alone is not sufficient.

100) Solution

Use Statement I:

Let the total work be considered as 1 unit.

The combined work rate of Akash and Ravi = $1 / 27$.

The work rate of Suresh = $1 / 54$.

Thus, the work done by Akash and Ravi together in one day = $1/27$,

and the work done by Suresh in one day = $1/54$.

Use Statement II:

The combined work rate of Ravi and Suresh = $1 / 24$.

The work rate of Akash = $1 / 72$.

Thus, the work done by Ravi and Suresh together in one day = $1/24$,





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and the work done by Akash in one day = $1/72$.

Use Statement III:

The ratio of the time taken by Akash and Ravi together to the time taken by Ravi and Suresh together is 8:9.

From this ratio, we can derive more information.

Combining Statements I, II, and III:

Work done by Akash in 15 days = $15 * (1 / 72) = 15 / 72 = 5 / 24$ of the work.

Work done by Ravi in 7.5 days = $7.5 * (1 / 27) = 7.5 / 27 = 5 / 18$ of the work.

The total work done by Akash and Ravi = $5 / 24 + 5 / 18$.

Taking the LCM of 24 and 18, which is 72, we get:

Work done by Akash and Ravi = $(15 / 72) + (20 / 72) = 35 / 72$ of the work.

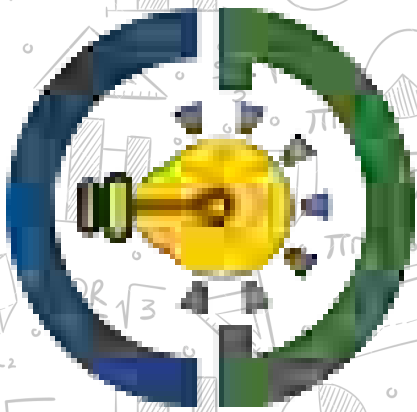
Thus, the remaining work = $1 - 35 / 72 = 37 / 72$.

Step 4: Calculate the number of days for Suresh to complete the remaining work:

Since Suresh's efficiency is $1 / 54$ per day, the time Suresh will take to complete the remaining work is:

Time taken by Suresh = $(37 / 72) / (1 / 54) = (37 / 72) * 54 = 37 * (54 / 72) = 37 * 3 / 4 = 27.75$ days.





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- 1) If 'P' and 'Q' can complete a work together in 20 days while Q alone takes 80 days to complete the work. Find the efficiency ratio of Q: P?
यदि 'P' और 'Q' मिलकर एक काम को 20 दिनों में पूरा कर सकते हैं जबकि Q को अकेले काम पूरा करने में 80 दिन लगते हैं। Q: P का दक्षता अनुपात ज्ञात कीजिए?
a) 3: 1
b) 1: 3
c) 2: 1
d) 1: 2
e) None of these
- 2) A and B can do work in 15 days. B takes twice time as A takes to do the same work. In how many days B alone can complete that work.
A और B 15 दिनों में काम कर सकते हैं। B को उसी कार्य को करने में A से दोगुना समय लगता है। B अकेले उस कार्य को कितने दिनों में पूरा कर सकता है?
a) 30 days
b) 35 days
c) 45 days
d) 40 days
e) None of these
- 3) Raushan can do a piece of work in 30 days while Diwakar can do the same piece of work in 40 days. If they work together for 8 days, how much fraction of work left?
रौशन एक काम को 30 दिनों में कर सकता है जबकि दिवाकर उसी काम को 40 दिनों में कर सकता है। यदि वे 8 दिनों तक एक साथ काम करते हैं, तो काम का कितना भाग शेष रह जाता है?
a) $\frac{6}{17}$
b) $\frac{9}{20}$
c) $\frac{7}{16}$
d) $\frac{8}{15}$
e) None of these
- 4) P, Q, and R together complete the work in 5 days, while Q alone can complete the work in 12 days, and R alone can complete the same work in 15 days. How many days can P take to complete the work alone?





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P, Q और R मिलकर काम को 5 दिनों में पूरा कर सकते हैं, जबकि Q अकेले काम को 12 दिनों में पूरा कर सकता है, और R अकेले उसी काम को 15 दिनों में पूरा कर सकता है। P को अकेले काम पूरा करने में कितने दिन लग सकते हैं?

- a) 20 days
- b) 16 days
- c) 18 days
- d) 22 days
- e) None of these

- 5) Prakash and Kunal can complete a work in 24 days and 32 days, respectively. In how many days can they complete the work if both of them work with twice of their original efficiencies?

प्रकाश और कुणाल एक काम को क्रमशः 24 दिन और 32 दिन में पूरा कर सकते हैं। यदि वे दोनों अपनी मूल क्षमता से दोगुनी क्षमता के साथ कार्य करते हैं तो वे कार्य को कितने दिनों में पूरा कर सकते हैं?

- a) 47/9 days
- b) 48/7 days
- c) 45/2 days
- d) 44/5 days
- e) None of these

- 6) Vishal and Vivek can finish a piece of work in 48 days and 60 days respectively. In how many days the work will be completed, if they work alternatively after every one day and Vishal started first?

विशाल और विवेक एक काम को क्रमशः 48 दिन और 60 दिन में पूरा कर सकते हैं। यदि वे प्रत्येक एक दिन के बाद वैकल्पिक रूप से कार्य करते हैं और विशाल पहले शुरू करता है, तो कार्य कितने दिनों में पूरा हो जाएगा?

- a) 52 days
- b) $58\frac{6}{5}$ days
- c) 58 days
- d) $52\frac{6}{5}$ days
- e) None of these

- 7) Manish can make a wall in 20 hours while Manjeet can destroy the completely build wall in 30 hours, of they work alternatively then in how many hours the wall will build?

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मनीष 20 घंटे में एक दीवार बना सकता है जबकि मंजीत 30 घंटे में पूरी तरह से बनी दीवार को नष्ट कर सकता है, यदि वे वैकल्पिक रूप से काम करते हैं तो दीवार कितने घंटे में बनेगी?

- a) 130 hours
- b) 105 hours
- c) 115 hours
- d) 90 hours
- e) None of these

8) P alone complete 60% of the work in 18 days and Q alone complete 55% of the work in 11 days. If P, Q and R together can complete the same work in 10 days, in how many days R alone complete the work?

P अकेले 60% काम 18 दिनों में पूरा करता है और Q अकेले 55% काम 11 दिनों में पूरा करता है। यदि P, Q और R मिलकर उसी कार्य को 10 दिनों में पूरा कर सकते हैं, तो R अकेले उस कार्य को कितने दिनों में पूरा करेगा?

- a) 60 days
- b) 40 days
- c) 48 days
- d) 72 days
- e) None of these

9) Vivek, Vishal and Vikram can do a piece of work in 20, 30 and 40 days, respectively. If Vikram assists Vivek and Vishal on alternate days starting from Vivek, then in how many days (integer value) the work will be completed.

विवेक, विशाल और विक्रम एक काम को क्रमशः 20, 30 और 40 दिनों में कर सकते हैं। यदि विक्रम विवेक से शुरू करके वैकल्पिक दिनों में विवेक और विशाल की सहायता करता है, तो कार्य कितने दिनों (पूर्णांक मान) में पूरा हो जाएगा।

- a) 14 days
- b) 10 days
- c) 15 days
- d) 16 days
- e) None of these

10) 150 men has done a piece of work in 30 days. After 20 days, they realized that only $\frac{4}{9}$ th of the total work was completed, to complete the work on time how many more men are needed?

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150 पुरुषों ने 30 दिनों में एक काम किया है। 20 दिनों के बाद, उन्हें एहसास हुआ कि कुल काम का केवल 4/9वां हिस्सा पूरा हुआ था, काम को समय पर पूरा करने के लिए कितने और लोगों की आवश्यकता है?

- a) 80
- b) 100
- c) 125
- d) 150
- e) None of these

11) 40 women planned to complete a certain work in 20 days working 6 hours per day. After 15 days only 3/5th of the work was completed, how many extra women required to completing the work on time working 8 hours per day?

40 महिलाओं ने प्रति दिन 6 घंटे काम करके 20 दिनों में एक निश्चित कार्य पूरा करने की योजना बनाई। 15 दिनों के बाद केवल 3/5 काम पूरा हुआ, प्रति दिन 8 घंटे काम करके समय पर काम पूरा करने के लिए कितनी अतिरिक्त महिलाओं की आवश्यकता होगी?

- a) 30
- b) 20
- c) 15
- d) 10
- e) None of these

12) X, Y and Z can do a piece of work in 6, 12 and 30 days respectively. They agreed to work together and finish the work for an amount of Rs 5100. What will be the share of the person B from the given amount?

X, Y और Z एक काम को क्रमशः 6, 12 और 30 दिनों में कर सकते हैं। वे एक साथ काम करने और 5100 रुपये की राशि पर काम खत्म करने के लिए सहमत हुए। दी गई राशि में से व्यक्ति B का हिस्सा क्या होगा?

- a) Rs 1500
- b) Rs 1200
- c) Rs 2000
- d) Rs 2500
- e) None of these

13) A contractor undertook to do a work in 42 days. He employed 25 workers to carry out the job, but after 32 days he found that only half work had been done. Now, how many workers should he employ to finish the work on time?

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एक ठेकेदार ने 42 दिनों में एक काम करने का वचन लिया। उन्होंने काम पूरा करने के लिए 25 श्रमिकों को नियुक्त किया, लेकिन 32 दिनों के बाद उन्हें पता चला कि केवल आधा काम ही हुआ था। अब, समय पर काम पूरा करने के लिए उसे कितने श्रमिक नियोजित करने चाहिए?

- a) 44
- b) 55
- c) 65
- d) 36
- e) None of these

14) 40 men can complete the work in 26 days. If 64 women started the work and after x days 16 more women joined the work and completed the remaining work in 7 days. Find the value of x .

40 आदमी काम को 26 दिनों में पूरा कर सकते हैं। यदि 64 महिलाओं ने काम शुरू किया और x दिन के बाद 16 और महिलाएं काम में शामिल हुईं और शेष काम 7 दिनों में पूरा किया। x का मान ज्ञात करें।

- a) 15 days
- b) 12 days
- c) 7.5 days
- d) 8 days
- e) None of these

15) A certain number of men can complete a job in 60 days. If there were 5 men more, it could be completed in 20 days less. How many men were working in the beginning?

एक निश्चित संख्या में आदमी एक काम को 60 दिनों में पूरा कर सकते हैं। यदि 5 आदमी अधिक होते, तो यह 20 दिन कम में पूरा हो सकता था। शुरुआत में कितने आदमी काम कर रहे थे?

- a) 20
- b) 35
- c) 30
- d) 40
- e) None of these

16) Shivam can finish 60% part of a piece of work in 24 days and Nidhi can finish the same work in 60 days and Sanket can finish $\frac{5}{8}$ th part of the work in 30 days. If Sanket work daily and assisted by Shivam and Nidhi alternatively. In how many days the work will be completed?

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शिवम किसी काम का 60% हिस्सा 24 दिनों में पूरा कर सकता है और निधि उसी काम को 60 दिनों में पूरा कर सकती है और संकेत काम का $\frac{5}{8}$ वां हिस्सा 30 दिनों में पूरा कर सकता है। यदि संकेत प्रतिदिन काम करता है और वैकल्पिक रूप से शिवम और निधि उसकी सहायता करते हैं। कार्य कितने दिनों में पूरा होगा?

- a) 28 days
- b) 24 days
- c) 12 days
- d) 36 days
- e) None of these

17) A starts a work and after 12 days he left and remaining work finished by B in 24 days. Had A worked for 18 days, B would have finished the remaining work in 16 days. Find the time taken by A to complete the work alone?

A एक काम शुरू करता है और 12 दिनों के बाद वह छोड़ देता है और शेष काम B 24 दिनों में पूरा करता है। यदि A ने 18 दिन काम किया होता, तो B ने शेष कार्य 16 दिनों में पूरा कर लिया होता। A द्वारा अकेले कार्य पूरा करने में लिया गया समय ज्ञात कीजिये?

- a) 30 days
- b) 40 days
- c) 50 days
- d) 20 days
- e) None of these

18) Aniket can complete $\frac{3}{5}$ th of the work in 18 days. Aniket and Badal together do $\frac{1}{5}$ th of the same work in $\frac{7}{2}$ days. Find time taken by Badal to complete the work alone.

अनिकेत कार्य का $\frac{3}{5}$ वां भाग 18 दिनों में पूरा कर सकता है। अनिकेत और बादल मिलकर उसी कार्य का $\frac{1}{5}$ भाग $\frac{7}{2}$ दिनों में करते हैं। बादल द्वारा अकेले कार्य पूरा करने में लिया गया समय ज्ञात कीजिए।

- a) 40 days
- b) 42 days
- c) 48 days
- d) 50 days
- e) None of these

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19) Prakash and Amit can make a wall in 24 days and 30 days respectively. While Badal can destroy the completely build wall in certain number of days. If they all start the work together then the wall is completed in 40 days. In how many days Badal can alone destroy the completely build wall?

प्रकाश और अमित क्रमशः 24 दिन और 30 दिन में एक दीवार बना सकते हैं। जबकि बादल निश्चित दिनों में पूरी बनी दीवार को नष्ट कर सकता है। यदि वे सभी एक साथ काम शुरू करते हैं तो दीवार 40 दिनों में पूरी हो जाती है। बादल अकेले कितने दिनों में पूरी बनी दीवार को नष्ट कर सकता है?

- a) 25 days
- b) 20 days
- c) 30 days
- d) 35 days
- e) None of these

20) A can finish a project in 18 days and B can finish the same project in 15 days. B worked for 10 days and left the job. How many days would it take A to finish the remaining work alone?

A एक परियोजना को 18 दिनों में पूरा कर सकता है और B उसी परियोजना को 15 दिनों में पूरा कर सकता है। B ने 10 दिनों तक काम किया और नौकरी छोड़ दी। शेष कार्य को अकेले पूरा करने में A को कितने दिन लगेंगे?

- a) 4 days
- b) 5 days
- c) 6 days
- d) 7 days
- e) None of these

21) 4 men and 6 boys complete a work in 8 days. 3 men and 7 boys complete the same work in 10 days So in how many days will 10 boys complete the same work?

4 आदमी और 6 लड़के एक काम को 8 दिनों में पूरा करते हैं। 3 आदमी और 7 लड़के उसी काम को 10 दिनों में पूरा करते हैं तो 10 लड़के उसी काम को कितने दिनों में पूरा करेंगे?

- a) 50 days
- b) 40 days
- c) 60 days
- d) 70 days
- e) None of these

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22) Kunal and Vivek can do a work together in 30 days while Vivek alone can do it in 60 days. Kunal started the work alone and left after 'x' days. If Vivek did the rest work alone in 45 days, then find the percentage of work done by Kunal.

कुणाल और विवेक एक साथ एक काम 30 दिनों में कर सकते हैं जबकि विवेक अकेले इसे 60 दिनों में कर सकता है। कुणाल ने अकेले काम शुरू किया और 'x' दिनों के बाद छोड़ दिया। यदि विवेक ने शेष कार्य अकेले 45 दिनों में किया, तो कुणाल द्वारा किये गये कार्य का प्रतिशत ज्ञात कीजिये।?

- a) 25%
- b) 30%
- c) 40%
- d) 50%
- e) None of these

23) A takes 18 days to complete a work while B completes the same work in 10 days. A and B working with C take only 6 days to complete the work. In how many days can C alone finish the work?

A को एक काम पूरा करने में 18 दिन लगते हैं जबकि B उसी काम को 10 दिनों में पूरा करता है। C के साथ काम करते हुए A और B को काम पूरा करने में केवल 6 दिन लगते हैं। C अकेले उस कार्य को कितने दिनों में पूरा कर सकता है?

- a) 80 days
- b) 84 days
- c) 72 days
- d) 90 days
- e) None of these

24) 'P' after working for 8 days left the work for 'Q' and 'R' which is completed by them together in 72 days. If 'P' had completed only 10% of the work, then find the number of days taken by 'P', 'Q' and 'R' together to complete the work.

'P' ने 8 दिन काम करने के बाद 'Q' और 'R' के लिए काम छोड़ दिया, जिसे उन्होंने मिलकर 72 दिनों में पूरा किया। यदि 'P' ने केवल 10% काम पूरा किया था, तो 'P', 'Q' और 'R' द्वारा काम पूरा करने में एक साथ लगे दिनों की संख्या ज्ञात कीजिए?

- a) 40 days
- b) 45 days
- c) 84 days
- d) 66 days
- e) None of these

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25) Sumit can complete a work in 'a' days while Anjali takes $(a + 10)$ days to complete the same work. If both of them together can complete the whole work in 12 days, then find the value of 'a'.

सुमित एक काम को 'a' दिनों में पूरा कर सकता है जबकि अंजलि को उसी काम को पूरा करने में $(a + 10)$ दिन लगते हैं। यदि वे दोनों मिलकर पूरा कार्य 12 दिनों में पूरा कर सकते हैं, तो 'a' का मान ज्ञात कीजिए।

- a) 15
- b) 20
- c) 10
- d) 25
- e) None of these

26) Twelve boys can complete $\frac{3}{5}$ part of a work in 8 days working 9 hours a day. How many women can complete rest of the work if they work for 6 hours a day for 12 days and efficiency of a boy is four times the efficiency of a woman.

17) बारह लड़के प्रतिदिन 9 घंटे काम करके एक काम का $\frac{3}{5}$ भाग 8 दिनों में पूरा कर सकते हैं। कितनी महिलाएं शेष कार्य पूरा कर सकती हैं यदि वे 12 दिनों तक प्रति दिन 6 घंटे काम करती हैं और एक लड़के की दक्षता एक महिला की दक्षता से चार गुना है।

- a) 32
- b) 28
- c) 38
- d) 24
- e) None of these

27) Arun and Himanshu can complete a piece of work in 56 days and 42 days, respectively, while Chandu can destroy the work in 32 days. If Arun and Himanshu started doing the work together and after 20 days, Chandu joins them, then find the time taken by them to complete the whole work.

अरुण और हिमांशु एक काम को क्रमशः 56 दिनों और 42 दिनों में पूरा कर सकते हैं, जबकि चंदू उस काम को 32 दिनों में नष्ट कर सकता है। यदि अरुण और हिमांशु एक साथ काम करना शुरू करते हैं और 20 दिनों के बाद, चंदू उनके साथ जुड़ जाता है, तो पूरा काम पूरा करने में उनके द्वारा लिया गया समय ज्ञात कीजिए।

- a) 48 days
- b) 32 days
- c) 42 days
- d) 36 days
- e) None of these





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- 28) (a + 5) women working 9 hours per day can complete 75% of a work in 24 days while (a - 4) women working $2\frac{1}{2}$ hours a day can complete the remaining work in 36 days. Find the number of women who completed 75% of the work.
(a + 5) प्रतिदिन 9 घंटे काम करने वाली महिलाएं 24 दिनों में 75% काम पूरा कर सकती हैं जबकि (a - 4) प्रतिदिन $2\frac{1}{2}$ घंटे काम करने वाली महिलाएं शेष काम 36 दिनों में पूरा कर सकती हैं। उन महिलाओं की संख्या ज्ञात कीजिये जिन्होंने 75% कार्य पूरा किया।
a) 54
b) 45
c) 38
d) 34
e) None of these
- 29) If 20 men or 25 women or 36 boys can do a piece of work in 15 days, working 9 hours a day, how many men must be associated with 10 women and 18 boys to do same work in 10 days working 10 hours in a day?
यदि 20 पुरुष या 25 महिलाएं या 36 लड़के एक काम को प्रतिदिन 9 घंटे काम करके 15 दिनों में कर सकते हैं, तो कितने पुरुषों को 10 महिलाओं और 18 लड़कों के साथ मिलकर उसी काम को 10 दिन में 10 घंटे काम करना होगा एक दिन?
a) 6
b) 4
c) 3
d) 9
e) None of these
- 30) Akash takes 8 days more than Diwakar while Sumit takes 4 days less than Diwakar, to complete a work. If Sumit is 60% more efficient than Akash, then find the time taken by Akash, Diwakar and Sumit, together to complete $73\frac{3}{4}\%$ of the work together.
एक कार्य को पूरा करने में आकाश को दिवाकर से 8 दिन अधिक लगते हैं जबकि सुमित को दिवाकर से 4 दिन कम लगते हैं। यदि सुमित आकाश से 60% अधिक कुशल है, तो आकाश, दिवाकर और सुमित द्वारा मिलकर $73\frac{3}{4}\%$ काम पूरा करने में लगा समय ज्ञात कीजिए।
a) 6 days
b) 5 days
c) 7 days
d) 9 days
e) None of these





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- 31) Badal alone can complete 80% of a work in 28 days. Rahul alone takes 7 more days than Badal alone to complete the same work. Rahul and Vijay can together complete the same work in 17.5 days. If Rahul, Badal and Vijay work on alternate days starting with Vijay followed by Rahul and then Badal, then how many days will it take to complete the entire work?

बादल अकेले 80% काम 28 दिनों में पूरा कर सकता है। उसी कार्य को पूरा करने में अकेले राहुल को बादल से 7 दिन अधिक लगते हैं। राहुल और विजय मिलकर समान कार्य को 17.5 दिनों में पूरा कर सकते हैं। यदि राहुल, बादल और विजय वैकल्पिक दिनों में काम शुरू करते हैं, तो विजय के बाद राहुल और फिर बादल काम करते हैं, तो पूरे काम को पूरा करने में कितने दिन लगेंगे?

- a) 50 days
- b) 40 days
- c) 35 days
- d) 45 days
- e) None of these

- 32) Anjali, Kartika and Prabha can do a piece of work in 12 days, 15 days and 20 days, respectively and the efficiency of Sapna is equal to the average of efficiencies of Anjali, Kartika and Prabha. If Anjali and Kartika work together on it for initial 4 days, then find the time taken by Prabha and Sapna together to complete the remaining work.

अंजलि, कार्तिका और प्रभा एक काम को क्रमशः 12 दिन, 15 दिन और 20 दिनों में कर सकती हैं और सपना की दक्षता अंजलि, कार्तिका और प्रभा की औसत दक्षता के बराबर है। यदि अंजलि और कार्तिका आरंभिक 4 दिनों तक इस पर एक साथ काम करती हैं, तो प्रभा और सपना द्वारा शेष कार्य को पूरा करने में लिया गया समय ज्ञात कीजिए।

- a) 60/11 days
- b) 50/9 days
- c) 21/8 days
- d) 24/7 days
- e) None of these

- 33) Puja while working 40% more efficiently can complete a piece of work in 25 days and Sonali while working 25% more efficiently can complete the same work in 21 days. While working with their original efficiencies Puja and Sonali together worked for 9 days, then find the time taken by Puja alone to complete the remaining work with his original efficiency.

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पूजा 40% अधिक कुशलता से काम करते हुए एक काम को 25 दिनों में पूरा कर सकती है और सोनाली 25% अधिक कुशलता से काम करते हुए उसी काम को 21 दिनों में पूरा कर सकती है। अपनी मूल दक्षता के साथ काम करते समय पूजा और सोनाली ने मिलकर 9 दिनों तक काम किया, तो शेष कार्य को अपनी मूल दक्षता के साथ अकेले पूरा करने में पूजा द्वारा लिया गया समय ज्ञात कीजिए।

- a) 18 days
- b) 14 days
- c) 12 days
- d) 10 days
- e) None of these

34) 'A', 'B' and 'C' together can do a work in 4 days which is 2 days less than time taken by 'B' and 'C' together to do it. 'A' and 'B' started doing the work together and after 2 days 'A' left and 'C' joined 'B'. 'B' left the work 2 days before the completion of the work and 'C' finished the remaining work alone. Find the total number of days taken to complete the work if the efficiency of 'C' is 120% of that of 'A'.

'ए', 'बी' और 'सी' मिलकर एक काम को 4 दिनों में कर सकते हैं जो कि 'बी' और 'सी' द्वारा इसे पूरा करने में लिए गए समय से 2 दिन कम है। 'ए' और 'बी' ने एक साथ काम करना शुरू किया और 2 दिनों के बाद 'ए' चला गया और 'सी' 'बी' में शामिल हो गया। 'बी' ने काम पूरा होने से 2 दिन पहले काम छोड़ दिया और 'सी' ने शेष काम अकेले पूरा किया। यदि 'C' की दक्षता 'A' की दक्षता 120% है, तो कार्य पूरा करने में लगने वाले दिनों की कुल संख्या ज्ञात कीजिए।

- a) 5 days
- b) 7 days
- c) 11 days
- d) 17 days
- e) None of these

35) Pankhuri and Kajal can do a work in 16 days and 36 days, respectively. For first few days Pankhuri and Kajal worked together and then Pankhuri left the work and Rashmi joined who individually can complete the work in 24 days. If the total work has been completed in overall 12 days then find for how many days did Pankhuri and Kajal work together?

पंखुड़ी और काजल एक काम को क्रमशः 16 दिन और 36 दिन में कर सकती हैं। पहले कुछ दिनों तक पंखुड़ी और काजल ने एक साथ काम किया और फिर पंखुड़ी ने काम छोड़ दिया और रश्मी शामिल हो गई जो व्यक्तिगत रूप से 24 दिनों में काम पूरा कर सकती है। यदि कुल कार्य कुल मिलाकर 12 दिनों में पूरा हो गया है तो पंखुड़ी और काजल ने एक साथ कितने दिनों तक कार्य किया?

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- a) 8 days
- b) 6 days
- c) 10 days
- d) 14 days
- e) None of these

36) Raghu, Mohan and Naveen are three Workers, employed by a contractor. They Completed the whole work in 7.5 days. Initially all of them worked together, but the last 60% of the work was completed by only Mohan and Naveen together. Raghu worked with Mohan and Naveen only for initial 3 days then he left the Work due to his poor health. Also, Mohan takes 40% less time to finish the Work alone than that of Naveen working alone. If they were paid Rs 4630 for the Entire work, then what is the share of the most efficient person?

रघु, मोहन और नवीन तीन श्रमिक हैं, जो एक ठेकेदार द्वारा नियोजित हैं। उन्होंने पूरा काम 7.5 दिनों में पूरा किया। प्रारंभ में सभी ने एक साथ काम किया, लेकिन अंतिम 60% काम केवल मोहन और नवीन ने मिलकर पूरा किया। रघु ने केवल शुरुआती 3 दिनों तक मोहन और नवीन के साथ काम किया, फिर अपने खराब स्वास्थ्य के कारण उसने काम छोड़ दिया। साथ ही, नवीन को अकेले काम पूरा करने में मोहन को अकेले काम पूरा करने में 40% कम समय लगता है। यदि उन्हें पूरे कार्य के लिए 4630 रुपये का भुगतान किया गया, तो सबसे कुशल व्यक्ति का हिस्सा कितना है?

- a) Rs.926
- b) Rs.1024
- c) Rs.926
- d) Rs.1024
- e) None of these

37) Working together Ragini and Shivani take 11.11% more number of days than Ragini, Shivangi and Payal together take and Shivangi and Payal working together, take $\frac{4}{3}$ more number of days than Ragini, Shivani and Payal take together. If Ragini, Shivani and Payal all have Worked together till the completion of the work and Ragini has received Rs 432 out of the total earning of Rs 1080, then in how many days did Ragini, Shivani and Payal together complete the whole work?





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रागिनी और शिवानी को एक साथ काम करने में रागिनी, शिवांगी और पायल को एक साथ काम करने में लगे दिनों से 11.11% अधिक दिन लगते हैं और शिवांगी और पायल को एक साथ काम करने में रागिनी, शिवानी और पायल को एक साथ काम करने में लगे दिनों से $\frac{4}{3}$ अधिक दिन लगते हैं। यदि रागिनी, शिवानी और पायल सभी ने काम पूरा होने तक एक साथ काम किया है और रागिनी को 1080 रुपये की कुल कमाई में से 432 रुपये मिले हैं, तो रागिनी, शिवानी और पायल ने मिलकर पूरा काम कितने दिनों में पूरा किया?

- a) 3
- b) 2
- c) 5
- d) 4
- e) None of these

38) Rahul, Vineet and Kishan are three friends. Vineet takes 2 Days less than Kishan to complete the Work. If Rahul started a work and 8 days Later Vineet joins him, then the work gets Completed in 4 more days. Working Together Rahul, Vineet and Kishan can Complete thrice the original work in $4\frac{2}{7}$ Days. In how many days Vineet can Complete twice the original work with four times the efficiency working alone?

रा राहुल, विनीत और किशन तीन दोस्त हैं। विनीत को कार्य पूरा करने में किशन से 2 दिन कम लगते हैं। यदि राहुल ने कोई काम शुरू किया और 8 दिन बाद विनीत उसके साथ जुड़ गया, तो काम 4 और दिनों में पूरा हो जाता है। एक साथ काम करते हुए राहुल, विनीत और किशन मूल कार्य का तीन गुना $4\frac{2}{7}$ दिनों में पूरा कर सकते हैं। विनीत अकेले काम करते हुए मूल कार्य को चार गुना दक्षता के साथ कितने दिनों में पूरा कर सकता है?

- a) 3
- b) 2
- c) 5
- d) 4
- e) None of these

39) A and B together work for 6 days and leave the job C can do the whole work in 30 days. B can complete the whole work alone in _____. If the efficiency ratio of A, B and C is 6: 3: 2, then in _____ days can C alone finish the remaining work?

The values given in which of the following options will fill the blanks in the same order in which is it given to make the above statement true:

A और B एक साथ 6 दिन काम करते हैं और काम छोड़ देते हैं C पूरा काम 30 दिनों में कर सकता है। B अकेले पूरा काम _____ में पूरा कर सकता है। यदि A, B और C का दक्षता अनुपात 6: 3: 2 है, तो क्या C अकेले शेष कार्य को _____ दिनों में पूरा कर सकता है?

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- A. 12, 20
- B. 10, 14
- C. 20, 3
- D. 18, 12
- a) Only A
- b) Both A and D
- c) Only C
- d) Both B and C
- e) All A, B, C and D

- 40) A, B and C can complete a work in 140 days, 112 days and 80 days respectively. All of them started working together but after ____% of the work is completed, C left the job and now A and B increased their efficiencies by 50% and ____% respectively. The whole work is completed in 39 days.

The values given in which of the following options will fill the blanks in the same order in which is it given to make the above statement true:

A, B और C एक काम को क्रमशः 140 दिन, 112 दिन और 80 दिन में पूरा कर सकते हैं। उन सभी ने एक साथ काम करना शुरू किया लेकिन ____% काम पूरा होने के बाद, C ने नौकरी छोड़ दी और अब A और B ने अपनी दक्षता क्रमशः 50% और ____% बढ़ा दी। पूरा कार्य 39 दिनों में पूरा हो जाता है।

निम्नलिखित में से किस विकल्प में दिए गए मान रिक्त स्थान को उसी क्रम में भरेंगे जिस क्रम में यह उपरोक्त कथन को सत्य बनाने के लिए दिया गया है:

- A. 20, 60
- B. 41, 20
- C. 58, 44
- D. 60, 100
- a) Only A and C
- b) Only D
- c) Only B and D
- d) Only C
- e) All A, B, C and D

- 41) P, Q and R can do a certain piece of work in 70 days, 80 days and 112 days, respectively. All of them started working together but after ____ days, P and Q left the job and the remaining work is done by R alone in ____ days.

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The values given in which of the following options will fill the blanks in the same order in which is it given to make the above statement true:

P, Q और R एक निश्चित कार्य को क्रमशः 70 दिन, 80 दिन और 112 दिन में कर सकते हैं। उन सभी ने एक साथ काम करना शुरू किया लेकिन ____ दिनों के बाद, P और Q ने काम छोड़ दिया और शेष काम R ने अकेले ____ दिनों में पूरा किया।

निम्नलिखित में से किस विकल्प में दिए गए मान रिक्त स्थान को उसी क्रम में भरेंगे जिस क्रम में यह उपरोक्त कथन को सत्य बनाने के लिए दिया गया है:

- A. 19, 34
- B. 21, 28
- C. 27, 14
- D. 20, 23
- a) Only A and C
- b) Only B
- c) Only B and D
- d) Only C
- e) All A, B, C and D

- 42) 24 women can complete ____% of the whole work in 12 days working 10 hours in a day. 30 women working 8 hours in a day can complete 30% of the whole work in ____ days.

The values given in which of the following options will fill the blanks in the same order in which is it given to make the above statement true:

24 महिलाएं एक दिन में 10 घंटे काम करके 12 दिनों में पूरे काम का ____% पूरा कर सकती हैं। एक दिन में 8 घंटे काम करने वाली 30 महिलाएं पूरे काम का 30% ____ दिनों में पूरा कर सकती हैं।

निम्नलिखित में से किस विकल्प में दिए गए मान रिक्त स्थान को उसी क्रम में भरेंगे जिस क्रम में यह उपरोक्त कथन को सत्य बनाने के लिए दिया गया है:

- A. 50, 14
- B. 60, 16
- C. 36, 10
- D. 20, 12
- a) Only A and C
- b) Only D
- c) Only B and D
- d) Only C
- e) All A, B, C and D

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- 43) Amit alone can paint a house in 15 days. Amit and Shashi together can paint the same house in ____ days. Tarun is ____% as efficient as Shashi. Tarun alone can paint the house in 11.25 days.

The values given in which of the following options will fill the blanks in the same order in which it is given so as to make the statement true?

अमित अकेले एक घर को 15 दिनों में पेंट कर सकता है। अमित और शशि मिलकर एक ही घर को ____ दिनों में पेंट कर सकते हैं। तरुण, शशि की तरह ____% कुशल है। तरुण अकेले घर को 11.25 दिनों में पेंट कर सकता है।

निम्नलिखित में से किस विकल्प में दिए गए मान रिक्त स्थान को उसी क्रम में भरेंगे जिस क्रम में दिए गए हैं ताकि कथन सत्य हो जाए?

- A. $5\frac{5}{8}$, 80
- B. $5\frac{3}{4}$, 100
- C. 6, 120
- D. 10, 120
- a) Only A
- b) Only C and D
- c) Only A and B
- d) Only C
- e) All A, B, C and D

- 44) The following questions are accompanied by two statements (I) and (II). You have to determine which statements(s) is/are sufficient/necessary to answer the questions.

The efficiency of Sanjay is what percent more or less than the efficiency of Manish?

Statement I: The time taken by Manish alone to complete 60% of the work is 2 days less than the time taken by Sanjay alone to complete the whole work.

Statement II: If Sanjay and Manish work together for only 8 days, then the remaining work is completed by Manish in 11 more days. In this way, they take 5 more days to do that work than the time taken by them together to do that whole work.

निम्नलिखित प्रश्नों के साथ दो कथन (I) और (II) दिए गए हैं। आपको यह निर्धारित करना होगा कि प्रश्नों का उत्तर देने के लिए कौन से कथन पर्याप्त/आवश्यक हैं।

संजय की दक्षता मनीष की दक्षता से कितने प्रतिशत अधिक या कम है?

कथन I: अकेले मनीष द्वारा 60% काम पूरा करने में लिया गया समय, अकेले संजय द्वारा पूरा काम पूरा करने में लिए गए समय से 2 दिन कम है।

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कथन II: यदि संजय और मनीष केवल 8 दिनों के लिए एक साथ काम करते हैं, तो शेष कार्य मनीष द्वारा 11 दिनों में पूरा किया जाता है। इस प्रकार, उस कार्य को करने में उन्हें उस पूरे कार्य को करने में लगे समय से 5 दिन अधिक लगते हैं।

- a) Both the statements taken together are necessary to answer the question, but neither of the statements alone is sufficient to answer the question.
- b) Either statement I or statement II by itself is sufficient to answer the question.
- c) Statements I and II taken together are not sufficient to answer the question.
- d) Statement I alone is sufficient to answer the question, but statement II alone is not sufficient to answer the questions.
- e) Statement II alone is sufficient to answer the question but statement I alone is not sufficient to answer the question.

45) The following questions are accompanied by two statements (I) and (II). You have to determine which statements(s) is/are sufficient/necessary to answer the questions.

Time taken by Shubham to complete the work alone.

Statement I: Nakul and Shubham can do a piece of work in 20 days together and Harsh can do the same piece of work in 30 days alone.

Statement II: Harsh works for 11 days and Shubham works for 13 days and remaining work was done by Nakul in 12-days.

निम्नलिखित प्रश्नों के साथ दो कथन (I) और (II) दिए गए हैं। आपको यह निर्धारित करना होगा कि प्रश्नों का उत्तर देने के लिए कौन से कथन पर्याप्त/आवश्यक हैं।

शुभम द्वारा अकेले कार्य पूरा करने में लिया गया समय।

कथन I: नकुल और शुभम एक साथ एक काम को 20 दिनों में कर सकते हैं और हर्ष अकेले उसी काम को 30 दिनों में कर सकता है।

कथन II: हर्ष 11 दिनों तक कार्य करता है और शुभम 13 दिनों तक कार्य करता है और शेष कार्य नकुल द्वारा 12 दिनों में पूरा किया जाता है।

- a) Statement (I) alone is sufficient to answer the question but statement (II) alone is not sufficient to answer the question.
- b) Statement (II) alone is sufficient to answer the question but statement (I) alone is not sufficient to answer the question.
- c) Both the statements taken together are necessary to answer the question, but neither of the statements alone is sufficient to answer the question.
- d) Either statement (I) or statement (II) by itself is sufficient to answer the question.
- e) Statements (I) and (II) taken together are not sufficient to answer the Question

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46) In the following questions three statements either or I, II and III are given. You have to use your knowledge of mathematics to answer which statement(s) is/are sufficient to answer the question.

Find the time taken by Sanjay, Manish, and Anit together to complete a work.

Statement I: Sanjay and Manish together can complete the work in $8/3$ hours. The efficiency of Sanjay is twice the efficiency of Manish.

Statement II: Anit alone can complete the work in 3 hours.

Statement III: Sanjay and Anit together can complete the work in $12/7$ hours. The ratio of efficiency of Sanjay and Anit is 3:4.

निम्नलिखित प्रश्नों में तीन कथन या तो I, II और III दिए गए हैं। आपको यह उत्तर देने के लिए गणित के अपने ज्ञान का उपयोग करना होगा कि कौन सा कथन प्रश्न का उत्तर देने के लिए पर्याप्त है/हैं।

किसी कार्य को पूरा करने में संजय, मनीष और अनित द्वारा मिलकर लिया गया समय ज्ञात कीजिए।

कथन I: संजय और मनीष मिलकर काम को $8/3$ घंटे में पूरा कर सकते हैं। संजय की दक्षता मनीष की दक्षता से दोगुनी है।

कथन II: एनित अकेले 3 घंटे में काम पूरा कर सकता है।

कथन III: संजय और अनित मिलकर काम को $12/7$ घंटे में पूरा कर सकते हैं। संजय और अनित की दक्षता का अनुपात 3:4 है।

a) Either statement II alone or statement I alone is sufficient.

b) Statement I alone is sufficient and Statement II alone is not sufficient

c) Statement I and II together are sufficient and Statement III alone is not sufficient.

d) All the statements are necessary.

e) Any two of the statements are sufficient

47) In the following questions three statements either or I, II and III are given. You have to use your knowledge of mathematics to answer which statement(s) is/are sufficient to answer the question

A and C began to work but A and Clef the work 8 hours and 16 hours before the completion of the work respectively. B worked for 12 hours. Find the total time taken for the completion of work. to complete the work is 24 hours, 30 hours and

Statement I: The time taken for A, B and C alone 40 hours respectively.

Statement II: The time taken for A and B together to complete the work is $40/3$ hours Statement III: The time taken for B and C together to complete the work is

$120/7$ days.

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निम्नलिखित प्रश्नों में तीन कथन या तो I, II और III दिए गए हैं। आपको यह उत्तर देने के लिए गणित के अपने ज्ञान का उपयोग करना होगा कि कौन सा कथन प्रश्न का उत्तर देने के लिए पर्याप्त है/हैं

A और C ने काम करना शुरू किया लेकिन A और काम पूरा होने से क्रमशः 8 घंटे और 16 घंटे पहले काम छोड़ दिया। B ने 12 घंटे तक काम किया। कार्य पूरा होने में लगा कुल समय ज्ञात कीजिये। काम पूरा करने के लिए 24 घंटे, 30 घंटे और का समय लगता है

कथन I: अकेले A, B और C के लिए लिया गया समय क्रमशः 40 घंटे है।

कथन II: A और B द्वारा मिलकर काम पूरा करने में लिया गया समय $40/3$ घंटे है। कथन III: B और C द्वारा काम पूरा करने में लिया गया समय $120/7$ दिन है।

- a) Statement I alone is sufficient.
- b) Statement II and III together are sufficient
- c) Statement I and II together are sufficient
- d) Statement III alone is sufficient
- e) Any two of the statements are sufficient

48) Three workers, A, B, and C, work in a furniture store, where they make three different types of furniture: a bed, an almirah, and a sofa.

A, B, and C together can make a bed in 8 days. B makes a bed three times faster than he makes an almirah. B alone takes 50% more time than A alone to make an almirah. C is three times as efficient as B in almirah-making. A's sofa-making speed is half of his almirah-making speed. B is 150% as efficient as A in sofa-making. C alone can make a sofa in $320/11$ days. A, B, and C can together make a sofa in 20 days. A alone takes 25 days to make a bed.

Quantity I: Average time taken for A and C together to make a bed and then a sofa.

Quantity II: Average time taken for B and C together to make an almirah and then a sofa.

तीन कर्मचारी, ए, बी और सी, एक फर्नीचर स्टोर में काम करते हैं, जहां वे तीन अलग-अलग प्रकार के फर्नीचर बनाते हैं: एक बिस्तर, एक अलमारी और एक सोफा।

A, B और C मिलकर 8 दिनों में एक बिस्तर बना सकते हैं। B, अलमारी बनाने की तुलना में तीन गुना तेजी से बिस्तर बनाता है। B को एक अलमारी बनाने में अकेले A से 50% अधिक समय लगता है। अलमारी बनाने में C, B से तीन गुना अधिक कुशल है। A की सोफा बनाने की गति उसकी अलमारी बनाने की गति की आधी है। सोफा बनाने में B, A से 150% अधिक कुशल है। C अकेले $320/11$ दिनों में एक सोफा बना सकता है। A, B और C मिलकर 20 दिनों में एक सोफा बना सकते हैं। A को एक बिस्तर बनाने में अकेले 25 दिन लगते हैं।

मात्रा I: A और C द्वारा मिलकर एक बिस्तर और फिर एक सोफा बनाने में लिया गया औसत समय।

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मात्रा II: B और C द्वारा मिलकर एक अलमारी और फिर एक सोफा बनाने में लिया गया औसत समय।

- a) $QI > QII$
- b) $QI < QII$
- c) $QI = QII$
- d) $QI \leq QII$
- e) $QI \geq QII$

- 49) Quantity I: After completing 25% of a work, Sumit worked for 24 more days to complete the remaining work alone. Vivek is 20% less efficient than Sumit. Both of them started the work together and completed 45% of the work. The remaining work is completed by Vivek alone. Find the total number of days taken to complete the work.

Quantity II: Aman, Ashutosh and Anurag alone can complete a work in 40 days, 48 days and 60 days respectively. All of them started working together but after 8 days Aman left the job and 6 days before the completion of the work Anurag also left. In how many days the work is completed?

मात्रा I: किसी काम का 25% पूरा करने के बाद, सुमित ने शेष काम को अकेले पूरा करने के लिए 24 और दिनों तक काम किया। विवेक, सुमित से 20% कम कुशल है। दोनों ने एक साथ काम शुरू किया और 45% काम पूरा कर लिया। शेष कार्य विवेक अकेले पूरा करता है। कार्य पूरा करने में लगे कुल दिनों की संख्या ज्ञात कीजिए।

मात्रा II: अमन, आशुतोष और अनुराग अकेले एक काम को क्रमशः 40 दिन, 48 दिन और 60 दिन में पूरा कर सकते हैं। उन सभी ने एक साथ काम करना शुरू किया लेकिन 8 दिन बाद अमन ने नौकरी छोड़ दी और काम पूरा होने से 6 दिन पहले अनुराग ने भी नौकरी छोड़ दी। कार्य कितने दिनों में पूरा होगा?

- a) $QI > QII$
- b) $QI < QII$
- c) $QI = QII$
- d) $QI \leq QII$
- e) $QI \geq QII$

- 50) Quantity I: Ankit alone can complete a work in 25 days and Badal alone can do the same work in 20 days. Ankit started the work and after working 7 days Badal joined Ankit to finish the remaining work. In how many days, the total work will be finished?

Quantity II: Deepak, Farahan and Himesh can do a piece of work in 12 days, 24 days and 36 days respectively. All of them started working together. If Deepak

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works for only 4 days and Himesh left the work 6 days before its completion then in how many days the work is completed.

Quantity III: P and Q individually can complete the work in 48 days and 72 days respectively. If the time taken by R alone to complete the work is 1.25 times the time taken by P and Q together to complete the work, find the time taken by R to complete the work while working alone.

मात्रा I: अंकित अकेले एक काम को 25 दिनों में पूरा कर सकता है और बादल अकेले उसी काम को 20 दिनों में पूरा कर सकता है। अंकित ने काम शुरू किया और 7 दिन काम करने के बाद बादल शेष काम को पूरा करने के लिए अंकित के साथ शामिल हो गया। कुल कार्य कितने दिनों में समाप्त हो जायेगा?

मात्रा II: दीपक, फरहान और हिमेश एक काम को क्रमशः 12 दिन, 24 दिन और 36 दिन में कर सकते हैं। वे सभी एक साथ काम करने लगे। यदि दीपक केवल 4 दिन काम करता है और हिमेश काम पूरा होने से 6 दिन पहले काम छोड़ देता है तो काम कितने दिनों में पूरा होगा?

मात्रा III: P और Q व्यक्तिगत रूप से क्रमशः 48 दिनों और 72 दिनों में कार्य पूरा कर सकते हैं। यदि R द्वारा अकेले काम पूरा करने में लिया गया समय, P और Q द्वारा मिलकर काम पूरा करने में लिए गए समय का 1.25 गुना है, तो R द्वारा अकेले काम करते हुए काम पूरा करने में लिया गया समय ज्ञात कीजिए।

- a) $QI > QII > QIII$
- b) $QI < QII < QIII$
- c) $QI > QII < QIII$
- d) $QI = QII = QIII$
- e) $QI > QII = QIII$

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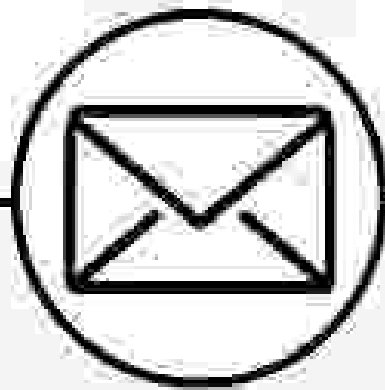
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SOLUTION

1) Answer - A

$$\text{Time taken of P} = (1/20) - (1/80) = 3/80$$

$$\text{Time ratio} = 3:1$$

$$\text{Efficiency ratio} = 1:3$$

2) Answer - C

$$\text{Let efficiency of B} = x \text{ unit/day}$$

$$\text{A's efficiency} = 2x \text{ unit/day}$$

$$\text{Total work} = 15 \times (2x + x) = 45x$$

$$\text{Time taken by B} = 45x/x = 45 \text{ days}$$

3) Answer - D

$$\text{Raushan} \quad 30 \quad 4$$

$$120$$

$$\text{Diwakar} \quad 40 \quad 3$$

$$4\text{-day work} = (4+3) \times 8 = 56 \text{ units}$$

$$\text{Remaining work} = 120 - 28 = 64 \text{ units}$$

$$\text{fraction of work left} = 64/120 = 8/15$$

4) Answer - A

$$(P + Q + R)\text{'s one day work} = 1/5$$

$$Q\text{'s one day work} = 1/12$$

$$R\text{'s one day work} = 1/15$$

$$P\text{'s one day work} = (1/5) - (1/12 + 1/15)$$

$$= > 1/5 - 3/20 = 1/20$$

$$\text{Req. no. of days} = 20$$

5) Answer - B

$$\text{Number of days taken to complete the work}$$

$$= 1 / (2 \times (1/24) + 2 \times (1/32))$$

$$= 1 / (1/12 + 1/16)$$

$$= 1 / (7/48)$$

$$= 48/7 \text{ days}$$

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6) Answer - D

Vishal	48	5
	240	
Vivek	90	4
5 + 4 = 9 units		2 days
× 26		× 26
234 + 6/5		52 days + 6/5 days = 52(6/5) days

7) Answer - C

Manish	20	3
	60	
Manjeet	30	(-2)
2 days		1 unit
× 56		× 56
112		56
+1		+3
113		59
+1		-2
114		57
+1		+3
115		60

8) Answer - A

A alone complete the whole work = $100/60 \times 18 = 30$

B alone complete the whole work = $100/55 \times 11 = 20$

$c = 1/10 - 1/30 - 1/20$

$= (6 - 2 - 3)/60 = 1/60$

Req. no. days = 60

9) Answer - C

Vivek	20	6
Vishal	30	120
Vikram	40	3

Vivek + Vikram = $6 + 3 = 9$

Vishal + Vikram = $4 + 3 = 7$

In 2 days, Vivek and Vishal with assistance of Vikram alternately can complete 16 units of work.

So, 112 units of work will be completed in 14 days.

Remaining work = $120 - 112 = 8$ units

8 units of work will be completed by Vivek and Vikram in approximate one day.

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Total work will be completed in 15 days.

10) Answer - B

$$150 \times 30 \times (5/9) = (150+x) \times 10$$
$$x = 100$$

11) Answer - B

$$(40 \times 15 \times 6)/(3/5) = ((40+x) \times 5 \times 8)/(2/5)$$
$$x = 20$$

12) Answer - A

$$[1/6 + 1/12 + 1/30] \times 60$$
$$x : y : z = 10 : 5 : 2$$
$$y = (5/17) \times 5100 = 1500$$

13) Answer - B

$$32 \times 25/0.5 = 10(25+x)/0.5$$
$$x = 55 \text{ men}$$

14) Answer - C

$$\text{Total work} = 40 \times 26 = 1040$$
$$\text{Total work done by 40 women in 7 days} = 80 \times 7 = 560$$
$$x = (1040 - 560)/64 = 7.5 \text{ days}$$

15) Answer - A

$$x \times 60 = (x+5) \times 40$$
$$x = 20$$

16) Answer - B

Shivam = $3/5$	$24 = 40 \text{ days}$	
Sanket = $5/8$	$30 = 48 \text{ days}$	
Shivam 40	6	
Nidhi 60	240 5	
Sanket 48	4	
(Shivam + Sanket)	(Sanket + Nidhi)	
(5 + 6)	+	(4 + 5)
11 + 9 = 20 units	-----	2 days
×12		×12
240 units	-----	24 days

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17) Answer - A

let efficiency of A and B be x units/day and y units/day respectively.

$$12x + 24y = 18x + 16y$$

$$3x = 4y$$

$$x/y = 4/3$$

$$\text{Let } x = 4a \text{ and } y = 3a$$

$$\text{Required time} = (12 \times 4a + 24 \times 3a) / 4a = 30 \text{ days}$$

18) Answer - B

$$\text{Aniket} = 18 \times 5/3 = 30 \text{ days}$$

$$\text{Aniket} + \text{Badal} = 5 \times 7/2 = 35/2 \text{ days}$$

$$\text{Let total work} = 210 \text{ units}$$

$$\text{efficiency of Aniket} = 210 / 30 = 7 \text{ units/day}$$

$$\text{efficiency of Aniket} + \text{Badal} = 105 / (35/4) = 12 \text{ units/day}$$

$$\text{efficiency of Badal} = 12 - 7 = 5 \text{ units/day}$$

$$\text{Time taken by Badal alone to complete the work alone} = 210 / 5 = 42 \text{ days}$$

19) Answer - B

$$\begin{array}{ccc} \text{Prakash} & 24 & 5 \\ \text{Amit} & 30 & 120 \\ \text{Badal} & 40 & 3 \end{array}$$

$$\text{Prakash} + \text{Amit} = 9$$

$$\text{Prakash} + \text{Amit} - \text{Badal} = 9 - 3 = 6$$

$$= 120 / 6 = 20 \text{ days}$$

20) Answer - C

$$\text{B's 10 days work} = 1/15 \times 10 = 2/3$$

$$\text{Remaining work} = 1 - (2/3) = 1/3$$

$$\text{A can finish a project in} = 18 \text{ days}$$

$$1/18 \text{ Work done by A in} = 1 \text{ day}$$

$$1/3 \text{ works can be done by A in} = 18 \times 1/3 = 6 \text{ days}$$

21) Answer - B

$$(4M + 6B) \times 8 = (3M + 7B) \times 10$$

$$32M + 48B = 30M + 70B$$

$$2M = 22B$$

$$M/B = 11/1$$

$$\text{Ratio of efficiencies of men and women} = 11 : 1$$

$$\text{Total work} = (4 \times 11 + 6) \times 8 = 400$$

$$\therefore 10 \text{ boys complete the same work in } 40 \text{ days.}$$

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22) Answer - A

Kunal + Vivek	30	2
	60	

Vivek	60	1
-------	----	---

Amount of work done by 'Vivek' = $45 \times 1 = 45$ units

Percentage of work done by 'Kunal' = $\{(60 - 45)/60\} \times 100 = 25\%$

23) Answer - D

Let, the total work be L.C.M of (18,10,6) = 90 units

Efficiency of A = 90 units/18 days = 5 units/day

Efficiency of B = 90 units/10 days = 9 units/day

Efficiency of A, B and C = 90 units/6 days = 15 units/day

Efficiency of C = 15 - (5 + 9) = 1 unit/day

Time taken by C to complete the whole work alone = $90/1 = 90$ days

24) Answer - A

P = $8/0.1 = 80$ days

Q + R = $72/0.9 = 80$ days

P	80	1
---	----	---

	80	
--	----	--

Q + R	80	1
-------	----	---

Required time taken = $80/(1 + 1) = 40$ days

25) Answer - B

$\Rightarrow (x + x + 10) \times 12 = x(x + 10)$

$\Rightarrow 24x + 120 = x^2 + 10x$

$\Rightarrow x^2 - 14x - 120 = 0$

$\Rightarrow x^2 - 20x + 6x - 120 = 0$

$\Rightarrow x(x - 20) + 6(x - 20) = 0$

$\Rightarrow (x - 20)(x + 6) = 0$

$\Rightarrow x = 20$

26) Answer - A

$\{12 \times 8 \times 9 \times 4\}/(3/5) = \{W \times 12 \times 6 \times 1\}/(2/5)$

W = 32

27) Answer - D

Arun	56	12
------	----	----

Himanshu	42	672	16
----------	----	-----	----

Chandu	32	21
--------	----	----

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Work done by Arun and Himanshu in 20 days = $(12 + 16) \times 20 = 560$ units
Time taken to complete the remaining work = $(672 - 560)/(12 + 16 - 21) = 16$ days
Total time taken = $16 + 20 = 36$ days

28) Answer - B

$$\{(x + 5) \times 9 \times 24\}/0.75 = \{(x - 4) \times 2.5 \times 36\}/0.25$$

$$2(x + 5) = 2.5(x - 4)$$

$$0.5x = 20$$

$$x = 20/0.5 = 40$$

$$\text{Number of men who completed 75\% of the work} = (x + 5) = 45$$

29) Answer - D

$$20 \text{ men} = 25 \text{ women}$$

$$4 \text{ men} = 5 \text{ women}$$

$$8 \text{ men} = 10 \text{ women}$$

$$20 \text{ men} = 36 \text{ boys}$$

$$5 \text{ men} = 9 \text{ boys}$$

$$10 \text{ men} = 18 \text{ boys}$$

Let x men must be associated.

$$20 \times 15 \times 9 = (x + 8 + 10) \times 10 \times 10$$

$$27 = x + 18$$

$$x = 9$$

30) Answer - A

Let Diwakar takes ' x ' days to complete the work

$$\text{Time taken by Akash to complete the work} = (x + 8) \text{ days}$$

$$\text{Time taken by Sumit to complete the work} = (x - 4) \text{ days}$$

$$\{1/(x - 4)\} = 1.6\{1/(x + 8)\}$$

$$x + 8 = 1.6x - 6.4$$

$$0.6x = 14.4$$

$$x = 24$$

$$\text{Time taken by Akash to complete the work} = x + 8 = 32 \text{ days}$$

$$\text{Time taken by Diwakar to complete the work} = x = 24 \text{ days}$$

$$\text{Time taken by Sumit to complete the work} = x - 4 = 20 \text{ days}$$

Let the total work = 480 units

$$\text{Efficiency of Akash} = 480/32 = 15 \text{ units/day}$$

$$\text{Efficiency of Diwakar} = 480/24 = 20 \text{ units/day}$$

$$\text{Efficiency of Sumit} = 480/20 = 24 \text{ units/day}$$

$$\text{Required time taken} = 0.7375 \times 480/(15 + 20 + 24) = 6 \text{ days}$$

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31) Answer - C

Time taken by Badal alone to complete the entire work = $28 / 0.8 = 35$ days

Time taken by Rahul alone to complete the entire work = $35 + 7 = 42$ days

Let total work = L.C.M of 35 and 42 = 210 units

Efficiency of Badal alone = $210 / 35 = 6$ units/day

Efficiency of Rahul alone = $210 / 42 = 5$ units/day

Combined efficiency of Rahul and Vijay = $210 / 17.5 = 12$ units/day

Efficiency of Vijay alone = $12 - 5 = 7$ units/day

Total work completed by Vijay, Rahul and Badal in 3 days, working alternately for one day each = $7 + 5 + 6 = 18$ units

Work completed in 33 days = $18 \times 11 = 198$ units

Remaining work = $210 - 198 = 12$ units

Work done by Vijay and Rahul in next two days = $7 + 5 = 12$ units

Total time taken = $33 + 2 = 35$ days

32) Answer - D

Let, the total work be 60 units.

Efficiency of Anjali = $60 / 12 = 5$ units/day

Efficiency of Kartika = $60 / 15 = 4$ units/day

Efficiency of Prabha = $60 / 20 = 3$ units/day

Efficiency of Sapna = $(5 + 4 + 3) / 3 = 4$ units/day

Work done by Anjali and Kartika together in 4 days = $(5 + 4) \times 4 = 36$ units

Time taken by Prabha and Sapna together to complete the remaining work = $(60 - 36) / (3 + 4) = 24 / 7$ days

33) Answer - B

Let total amount of work be 525 units

Amount of work done by Puja in one day = $525 / 25 = 21$ units

Original efficiency of Puja = $21 / 1.40 = 15$ units per day

Amount of work done by Sonali in one day = $525 / 21 = 25$ units

Original efficiency of Sonali = $25 / 1.25 = 20$ units per day

Amount of work completed by Anjali and Sonali together in 9 days = $35 \times 9 = 315$ units

Remaining work = $525 - 315 = 210$ units

Desired time = $210 / 15 = 14$ days

34) Answer - B

Number of days taken by 'B' and 'C' together to finish the work = $(4 + 2) = 6$ days

Let total work = 60 units

Combined efficiency of 'B' and 'C' = $60 / 6 = 10$ units/day

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Combined efficiency of 'A', 'B' and 'C' = $60/4 = 15$ units/day

Efficiency of 'A' = $(15 - 10) = 5$ units/day

Efficiency of C = $1.2 \times 5 = 6$ units/day

Efficiency of 'B' = $\{10 - 6\} = 4$ units/day

Let C worked for 'x' days

$$(5 + 4) \times 2 + 4 \times (x - 2) + 6 \times x = 60$$

$$18 + 4 \times (x - 2) + 6x = 60$$

$$4x + 6x - 8 = 60 - 18$$

$$10x = 42 + 8$$

$$x = 5$$

Total number of days taken to complete the work = $5 + 2 = 7$ days

35) Answer - A

Let total work = LCM of 16, 24 and 36 = 144 units

Work done by Pankhuri in one day = $144/16 = 9$ units

Work done by Kajal in one day = $144/36 = 4$ units

Work done by Rashmi in one day = $144/24 = 6$ units

Let, number of days Pankhuri and Kajal worked together = x

And, number of days Kajal and Rashmi worked together = y

$$13x + 10y = 144$$

$$x + y = 12$$

$$x = 12 - y$$

$$13 \times (12 - y) + 10y = 144$$

$$156 - 13y + 10y = 144$$

$$3y = 12$$

$$y = 4$$

$$x = 12 - y = 12 - 4 = 8$$

36) Answer - A

R+M+N ----- > work 40% in 3days

M+N = 60% in 9days

M : N = time = 3:5

Eff. = 5:3

$$(5+3)*9 = 72 \text{ (60\%)}$$

$$40\% = 48$$

$$48/3 = 16 \text{ (eff. of R = } 16 - 5 - 3 = 8)$$

$$R = 8 \quad M = 5 \quad N = 3$$

$$8*3 \quad 5*12 \quad 12*3$$

$$24 \quad 60 \quad 36$$

$$= 120$$

$$24/120 = 1/5 = 4630/5 = 926$$

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37) Answer - B

	Time	eff.
$(R+5)/(R+S+P) =$	10/9	9/10
$(R)/(R+S+P) =$	432/1080	4/10
Eff. of R = 4	Eff. of S = 5	Eff. of P = 1
$(S+P) :$	$(R+S+P)$	
6 :	10	
3 :	5 (eff.)	
5 :	3 (time.)	
$2=4/3 \Rightarrow 3 = 2\text{days}$		

38) Answer - C

Rahul	Vineet	Kishan
	$(x-2)$	x

By option C $\rightarrow$ Vineet can due 2 times of work in 5 days with 4 times of his efficiency. Then one time can do I 2.5day with 4 times of his efficiency.

If he works with his actual efficiency then time-

$$v/4 = 2.5$$

$$v=10$$

Vineet	Kishan
10	12
60	
6	5

$$60/14 = 30/7 = 4(2/7) \text{ [by all three if they work together]}$$

$$60/(60/14) = 14 - 6 - 5 = 3 \text{ (Rahul's eff.)}$$

Time by Rahul = 20days

Rahul	Vineet	Kishan
20	10	12
	60	
3	6	5

$$3*8 + 9*4 = 60$$

Hence option C

39) Answer - C

From option C

Ratio of their efficiency = 6: 3: 2

Days of completion ratio = $1/3 : 1/6 : 1/2 = 2: 1: 3$

3 unit -----> 30 days

Work done by A alone= 1 unit-----> 10 days

And work done by B alone= 2 unit-----> 20 days

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Part of work done by A and B in 6 days = $(6/10 + 6/20) = 18/20 = 9/10$

Remaining work = $(1 - 9/10) = 1/10$

∴ 1/10 of work will be completed by C in 3 days.

Hence, only C

40) Answer – B

From option A

Let the total work = 560 units (LCM of 140, 112 and 80)

Amount of work done by A in one day = $560/140 = 4$ units

Amount of work done by B in one day = $560/112 = 5$ units

Amount of work done by C in one day = $560/80 = 7$ units

Amount of work done by A with increased efficiency = $1.50 \times 4 = 6$ units

Time taken by all of them together to complete 20% work = $0.20 \times 560 / (4 + 5 + 7) = 7$ days

Amount of work done by B with increased efficiency = $1.60 \times 5 = 8$ units

Time taken by A and B to complete the remaining work = $0.80 \times 560 / (8 + 6) = 32$ days

Total time taken to complete the work = $32 + 7 = 39$ days

Hence, only A

41) Answer – B

Let the total work = 560 units (LCM of 70, 80 and 112)

Amount of work done by P alone in one day = $560/70 = 8$ units

Amount of work done by Q alone in one day = $560/80 = 7$ units

Amount of work done by R alone in one day = $560/112 = 5$ units

Amount of work done by P, Q and R together in one day = $8 + 7 + 5 = 20$ units

From option B:

Amount of work done by P, Q and R together in 21 days = $20 \times 21 = 420$ units

Amount of work done by R alone in 28 days = $28 \times 5 = 140$ units

Total work done = $420 + 140 = 560$ units

Hence, Only B

42) Answer – D

From option C:

$24 \times 12 \times 10 / 36\% = 30 \times 10 \times 8 / 30\%$

$1 = 1$

Hence, Only C

43) Answer – A

From option A:

Let total work = 45 units

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Efficiency of Amit = $45/15 = 3$ units/day

Combined efficiency of Amit and Shashi = $\{45/(45/8)\} = 8$ units/day

Combined efficiency of Amit and Shashi = $\{45/(45/8)\} = 8$ units/day

Efficiency of Tarun = $(8 - 3) \times (80/100) = 4$ units/day

Time taken by Tarun alone to complete the work = $45/4 = 11.25$ days

Hence option A

44) Answer - E

Statement I:

Let total work be $100x$ units and efficiency of Sanjay and Manish be a unit/day and b units/day respectively. $\therefore$

$$60x/b - 100x/a = 2$$

Hence it can't be solved.

Statement II:

Let the efficiency of Sanjay be ' a ' unit/day and that of Manish be ' b ' units/day.

Work done by Sanjay in 8 days is $8a$ units.

Work done by Manish in 19 days is $19b$ units.

Now the total days taken by them is 19 days.

If they had worked for $19 - 5 = 14$ days together they would complete whole work

Hence,

$$8a + 19b = 14(a + b)$$

$$6a = 5b$$

$$a/b = 5/6$$

$$\text{Req. \%} = (6-5)/6 \times 100 = 16.66\%$$

Statement II alone is sufficient to answer the question

45) Answer - C

Exp: From (I)

Let total work be 60 units (l.c.m. of 20 and 30)

Then efficiency of Harsh = $60/30 = 2$ units/day

Efficiency of Shubham and Nakul together = $60/20 = 3$ units

Let efficiency of Shubham = x units/day

From (II)

ATQ,

$$11 \text{ days (Harsh + Shubham)} + 2 \text{ days (Shubham + Nakul)} + 10\frac{1}{4} \text{ days (Nakul)}$$

= total work

$$11(2 + x) + 2(3) + 41/4 * (3 - x) = 60$$

$$x = 5/3$$

Time taken by Shubham = $60/5 \times 3 = 36$ days

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So, statement (I) and (II) together necessary to answer the question

46) Answer – C

From statement I and II together:

Let the time taken by Sanjay, Manish, and Anit alone to complete the work be A, B, and C respectively. $\frac{1}{A} + \frac{1}{B} = \frac{3}{8}$

Efficiency ratio of A and B = $\frac{2}{1}$

Time ratio of A and B = $\frac{1}{2}$

$A/B = 1/2$

$A = B/2$

$\frac{2}{B} + \frac{1}{B} = \frac{3}{8}$

$B = 8$ days

$A = 4$ days

Thus, Sanjay and Manish alone can complete the work in 4 days and 8 days respectively.

Time taken by all of them to complete the work:

$\frac{1}{4} + \frac{1}{8} + \frac{1}{3} = \frac{(6 + 3 + 8)}{24} = \frac{17}{24}$

Time taken by Sanjay Manish and Anit together to complete the work = $\frac{24}{17}$ days

47) Answer – A

Let the total time taken for A, B, and C to complete the work be x hours

$\frac{(x - B)}{A} + \frac{12}{B} + \frac{(x - 16)}{C} = 1$

From statement I:

$A = 24$ hours

$B = 30$ hours

$C = 40$ hours

$\frac{(x - 8)}{24} + \frac{12}{30} + \frac{(x - 16)}{40} = 1$

$\frac{[5(x-8)+3(x-16)]}{120} = 1 - \frac{(2)}{5} = \frac{3}{5}$

$5x - 40 + 3x - 48 = \frac{3}{5} * 120 = 72$

$8x - 88 = 72$

$8x = 160$

$x = 20$ hours

The total time taken for A, B and C to complete the work is 20 hours

48) Answer – A

C alone can make a Sofa in $\frac{320}{11}$ days.

So, C's Sofa making efficiency = $\frac{11}{320}$ units/day

Since a Sofa is made by A, B and C together in 20 days, and B is 150% as efficient A, we get the following

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	Bed- making	Almirah- making	Sofa- making
A			$2/320$
B			$3/320$
C			$11/320$

A's almirah- making speed is twice of his Sofa - making speed. B takes 50% more time than A to make an almirah and C is 3 times as efficient as B in almirah - making.

	Bed- making	Almirah- making	Sofa- making
A		$1/80$	$2/320$
B		$1/120$	$3/320$
C		$1/40$	$11/320$

B makes a bed 3 times faster than he makes a almirah and A takes 25 days to make a bed alone. From the statement that A, B, and C together can make a bed in 8 days, we obtain the following:

	Bed- making	Almirah- making	Sofa- making
A	$1/25$	$1/80$	$2/320$
B	$1/40$	$1/120$	$3/320$
C	$3/50$	$1/40$	$11/320$

Quantity I

Time taken for A and C to make a bed = 40 days

Time taken for A and C to make a Sofa = $320/13$ days.

Hence, average time taken = $840/13$ days

Quantity II

Time taken for B and C to make an almirah = 30 days

time taken for B and C to make a Sofa = $160/7$ days

Hence, average time taken = $370/7$ days

49) Answer - A

Quantity I:

Sumit = $24/0.75 = 32$ days

Vivek = $32/0.8 = 40$ days

Sumit 32 5

160

Vivek 40 4

Time taken by both Arun and Himanshu to complete 45% of the work = $0.45 \times 160/(5 + 4) = 8$ days

Time taken by Himanshu to complete 55% of the work = $0.55 \times 160/4 = 22$ days

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Total time taken = $22 + 8 = 30$ days

Quantity II:

Aman	40		6
Ashutosh	48	240	5
Anurag	60		4

Amount of work done by all of them together in 8 days = $8 \times \{6 + 5 + 4\} = 120$ units

Amount of work done by Ashutosh alone in 6 days = $6 \times 5 = 30$ units

Amount of work done by Ashutosh and Anurag together in middle = $240 - 120 - 30 = 90$ units

Time taken by Ashutosh and Anurag together to complete 90 units work = $90 / (5 + 4) = 90 / 9 = 10$ units

Total time taken to complete the work = $10 + 8 + 6 = 24$ days

50) Answer - C

Quantity I:

Ankit	25		4
		100	

Badal	20		5
-------	----	--	---

Work done by Ankit in 7 days = $4 \times 7 = 28$ units

Remaining work = $100 - 28 = 72$ units

Time taken by Ankit and Badal to complete remaining work = $72 / 9 = 8$

Total Days = $7 + 8 = 15$ days

Quantity II:

Deepak	12		6
Farhan	24	72	3
Himesh	36		2

$$6 \times 4 + 3 \times x + 2 \times (x - 6) = 72$$

$$3x + 2x - 12 = 48$$

$$5x = 60$$

$$x = 12 \text{ days}$$

Quantity III:

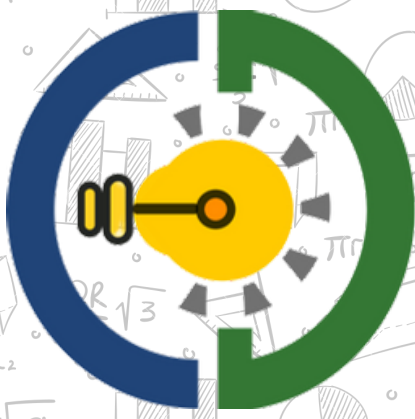
$$P + Q = (48 \times 72) / (48 + 36) = 14.4 \text{ days}$$

$$R = 1.25 \times 14.4 = 18 \text{ days}$$

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- 1) 6 men and 8 boys together can do a work in 40 days while 9 men and 4 boys together can do the same work in 30 days. Find the time taken by 3 men and 40 boys together to do the same work.
6 आदमी और 8 लड़के मिलकर एक काम को 40 दिनों में कर सकते हैं जबकि 9 आदमी और 4 लड़के एक साथ उसी काम को 30 दिनों में कर सकते हैं। 3 पुरुषों और 40 लड़कों द्वारा एक साथ समान कार्य करने में लिया गया समय ज्ञात कीजिए।
a) 70 days
b) 60 days
c) 32 days
d) 56 days
e) 35 days
- 2) Ratio of efficiencies of Ganesh and Ramesh is 8:11, respectively. If Ganesh alone can do the work in 30 days, then find the time taken by Ramesh alone to do $(11/12)$ th part of the work.
गणेश और रमेश की दक्षताओं का अनुपात क्रमशः 8:11 है। यदि गणेश अकेले काम को 30 दिनों में कर सकता है, तो रमेश द्वारा अकेले काम का $(11/12)$ भाग पूरा करने में लिया गया समय ज्ञात कीजिए।
a) 42 days
b) 44 days
c) 20 days
d) 48 days
e) 15 days
- 3) 10 women or 6 men can do a piece of work in 20 days. 5 men started the work, and after 12 days they left the work. If the remaining work was completed by 10 women, then in how many days would 10 women complete the remaining work?
10 महिलाएं या 6 पुरुष एक काम को 20 दिनों में कर सकते हैं। 5 आदमियों ने काम शुरू किया और 12 दिनों के बाद उन्होंने काम छोड़ दिया। यदि शेष कार्य 10 महिलाओं द्वारा पूरा किया जाता है, तो 10 महिलाएँ शेष कार्य को कितने दिनों में पूरा करेंगी?
a) 9 days
b) 8 days
c) 12 days
d) 10 days
e) None of these
- 4) A and B alone can complete a work in 90 days and 60 days, respectively. They together can complete the work in x days. By how much percentage A has to increase his efficiency so that they can complete the work in $x/2$ days?

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A और B अकेले एक काम को क्रमशः 90 दिन और 60 दिन में पूरा कर सकते हैं। वे मिलकर कार्य को x दिनों में पूरा कर सकते हैं। A को अपनी कार्यकुशलता कितने प्रतिशत बढ़ानी होगी ताकि वे कार्य को $x/2$ दिन में पूरा कर सकें?

- a) 250%
- b) 200%
- c) 350%
- d) 150%
- e) None of these

- 5) Ankita and Monika together can do a piece of work in 10 days, while Ankita alone can do the same work in 18 days. In how many days can Monika alone do the same work?

अंकिता और मोनिका मिलकर एक काम को 10 दिनों में कर सकती हैं, जबकि अंकिता अकेले उसी काम को 18 दिनों में कर सकती है। मोनिका उसी कार्य को अकेले कितने दिनों में कर सकती है?

- a) 42 days
- b) 30 days
- c) 22.5 days
- d) 48 days
- e) None of these

- 6) Ratio of efficiencies of A and B is 15:17, respectively. A and B together can complete total work in 15 days. In how many days will A alone complete the whole work?

A और B की दक्षताओं का अनुपात क्रमशः 15:17 है। A और B मिलकर कुल कार्य 15 दिनों में पूरा कर सकते हैं। A अकेले पूरा कार्य कितने दिनों में पूरा करेगा?

- a) 54 days
- b) 48 days
- c) 32 days
- d) 46 days
- e) None of these

- 7) 16 men or 20 women can do a piece of work in 21 days. 10 women and 4 men together can do the same work in how many days?

16 पुरुष या 20 महिलाएं एक काम को 21 दिनों में कर सकते हैं। 10 महिलाएं और 4 पुरुष मिलकर उसी काम को कितने दिनों में कर सकते हैं?

- a) 28 days
- b) 15 days
- c) 10 days

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- d) 18 days
- e) 16 days

- 8) Subham can complete one-fifth of a work in 8 days, while lucky can complete the entire work in 32 days. Both of them started working together, but Lucky worked with 20% of his efficiency. In how many days the whole work will be completed?

सुभम किसी काम का पांचवां हिस्सा 8 दिनों में पूरा कर सकता है, जबकि भाग्यशाली पूरे काम को 32 दिनों में पूरा कर सकता है। दोनों ने एक साथ काम करना शुरू किया, लेकिन लकी ने अपनी 20% दक्षता के साथ काम किया। पूरा काम कितने दिनों में पूरा हो जाएगा?

- a) 30 days
- b) 32 days
- c) 36 days
- d) 40 days
- e) None of these

- 9) 'A' can complete a piece of work in 36 days while 'A' and 'B' together can complete the same work in 18 days. If 'A' and 'B' together started the work and 'A' left after 10 days then rest of the work will be completed by 'B' in:

'ए' एक काम को 36 दिनों में पूरा कर सकता है जबकि 'ए' और 'बी' मिलकर उसी काम को 18 दिनों में पूरा कर सकते हैं। यदि 'ए' और 'बी' ने एक साथ काम शुरू किया और 'ए' ने 10 दिनों के बाद छोड़ दिया तो शेष काम 'बी' द्वारा पूरा किया जाएगा:

- a) 16 days
- b) 54 days
- c) 50 days
- d) 56 days
- e) None of these

- 10) 'A' can complete 40% of work in 24 days while 'A' and 'B' together can complete the whole work in 35 days. If A and 'B' started the Work and worked for 10 days then rest of the work will be completed by 'B' alone in:

'ए' 40% काम 24 दिनों में पूरा कर सकता है जबकि 'ए' और 'बी' मिलकर पूरा काम 35 दिनों में पूरा कर सकते हैं। यदि ए और 'बी' काम शुरू करते हैं और 10 दिनों तक काम करते हैं तो शेष कार्य अकेले 'बी' द्वारा पूरा किया जाएगा:

- a) 32 days
- b) 40 days
- c) 60 days
- d) 34 days
- e) None of these





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11) 'A' is 4 times as efficient as 'B' and both can complete a work together in 24 days. If efficiency of 'C' is equal to average of efficiencies of 'A' and B, then find the time taken by 'C' to do the same work alone.

'A', 'B' से 4 गुना कुशल है और दोनों मिलकर एक काम 24 दिनों में पूरा कर सकते हैं। यदि 'C' की दक्षता 'A' और 'B' की औसत दक्षता के बराबर है, तो 'C' द्वारा उसी कार्य को अकेले करने में लगा समय ज्ञात कीजिए।

- a) 30 days
- b) 20 days
- c) 48 days
- d) 32 days
- e) None of these

12) 'A' and B' together can complete a work in 10 days while 'A' takes 18 days to complete the same work alone. If 'C' is 50% more efficient than 'B', then find the time taken by 'C' alone to complete the whole work.

'ए' और 'बी' मिलकर एक काम को 10 दिनों में पूरा कर सकते हैं जबकि 'ए' को अकेले उसी काम को पूरा करने में 18 दिन लगते हैं। यदि 'C', 'B' से 50% अधिक कुशल है, तो अकेले 'C' द्वारा पूरे कार्य को पूरा करने में लिया गया समय ज्ञात कीजिए।

- a) 36 days
- b) 15 days
- c) 32 days
- d) 24 days
- e) 28 days

13) 'A' and 'B' together can complete a piece of work in 34 days while 'A' alone can complete the same work in 50 days. If B' started the work and was joined by 'A' after 70 days, then total work will be completed in:

'ए' और 'बी' मिलकर एक काम को 34 दिनों में पूरा कर सकते हैं जबकि 'ए' अकेले उसी काम को 50 दिनों में पूरा कर सकता है। यदि 'B' ने काम शुरू किया और 70 दिनों के बाद 'A' से जुड़ गया, तो कुल काम कितने समय में पूरा हो जाएगा:

- a) 78.8 days
- b) 81.6 days
- c) 42.3 days
- d) 69.8 days
- e) None of these

14) 36 men can complete a piece of work in 80 days. 'x' men started the work. After 30 days, 56 men took 30 days to complete the remaining work. Find the value of x.

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36 आदमी एक काम को 80 दिनों में पूरा कर सकते हैं। 'x' पुरुषों ने काम शुरू किया। 30 दिनों के बाद, 56 पुरुषों को शेष कार्य पूरा करने में 30 दिन लगे। एक्स का मान ज्ञात करें।

- a) 40
- b) 50
- c) 30
- d) 60
- e) None of these

- 15) 'A' can complete a piece of work in 70 days while with the help of B' and 'C' he can complete the same work in 42 days and 30 days respectively. If 'A' started the work and left after 40 days, then remaining work will be completed by 'B' and C' together in:

'A' एक काम को 70 दिनों में पूरा कर सकता है जबकि B' और 'C' की मदद से वह उसी काम को क्रमशः 42 दिनों और 30 दिनों में पूरा कर सकता है। यदि 'A' ने कार्य शुरू किया और 40 दिनों के बाद छोड़ दिया, तो शेष कार्य 'B' और C' द्वारा मिलकर पूरा किया जाएगा:

- a) 20 days
- b) 35 days
- c) 10 days
- d) 15 days
- e) 25 days

- 16) Chirag can complete a work in 16 days. He worked for 12 days and left such that remaining work was completed by Hemant in 5 days. Find the time taken by Hemant and Chirag to complete the work together.

चिराग एक काम को 16 दिनों में पूरा कर सकता है। उसने 12 दिन काम किया और ऐसे चला गया कि शेष काम हेमंत ने 5 दिनों में पूरा कर लिया। हेमंत और चिराग द्वारा एक साथ काम पूरा करने में लिया गया समय ज्ञात कीजिए।

- a) 80/9 days
- b) 124/5 days
- c) 145/9 days
- d) 132/7 days
- e) None of these

- 17) 'A' can complete a piece of work in 16 days while 'A' and B' together can complete the whole work in 9 days. If 'B' started the work and left after 18 days, then time taken by 'A' to complete the remaining work is:

'A' एक काम को 16 दिनों में पूरा कर सकता है जबकि 'A' और B' मिलकर पूरे काम को 9 दिनों में पूरा कर सकते हैं। यदि 'बी' ने काम शुरू किया और 18 दिनों के बाद छोड़ दिया, तो शेष काम को पूरा करने में 'ए' द्वारा लिया गया समय है:

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- a) 2 days
- b) 6 days
- c) 4 days
- d) 8days
- e) None of these

18) 'A' and B' can do a piece of work in 50 days and 60 days, respectively. They started the work and were joined by 'C' after 10 days. If the whole work was completed in 10 days then find the time taken by 'C' alone to complete the whole work.

'ए' और बी' क्रमशः 50 दिनों और 60 दिनों में एक काम कर सकते हैं। उन्होंने काम शुरू किया और 10 दिनों के बाद 'C' से जुड़ गए। यदि पूरा काम 10 दिनों में पूरा हो गया तो अकेले 'C' द्वारा पूरा काम पूरा करने में लगा समय ज्ञात कीजिए।

- a) 26.75 days
- b) 32.5 days
- c) 37.5 days
- d) 49.75 days
- e) None of these

19) 'A' and B' can complete a work in 40 days and 48 days, respectively. They started working together but 'A' left after 18 days such that remaining work is completed by B' along with C in 6 days. Find the time taken by 'C' to complete the whole work alone.

'ए' और बी' एक काम को क्रमशः 40 दिन और 48 दिन में पूरा कर सकते हैं। उन्होंने एक साथ काम करना शुरू किया लेकिन 'ए' ने 18 दिनों के बाद इस तरह काम छोड़ दिया कि शेष काम बी' ने सी के साथ मिलकर 6 दिनों में पूरा किया। 'C' द्वारा अकेले पूरा कार्य पूरा करने में लिया गया समय ज्ञात कीजिए।

- a) 140days
- b) 100 days
- c) 180 days
- d) 120 days
- e) 125 days

20) 'B' completed 60% of work alone in 24 days and left the rest Work for 'A' who completed the remaining work in 24 days. find the time taken by 'A' and 'B' together to complete the whole work.

'बी' ने अकेले 60% काम 24 दिनों में पूरा किया और बाकी काम 'ए' के लिए छोड़ दिया जिसने शेष काम 24 दिनों में पूरा किया। पूरे कार्य को पूरा करने में 'ए' और 'बी' द्वारा मिलकर लिया गया समय ज्ञात कीजिए।

- a) 24days

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- b) 20 days
- c) 16 days
- d) 10 days
- e) 15 days

21) 'A' is 70% more efficient than B'. Working together, they can complete a work in $7\frac{7}{9}$ days. If 'A' started the work alone and left after 10 days and then 'B' completed the remaining work, then find the total time taken by them to complete the work in this way.

'A', 'B' से 70% अधिक कुशल है। एक साथ काम करते हुए, वे एक काम को $7\frac{7}{9}$ दिनों में पूरा कर सकते हैं। यदि 'ए' अकेले काम शुरू करता है और 10 दिनों के बाद छोड़ देता है और फिर 'बी' शेष काम पूरा करता है, तो इस प्रकार काम पूरा करने में उनके द्वारा लिया गया कुल समय ज्ञात कीजिए।

- a) 14 days
- b) 18 days
- c) 15 days
- d) 4 days
- e) None of these

22) 4 men or 6 boys can complete a work in 36 days. In how many days 6 men and 7 boys can complete the same work together?

4 आदमी या 6 लड़के एक काम को 36 दिनों में पूरा कर सकते हैं। 6 आदमी और 7 लड़के मिलकर उसी काम को कितने दिनों में पूरा कर सकते हैं?

- a) 13.5 days
- b) 5.25 days
- c) 8.5 days
- d) 10.5 days
- e) None of these

23) 'A', 'B' and 'C' can complete a piece of work in 80 days, 144 days and 90 days, respectively. All of them started the work together but 'A' left the work 40 days before completion of the work while C left the work 10 days after 'A' left the work. Find the time taken to complete the whole work.

'ए', 'बी' और 'सी' एक काम को क्रमशः 80 दिन, 144 दिन और 90 दिन में पूरा कर सकते हैं। उन सभी ने एक साथ काम शुरू किया लेकिन 'ए' ने काम पूरा होने से 40 दिन पहले काम छोड़ दिया जबकि 'सी' ने 'ए' के काम छोड़ने के 10 दिन बाद काम छोड़ दिया। संपूर्ण कार्य को पूरा करने में लगने वाला समय ज्ञात कीजिए।

- a) 45
- b) 75
- c) 60

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- d) 50
- e) None of these

24) 8 men and 5 boys, together can earn Rs.3240 in 3 days while 3 men and 2 boys, together takes 6 days to earn Rs. 2484. In how many days 5 men and 4 boys, together can earn Rs.1476?

8 आदमी और 5 लड़के मिलकर 3 दिन में 3240 रुपये कमा सकते हैं जबकि 3 आदमी और 2 लड़के मिलकर 6 दिन में 2484 रुपये कमा सकते हैं। 5 आदमी और 4 लड़के मिलकर कितने दिनों में 1476 रुपये कमा सकते हैं?

- a) 5 days
- b) 6 days
- c) 8 days
- d) 2 days
- e) 4 days

25) A and B together can complete a piece of work in 'x' days while A alone can complete the same work in 'x + 32' days. B alone can complete the whole work in 90 days. Find the time taken by C alone to complete the whole work if C is 20% more efficient than A.

A और B मिलकर एक काम को 'x' दिनों में पूरा कर सकते हैं जबकि A अकेले उसी काम को 'x + 32' दिनों में पूरा कर सकता है। B अकेले पूरा कार्य 90 दिनों में पूरा कर सकता है। यदि C, A से 20% अधिक कुशल है तो C द्वारा अकेले पूरा कार्य पूरा करने में लिया गया समय ज्ञात कीजिए।

- a) 50 days
- b) 54 days
- c) 60 days
- d) 48 days
- e) None of these

26) A, B and C alone can complete a piece of work in 90 days, 70 days and 126 days respectively. They started the work together but A left the work 48 days before its completion and B left the work 20 days after A left. Find the total time taken to complete the whole work.

A, B और C अकेले एक काम को क्रमशः 90 दिन, 70 दिन और 126 दिन में पूरा कर सकते हैं। उन्होंने एक साथ काम शुरू किया लेकिन A ने काम पूरा होने से 48 दिन पहले काम छोड़ दिया और B ने A के छोड़ने के 20 दिन बाद काम छोड़ दिया। संपूर्ण कार्य को पूरा करने में लगा कुल समय ज्ञात कीजिये।

- a) 58 days
- b) 62 days
- c) 50 days

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- d) 54 days
- e) None of these

27) 'A' completed $\frac{1}{3}$ rd of work in 8 days and the rest work is done by 'B' in 16 days. If 'B' is 25% less efficient than 'C', then in how many days 'A', 'B' and 'C' together can complete $\frac{5}{6}$ th of the work?

'ए' ने $\frac{1}{3}$ काम 8 दिनों में पूरा किया और शेष काम 'बी' ने 16 दिनों में पूरा किया। यदि 'B', 'C' से 25% कम कुशल है, तो 'A', 'B' और 'C' मिलकर कितने दिनों में काम का $\frac{5}{6}$ भाग पूरा कर सकते हैं?

- a) 2.5 days
- b) 6 days
- c) 2 days
- d) 4 days
- e) 4.5 days

28) 'x' number of men were asked to complete a work in 48 days. But, 7 men didn't join the work so that work gets delayed by 42 days. Find the value of 'x'.

'x' संख्या में पुरुषों को 48 दिनों में एक काम पूरा करने के लिए कहा गया था। लेकिन, 7 आदमी काम में शामिल नहीं हुए जिससे काम में 42 दिन की देरी हो गई। एक्स का मान ज्ञात करें।

- a) 14
- b) 10
- c) 20
- d) 12
- e) 15

29) 'A' alone can complete 40% of work in 48 days, 'A' and 'B' together can complete 62.5% of the work in 45 days while 'A', 'B' and 'C' together can complete 50% of the work in 27 days. 'B' is how much percent more efficient than 'C'?

'A' अकेले 40% काम 48 दिनों में पूरा कर सकता है, 'A' और 'B' मिलकर 62.5% काम 45 दिनों में पूरा कर सकते हैं जबकि 'A', 'B' और 'C' मिलकर 50% काम पूरा कर सकते हैं 27 दिनों में काम का पूरा. 'B', 'C' से कितने प्रतिशत अधिक कुशल है?

- a) 10%
- b) 20%
- c) 25%
- d) 60%
- e) None of these

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30) A 10 metre high wall is to be built in 300 days. 60 men started to build the wall. After 180 days, they realized that only 6 metres of wall is built. Find the additional number of men that should be employed to completely build the wall on time.

10 मीटर ऊंची दीवार 300 दिन में बनानी है। 60 लोगों ने दीवार बनानी शुरू की। 180 दिन बाद उन्हें एहसास हुआ कि दीवार तो सिर्फ 6 मीटर ही बनी है। समय पर दीवार को पूरी तरह से बनाने के लिए नियोजित किए जाने वाले अतिरिक्त लोगों की संख्या ज्ञात कीजिए।

- a) 15
- b) 12
- c) 25
- d) 10
- e) None of these

31) A can do a piece of work in 10 days; R in 15 days. They work together for 5 days. The rest of the work was finished by N in 2 days. If they get Rs. 1,500 for the whole work, the sum of the daily wages of R and N are

A 10 दिनों में एक काम कर सकता है; 15 दिन में 'आर'. वे 5 दिनों तक एक साथ काम करते हैं। शेष कार्य 'N' द्वारा 2 दिनों में समाप्त कर दिया गया। यदि उन्हें रु. पूरे काम के लिए 1,500, 'R' और 'N' की दैनिक मजदूरी का योग है

- a) 250
- b) 275
- c) 225
- d) 200
- e) None of these

32) $(a+10)$ persons can complete a piece of work in $(a-11)$ days while $(a+5)$ persons can complete the same work in $(a-10)$ days. Then, find the time taken by $(a-13)$ persons to complete $1/10^{\text{th}}$ of the same work?

$(a+10)$ व्यक्ति किसी कार्य को $(a-11)$ दिनों में पूरा कर सकते हैं जबकि $(a+5)$ व्यक्ति उसी कार्य को $(a-10)$ दिनों में पूरा कर सकते हैं। तो, उसी कार्य का $1/10$ वां भाग पूरा करने में $(a-13)$ व्यक्तियों द्वारा लिया गया समय ज्ञात कीजिए?

- a) 5 days
- b) 4 days
- c) 6 days
- d) 3 days
- e) 9 days

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- 33) AA and BB together do some work in 100 days, BB and CC together do the same work in 250 days and CC and AA together do the same work in 500 days, then, in how many days will the three of them together complete same the work?
AA और BB मिलकर किसी काम को 100 दिन में करते हैं, BB और CC मिलकर उसी काम को 250 दिन में करते हैं और CC और AA मिलकर उसी काम को 500 दिन में करते हैं, तो तीनों मिलकर उसे कितने दिन में पूरा करेंगे? वही काम?
- a) 250 days
 - b) 125 days
 - c) $250/8$ days
 - d) $250/12$ days
 - e) None of these
- 34) A can do a work in 30 hours and B is 120% as efficiency as A . If they do the work on alternate hours starting with A, find time to complete the work in hours.
A एक काम को 30 घंटे में कर सकता है और B की दक्षता A से 120% है। यदि वे A से शुरू करके वैकल्पिक घंटों पर काम करते हैं, तो काम को घंटों में पूरा करने के लिए समय निकालें।
- a) 27.33
 - b) 28.4
 - c) 26.6
 - d) 26.33
 - e) None of these
- 35) A, b, c take a contract work for Rs. 4,465 lakhs. A and C take up $12/19$ of the work and complete it by themselves. The bill was submitted by them to the authorities and they suggested separate payment for B may be made for the work done by him. How much will be paid to B ?
ए, बी, सी रुपये के लिए अनुबंध कार्य लेते हैं। 4,465 लाख. A और C कार्य का $12/19$ भाग लेते हैं और इसे स्वयं पूरा करते हैं। बिल उनके द्वारा अधिकारियों को प्रस्तुत किया गया था और उन्होंने सुझाव दिया कि बी के लिए उसके द्वारा किए गए कार्य के लिए अलग से भुगतान किया जा सकता है। B को कितना भुगतान किया जाएगा?
- a) Rs. 1545 lakhs
 - b) Rs. 1700 lakhs
 - c) Rs. 2000 lakhs
 - d) Rs. 1645 lakhs
 - e) None of these

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- 36) 21 boys and 13 girls can finish a job in 1 day while 9 boys and 18 girls finish the same job in same time . Then , in how many days 3 girls can finish the job ?
21 लड़के और 13 लड़कियाँ एक काम को 1 दिन में पूरा कर सकते हैं जबकि 9 लड़के और 18 लड़कियाँ उसी काम को एक ही समय में पूरा कर सकते हैं। तो फिर, 3 लड़कियाँ उस काम को कितने दिनों में पूरा कर सकती हैं?
- a) $21/4$ days
b) $25/4$ days
c) $17/4$ days
d) $33/4$ days
e) $29/4$ days
- 37) A can do a piece of work in 40 days while B can do that work in 60 days , if both worked for 4 days , then the fraction of work left is –
A एक काम को 40 दिनों में कर सकता है जबकि B उस काम को 60 दिनों में कर सकता है, यदि दोनों 4 दिनों तक काम करते हैं, तो बचा हुआ काम का अंश है -
- a) $6/7$
b) $7/6$
c) $1/9$
d) $5/6$
e) $3/2$
- 38) IF S can build a part of a car in 4 days and R takes 10 days to build the same part. The work is increased by 6 times more when owner wants the complete car build . then how many days will it take for both to finish the car if they work together ?
यदि S एक कार का एक हिस्सा 4 दिनों में बना सकता है और R को उसी हिस्से को बनाने में 10 दिन लगते हैं। जब मालिक पूरी कार चाहता है तो काम 6 गुना अधिक बढ़ जाता है। यदि वे एक साथ काम करते हैं तो कार को पूरा करने में दोनों को कितने दिन लगेंगे?
- a) 12 days
b) 15 days
c) 10 days
d) 5 days
e) 20 days





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39) T completes 30% of work in 7.5 days. U is 50% as efficient as T, V is 50% as efficient as U. After 30% now U and V joined with T for the rest of work then in how many days will take to complete the remaining work.

T 30% काम 7.5 दिनों में पूरा करता है। U, T से 50% कुशल है, V, U से 50% कुशल है। 30% के बाद अब U और V शेष कार्य के लिए T के साथ जुड़ गए, तो शेष कार्य को पूरा करने में कितने दिन लगेंगे?

- a) 10 days
- b) 5 days
- c) 9 days
- d) 12 days
- e) None of these

40) 2 men and 1 women can finish a piece of work in 12 days while 4 women and 2 men can do the same work in 8 days. If a man gets Rs 75 per day and wages per day for a women is Rs x, then what will be the value of $(x^2 + 100)$?

2 पुरुष और 1 महिला एक काम को 12 दिनों में पूरा कर सकते हैं जबकि 4 महिलाएं और 2 पुरुष उसी काम को 8 दिनों में पूरा कर सकते हैं। यदि एक पुरुष को प्रति दिन 75 रुपये मिलते हैं और महिलाओं के लिए प्रति दिन मजदूरी x रुपये है, तो $(x^2 + 100)$ का मूल्य क्या होगा?

- a) 780
- b) 875
- c) 650
- d) 1000
- e) 1170

41) 2 men and 5 women can finish work in 8 days while 6 women and 1 child can finish the same work in 10 days. 6 men and 5 children can do the same work in 5 days. Find the time taken by 4 men, 4 women, and 4 children to complete half of the work.

2 पुरुष और 5 महिलाएं काम को 8 दिनों में पूरा कर सकते हैं जबकि 6 महिलाएं और 1 बच्चा उसी काम को 10 दिनों में पूरा कर सकते हैं। 6 आदमी और 5 बच्चे उसी काम को 5 दिनों में कर सकते हैं। 4 पुरुषों, 4 महिलाओं और 4 बच्चों द्वारा आधा काम पूरा करने में लिया गया समय ज्ञात कीजिए।

- a) 3.5
- b) 2.5
- c) 4.5
- d) 1.5

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e) None of these

42) A, B, and C can complete a work in $(2x - 4)$ days, $(x+2)$ days, and $(x+5)$ days respectively. B and C together can do the work in 30% of the work in 2 days. Now A and B started the work after 2 days, and C and D joined them and worked after 2 more days A, B, and C left and D finished the work in 3.1 days. Find the time required by D to complete the work alone.

A, B और C एक काम को क्रमशः $(2x - 4)$ दिन, $(x+2)$ दिन और $(x+5)$ दिन में पूरा कर सकते हैं। B और C मिलकर उस कार्य का 30% भाग 2 दिनों में पूरा कर सकते हैं। अब A और B ने 2 दिनों के बाद काम शुरू किया, और C और D उनके साथ जुड़ गए और 2 दिनों के बाद काम किया, A, B और C ने काम छोड़ दिया और D ने 3.1 दिनों में काम पूरा कर लिया। D द्वारा अकेले कार्य पूरा करने के लिए आवश्यक समय ज्ञात कीजिए।

- a) $(2x - 1)$ days
- b) $(x - 2)$ days
- c) $(2x + 2)$ days
- d) $(2x - 8)$ days
- e) $(2x - 2)$ days

43) Initially 5 men are working on a project who were able to complete only 5% of the project in 2 days working 4 hours daily. How many more men are required so that the 60% of the project is completed in 12 days working 8 hours daily?

प्रारंभ में 5 आदमी एक परियोजना पर काम कर रहे हैं जो प्रतिदिन 4 घंटे काम करके 2 दिनों में परियोजना का केवल 5% ही पूरा कर पाए। प्रतिदिन 8 घंटे काम करके 12 दिनों में परियोजना का 60% पूरा करने के लिए कितने और पुरुषों की आवश्यकता है?

- a) 5
- b) 8
- c) 6
- d) 9
- e) None of these

44) A can type 72 pages in 24 hours. A and B together can type 96 pages in 12 hours. A and B started a project of typing 166 pages together, and 2 hours before completing the work B left. Find the difference between the wages of A and B if they got Rs. 2490 in total for the project

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A 24 घंटे में 72 पेज टाइप कर सकता है। A और B मिलकर 12 घंटे में 96 पेज टाइप कर सकते हैं। A और B ने मिलकर 166 पेज टाइप करने का प्रोजेक्ट शुरू किया और काम पूरा करने से 2 घंटे पहले B ने काम छोड़ दिया। यदि A और B को रु. मिले तो उनके वेतन के बीच अंतर ज्ञात कीजिए। परियोजना के लिए कुल 2490 रु

- a) 510
- b) 680
- c) 560
- d) 640
- e) None of these

- 45) Two friends Sunil and Aman can complete a task in 12 hours and 22 hours respectively and a third friend named Ankit can complete the whole task alone in X hours with three - fourth of his original efficiency. If all the three working together with their usual efficiency can complete the whole task in 4 hours, then find the value of X.

दो दोस्त सुनील और अमन एक कार्य को क्रमशः 12 घंटे और 22 घंटे में पूरा कर सकते हैं और अंकित नाम का तीसरा दोस्त अपनी मूल दक्षता के तीन-चौथाई के साथ अकेले पूरे कार्य को X घंटे में पूरा कर सकता है। यदि तीनों मिलकर अपनी सामान्य दक्षता से पूरा कार्य 4 घंटे में पूरा कर सकें, तो X का मान ज्ञात कीजिए।

- a) 22 hr
- b) 47 hr
- c) 35 hr
- d) 11 hr
- e) None of these

- 46) A, B and C can complete a piece of work in 24 days, 30 days and 36 days respectively. All of them complete the whole work together and after completing the work they received Rs.1184 together as wages then find the difference between amount received by A and C together and B alone.

A, B और C एक काम को क्रमशः 24 दिन, 30 दिन और 36 दिन में पूरा कर सकते हैं। वे सभी पूरा काम एक साथ पूरा करते हैं और काम पूरा करने के बाद उन्हें मजदूरी के रूप में 1184 रुपये मिलते हैं, तो A और C द्वारा एक साथ प्राप्त राशि और अकेले B द्वारा प्राप्त राशि के बीच अंतर ज्ञात करें।

- a) Rs, 198
- b) Rs. 416
- c) Rs. 184
- d) Rs. 248
- e) Rs. 232

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47) 'C' with the help of 'A' can complete a piece of work in 21 days while with the help of 'B' can complete the same work in 24 days. If all of them together can complete the whole work in 14 days, then find the time taken by 'C' alone to complete 25% of the whole work.

'C' 'A' की मदद से एक काम को 21 दिनों में पूरा कर सकता है जबकि 'B' की मदद से उसी काम को 24 दिनों में पूरा कर सकता है। यदि वे सभी मिलकर पूरे कार्य को 14 दिनों में पूरा कर सकते हैं, तो अकेले 'C' द्वारा पूरे कार्य का 25% पूरा करने में लिया गया समय ज्ञात कीजिए।

- a) 22.8 days
- b) 29.2 days
- c) 26.4 days
- d) 14 days
- e) 24.8 days

48) A and B together can complete 60% of a work in 54 days while B alone takes 60% more time than A and B together to complete the whole work. Find the time taken by A alone to complete 40% of the whole work.

A और B मिलकर किसी काम का 60% हिस्सा 54 दिनों में पूरा कर सकते हैं जबकि B अकेले पूरे काम को पूरा करने में A और B की तुलना में 60% अधिक समय लेता है। पूरे कार्य का 40% पूरा करने में A द्वारा अकेले लिया गया समय ज्ञात कीजिए।

- a) 120 days
- b) 116 days
- c) 96 days
- d) 84 days
- e) None of these

49) 25 men can complete a piece of work in 24 days. 16 men started working and after 'x' days, 6 more men joined the work such that the remaining work is completed in 20 days. Find the value of x.

25 आदमी एक काम को 24 दिनों में पूरा कर सकते हैं। 16 आदमी काम करना शुरू करते हैं और 'x' दिनों के बाद, 6 और आदमी काम में इस तरह शामिल होते हैं कि शेष काम 20 दिनों में पूरा हो जाता है। एक्स का मान ज्ञात करें।

- a) 8
- b) 4
- c) 10
- d) 6
- e) 9

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- 50) A and B undertook to complete a piece of work for Rs. 4500. A can do it in 24 days, B can do it in 32 days and with the help of C, they complete the work in $10\frac{2}{3}$ days. Find the share of C?
ए और बी ने रुपये के लिए एक काम पूरा करने का बीड़ा उठाया। 4500. A इसे 24 दिनों में कर सकता है, B इसे 32 दिनों में कर सकता है और C की मदद से, वे काम को $10\frac{2}{3}$ दिनों में पूरा करते हैं। C का हिस्सा ज्ञात कीजिये?
- a) Rs. 850
 - b) Rs. 1000
 - c) Rs. 600
 - d) Rs. 700
 - e) None of these
- 51) 25 men can complete a piece of work in 32 days. After 8 days from the start of the work, some men left. If the remaining work was completed by the remaining men in 30 days, then find the men left after 8 days from the start of the work?
25 आदमी एक काम को 32 दिनों में पूरा कर सकते हैं। काम शुरू होने के 8 दिन बाद कुछ आदमी चले गए। यदि शेष कार्य को शेष पुरुषों द्वारा 30 दिनों में पूरा किया गया, तो कार्य शुरू होने के 8 दिन बाद बचे हुए पुरुषों का पता लगाएं?
- a) 3 men
 - b) 4 men
 - c) 6 men
 - d) 5 men
 - e) None of these
- 52) P, Q and R can complete the whole work in 40 days. P starts the work and works for 'x' days while Q and R complete the remaining $\frac{2}{5}$ of the work in 28 days then find the value of x?
P, Q और R पूरा काम 40 दिनों में पूरा कर सकते हैं। P कार्य शुरू करता है और 'x' दिनों तक कार्य करता है जबकि Q और R शेष $\frac{2}{5}$ कार्य 28 दिनों में पूरा करते हैं तो x का मान ज्ञात कीजिए?
- a) 48 days
 - b) 56 days
 - c) 32 days
 - d) 40 days
 - e) None of these
- 53) P takes 16 days to complete $\frac{2}{3}$ of a work, Q takes 6 days to complete $\frac{1}{7}$ of the same work and R takes 16 days to complete $\frac{4}{5}$ of the same work. If they work for 6 days together then Q and R leaves the work. Find the number of days P will take to complete the remaining work?

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P को किसी कार्य का $\frac{2}{3}$ भाग पूरा करने में 16 दिन लगते हैं, Q को उसी कार्य का $\frac{1}{7}$ भाग पूरा करने में 6 दिन लगते हैं और R को उसी कार्य का $\frac{4}{5}$ भाग पूरा करने में 16 दिन लगते हैं। यदि वे 6 दिनों तक एक साथ काम करते हैं तो Q और R काम छोड़ देते हैं। शेष कार्य को पूरा करने में P को कितने दिन लगेंगे?

- a) $\frac{117}{28}$ days
- b) $\frac{95}{23}$ days
- c) $\frac{156}{25}$ days
- d) $\frac{258}{35}$ days
- e) None of these

54) 10 men can complete a piece of work in 16 days. After 5 days from the start of the work, some men left. If the remaining work was completed by the remaining men $18\frac{1}{3}$ days, how many men left after 5 days from the start of the work?

10 आदमी एक काम को 16 दिनों में पूरा कर सकते हैं। काम शुरू होने के 5 दिन बाद कुछ आदमी चले गए। यदि शेष कार्य शेष पुरुषों द्वारा $18\frac{1}{3}$ दिनों में पूरा कर लिया गया, तो कार्य शुरू होने के 5 दिन बाद कितने पुरुष बचे?

- a) 10 men
- b) 9 men
- c) 4 men
- d) 6 men
- e) None of these.

55) Two men undertook to do a piece of work for Rs. 6000. One alone could do it in 30 days and the other can do it in 24 days, while with the assistance of a boy, work gets completed in 10 days. Find the share of the boy?

दो लोगों ने रुपये के लिए एक काम करने का बीड़ा उठाया। 6000. एक अकेले इसे 30 दिनों में पूरा कर सकता है और दूसरा इसे 24 दिनों में कर सकता है, जबकि एक लड़के की सहायता से काम 10 दिनों में पूरा हो जाता है। लड़के का हिस्सा ज्ञात कीजिये?

- a) Rs.1500
- b) Rs.2750
- c) Rs.2500
- d) Rs.2000
- e) None of these

56) Kartik is 40% more efficient than Raju, and Kartik can alone do a piece of work in 20 days. If they working in alternative days starts with Kartik, then in how many days the work will get completed?

कार्तिक, राजू से 40% अधिक कुशल है, और कार्तिक अकेले एक काम को 20 दिनों में पूरा कर सकता है। यदि वे वैकल्पिक दिनों में काम करना कार्तिक से शुरू करते हैं, तो काम कितने दिनों में पूरा हो जाएगा?

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- a) 21 (1/7) days
- b) 26 (2/5) days
- c) 18 (6/17) days
- d) 23 (1/5) days
- e) 21 (3/5) days

57) A can complete the whole work in 'x' days, B can complete the whole work in '3x' days, and C can complete the whole work in '6x' days. The time taken by B and C together to complete the work is the same as the time taken by D alone to complete the whole work. If 70% of the work is completed by D in 7 days, then find the time taken for A, B, C, and D together to complete the work.

A पूरा काम 'x' दिनों में पूरा कर सकता है, B पूरा काम '3x' दिनों में पूरा कर सकता है, और C पूरा काम '6x' दिनों में पूरा कर सकता है। B और C द्वारा मिलकर काम पूरा करने में लिया गया समय, D द्वारा अकेले पूरा काम पूरा करने में लिया गया समय के बराबर है। यदि D द्वारा 7 दिनों में 70% कार्य पूरा कर लिया जाता है, तो A, B, C और D द्वारा मिलकर कार्य पूरा करने में लिया गया समय ज्ञात कीजिये।

- a) 3.5
- b) 2.5
- c) 5.5
- d) 4.5
- e) None of these

58) A and B alone can do a piece of work in x and (x+10) days respectively and they work together can finish the work in $9\frac{3}{8}$ days. Then find the efficiency ratio of A and B?

A और B अकेले एक काम को क्रमशः x और (x+10) दिनों में पूरा कर सकते हैं और वे एक साथ काम करके उस काम को $9\frac{3}{8}$ दिनों में पूरा कर सकते हैं। तो A और B का दक्षता अनुपात ज्ञात कीजिये?

- a) 2:7
- b) 5:3
- c) 2:3
- d) 1:3
- e) 4:5

59) The ratio of time taken by a man, a woman, and a child to complete the work is 2: 5: 8, respectively. If 12 men, 15 women, and 20 children together start the same work and receive a total wage of Rs. 10580. If 'x' number of men gets a wage of Rs. 6900. Find the total wages of 'x' men, 'x - 3' women, and 'x - 5' children.

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एक पुरुष, एक महिला और एक बच्चे द्वारा कार्य पूरा करने में लिए गए समय का अनुपात क्रमशः 2: 5: 8 है। यदि 12 पुरुष, 15 महिलाएं और 20 बच्चे एक साथ एक ही काम शुरू करते हैं और कुल मजदूरी रु। 10580. यदि 'x' संख्या के पुरुषों को रुपये का वेतन मिलता है। 6900. 'x' पुरुषों, 'x - 3' महिलाओं और 'x - 5' बच्चों की कुल मजदूरी ज्ञात कीजिए।

- a) Rs.9858
- b) Rs.10258
- c) Rs.14258
- d) Rs.12498
- e) Rs.8548

- 60) A, B and C gets a total wage of Rs.1800 if they work together. A and B gets a total wage of Rs. 1400 if they work together and A alone gets Rs.800. Find the efficiency ratio of A,B and C.

यदि ए, बी और सी एक साथ काम करते हैं तो उन्हें कुल 1800 रुपये का वेतन मिलता है। A और B को कुल मिलाकर रु. का वेतन मिलता है। यदि वे एक साथ काम करते हैं तो 1400 रुपये मिलते हैं और ए को अकेले 800 रुपये मिलते हैं। A,B और C का दक्षता अनुपात ज्ञात कीजिए।

- a) 1:2:3
- b) 2:4:3
- c) 4:3:2
- d) 2:5:4
- e) 1:5:3

- 61) Manu and Ragav can separately do a piece of work in 30 and 36 days respectively. They worked together for 12 days, after which Ragav was replaced by Ranjni. If the work was finished in next $8\frac{2}{3}$ days, then find the number of days in which Ranjni alone could do the work?

मनु और राघव अलग-अलग एक काम को क्रमशः 30 और 36 दिनों में कर सकते हैं। उन्होंने 12 दिनों तक एक साथ काम किया, जिसके बाद राघव की जगह रंजनी ने ले ली। यदि कार्य अगले $8\frac{2}{3}$ दिनों में समाप्त हो गया, तो ज्ञात कीजिए कि रंजनी अकेले कितने दिनों में कार्य कर सकती है?

- a) 8 days
- b) $9\frac{5}{6}$ days
- c) $10\frac{2}{5}$ days
- d) 15 days
- e) None of these

- 62) A piece of work has to be completed in 30 days, a number of men are employed but it is found that only half of the work is done in 20 days, then an additional 30 men were joined to complete the work on time. Initially how many men are there to work?

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एक काम को 30 दिनों में पूरा करना है, कई आदमी कार्यरत हैं लेकिन यह पाया गया कि 20 दिनों में केवल आधा काम हुआ है, फिर काम को समय पर पूरा करने के लिए अतिरिक्त 30 आदमी शामिल किए गए। प्रारंभ में काम करने के लिए कितने आदमी हैं?

- a) 30 men
- b) 26 men
- c) 24 men
- d) 34 men
- e) None of these

- 63) The ratio of efficiency of Aman and Neha is 6: 5. The ratio of number of days taken by Pranav to Neha is 3:2. Aman takes 3 days less than Neha, when Aman and Neha complete the work individually. Pranav and Neha started the work and left after 3 days. The number of days taken by Aman to finish the remaining work is?

अमन और नेहा की दक्षता का अनुपात 6: 5 है। प्रणव और नेहा द्वारा लिए गए दिनों की संख्या का अनुपात 3: 2 है। अमन को नेहा से 3 दिन कम समय लगता है, जब अमन और नेहा अकेले काम पूरा करते हैं। प्रणव और नेहा ने काम शुरू किया और 3 दिन बाद छोड़ दिया। शेष कार्य को पूरा करने में अमन को कितने दिन लगेंगे?

- a) 12 days
- b) 9 days
- c) 10 (5/6) days
- d) 11 (2/5) days
- e) None of these

- 64) A, B and C alone complete a piece of work in 12, 6 and 18 days respectively. If two persons work on each day (BC, AC and AB) starts from BC and continue until the work was completed, then find the no. of days taken to complete the whole work.

A, B और C अकेले एक काम को क्रमशः 12, 6 और 18 दिनों में पूरा करते हैं। यदि प्रत्येक दिन दो व्यक्ति (बीसी, एसी और एबी) काम करना शुरू करते हैं और काम पूरा होने तक जारी रखते हैं, तो संख्या ज्ञात करें। संपूर्ण कार्य को पूरा करने में लगने वाले दिनों की संख्या।

- a) 9 days
- b) 46/9 days
- c) 12 days
- d) 48/9 days
- e) 20 days

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- 65) The ratio of efficiency of Anu to Chintu is 5:3. The ratio of number of days taken by Chintu to Banu is 2:3. Anu takes 16 days less than Chintu, when Anu and Chintu complete the work individually. Banu and Chintu started the work and left after 4 days. The number of days taken by Anu to finish the remaining work is?

अनु और चिंटू की दक्षता का अनुपात 5:3 है। चिंटू और बानू द्वारा लिए गए दिनों की संख्या का अनुपात 2:3 है। अनु को चिंटू से 16 दिन कम समय लगता है, जब अनु और चिंटू व्यक्तिगत रूप से कार्य पूरा करते हैं। बानू और चिंटू ने काम शुरू किया और 4 दिन बाद छोड़ दिया। शेष कार्य को पूरा करने में अनु को कितने दिन लगेंगे?

- a) 24 days
- b) 15 days
- c) 16 days
- d) 20 days
- e) None of these

- 66) A can do a piece of work in 20 days, B in 30 days. They work together for 10 days, the rest of the work is finished by C in four days. If they get Rs. 6000 as wages for the whole work, what are the daily wages of A,B and C respectively (in Rs)?

A एक काम को 20 दिनों में कर सकता है, B 30 दिनों में। वे 10 दिनों तक एक साथ काम करते हैं, शेष कार्य C चार दिनों में पूरा करता है। यदि उन्हें रु. पूरे काम के लिए मजदूरी 6000 रुपये है, तो क्रमशः A, B और C की दैनिक मजदूरी (रुपये में) क्या है?

- a) 2400, 1600, 2000
- b) 2000, 1800, 2200
- c) 1800, 2600, 1600
- d) 1500, 2500, 2000
- e) 2800, 1400, 1800

- 67) A alone can do a piece of work in 40 days. B alone can do the same work in 10 days more than A. A and C together can do the same work in 24 days. B started working, after 2 days A joins with him for 2 days only and then C joins with him for last day only. If the remaining work completed by D alone in 10 days, then find the number of days taken by D alone to complete whole work?

A अकेले किसी काम को 40 दिनों में पूरा कर सकता है। B अकेले उसी कार्य को A से 10 दिन अधिक में कर सकता है। A और C मिलकर उसी कार्य को 24 दिनों में कर सकते हैं। B ने काम करना शुरू किया, 2 दिन बाद A केवल 2 दिनों के लिए उसके साथ जुड़ता है और फिर C केवल आखिरी दिन के लिए उसके साथ जुड़ता है। यदि शेष कार्य D अकेले 10 दिनों में पूरा करता है, तो D द्वारा अकेले पूरा कार्य पूरा करने में लगे दिनों की संख्या ज्ञात कीजिये?

- a) 8 days
- b) 12 days

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- c) 64 days
- d) 84 days
- e) None of these

68) A, B and C together can do a piece of work in $2\frac{2}{3}$ days. A and B together can do the same work in $3\frac{3}{7}$ days and B takes 2 days more than A. A and B started working together and every 3rd day C alone working until the work was completed. How many days are required to complete the whole work?

A, B और C मिलकर एक काम को $2\frac{2}{3}$ दिनों में पूरा कर सकते हैं। A और B मिलकर उसी काम को $3\frac{3}{7}$ दिनों में कर सकते हैं और B को A से 2 दिन अधिक लगते हैं। A और B एक साथ काम करना शुरू करते हैं और काम पूरा होने तक हर तीसरे दिन C अकेले काम करता है। संपूर्ण कार्य को पूरा करने में कितने दिन लगेंगे?

- a) 6 days
- b) $8\frac{2}{7}$ days
- c) $3\frac{1}{7}$ days
- d) 12 days
- e) 15 days

69) Shamli alone can do a piece of work in 24 days who is 50% and 200% more efficient Sahiba and Ahkil respectively. they started the work at a manner that Sahiba started working on first day, second day Shamli joins with her and Ahkil joins with her on third day, how many days required to completing the whole work?

अकेले शामली एक काम को 24 दिनों में पूरा कर सकता है जो क्रमशः साहिबा और अहकिल से 50% और 200% अधिक कुशल है। उन्होंने काम इस प्रकार शुरू किया कि पहले दिन साहिबा ने काम करना शुरू किया, दूसरे दिन शामली उसके साथ शामिल हो गई और तीसरे दिन अहकिल उसके साथ शामिल हो गई, पूरे काम को पूरा करने में कितने दिन लगेंगे?

- a) $8\frac{2}{3}$
- b) $7\frac{1}{2}$
- c) $8\frac{1}{3}$
- d) $7\frac{1}{3}$
- e) 17

70) A alone can do a piece of work with 50% of its efficiency in 60 days, which is two times of the no. of days taken by C alone can complete the whole work and five times of the no. of days taken by B and C together can complete the whole work. Then find the no. of days required by A and B together working 75% of their efficiency ?

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A अकेले किसी कार्य को अपनी 50% दक्षता के साथ 60 दिनों में कर सकता है, जो कि संख्या का दो गुना है। C अकेले पूरा काम पूरा कर सकता है और उससे पाँच गुना अधिक दिन ले सकता है। B और C द्वारा मिलकर लिया गया दिन पूरा काम पूरा कर सकता है। फिर नंबर ढूँढ़ें। A और B को अपनी कार्यकुशलता का 75% एक साथ काम करने में कितने दिन लगेंगे?

- a) 6 days
- b) 7 days
- c) 16 days
- d) 18 days
- e) 10 days

71) A and B together can do a piece of work in 4 days. B and C together can do the same work in $4\frac{4}{5}$ days. A and C together can do the same work in $3\frac{3}{7}$ days. If A and B started working together after every two days C joins with them for one day, then find the number of days required to complete the whole work. A और B मिलकर एक काम को 4 दिनों में पूरा कर सकते हैं। B और C मिलकर उसी कार्य को $4\frac{4}{5}$ दिनों में कर सकते हैं। A और C मिलकर उसी कार्य को $3\frac{3}{7}$ दिनों में कर सकते हैं। यदि A और B एक साथ काम करना शुरू करते हैं तो हर दो दिन के बाद C उनके साथ एक दिन के लिए जुड़ जाता है, तो पूरे काम को पूरा करने के लिए आवश्यक दिनों की संख्या ज्ञात कीजिए

- a) $7\frac{3}{4}$ days
- b) $8\frac{4}{5}$ days
- c) $9\frac{1}{3}$ days
- d) $8\frac{1}{2}$ days
- e) 3.5 days

72) A and B together can complete a job in 8 days. Both B and C, working alone can finish the same job in 6 days, A and B commence work on the job, and work for 2 days, where upon A leaves, B continues for 1 more days, and then he leaves too, C now starts working, and finishes the job if C can do the same job in 12 days. How many days will C work?

A और B मिलकर एक काम को 8 दिनों में पूरा कर सकते हैं। B और C दोनों अकेले काम करते हुए उसी काम को 6 दिनों में पूरा कर सकते हैं, A और B काम पर काम शुरू करते हैं, और 2 दिनों तक काम करते हैं, जहाँ A के जाने के बाद, B 1 दिन और काम करता है और फिर वह भी काम छोड़ देता है, C अब कार्य प्रारंभ करता है, और कार्य समाप्त करता है यदि C उसी कार्य को 12 दिनों में कर सकता है। C कितने दिन काम करेगा?

- a) 5 days
- b) 7.5 days
- c) 3 days
- d) 8 days
- e) None of these





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73) A man is 20% more efficient than a woman and a woman is 25% more efficient than a child. 'm' men can complete in 10 days, 18 women can do it in 'n' days and 'p' children can do it in 18 days. If the value of 'p' is 2 more than that of 'n', then how many days '2n' women and 'p' children work together to complete the work?

एक पुरुष एक महिला की तुलना में 20% अधिक कुशल है और एक महिला एक बच्चे की तुलना में 25% अधिक कुशल है। 'एम' पुरुष इसे 10 दिनों में पूरा कर सकते हैं, 18 महिलाएं इसे 'एन' दिनों में कर सकती हैं और 'पी' बच्चे इसे 18 दिनों में पूरा कर सकते हैं। यदि 'p' का मान 'n' से 2 अधिक है, तो '2n' महिलाएं और 'p' बच्चे मिलकर काम पूरा करने में कितने दिन लगेंगे?

- a) 10 days
- b) 8 days
- c) 6 days
- d) 12 days
- e) None of these

74) The efficiency of a woman is 35 percent more than the efficiency of a man. $(X + 5)$ men can complete a work in 9 days while $3X$ women can complete the same work in 5 days. If the efficiency of a man is increased by 20% then $(5Y + 6)$ men can complete the same work in 3 days, while $(Y + 4)$ women can complete the same work in 8 days with the same efficiency. Find the value of $(3X + 2Y)$.

एक महिला की कार्यकुशलता पुरुष की कार्यकुशलता से 35 प्रतिशत अधिक होती है। $(X + 5)$ पुरुष एक काम को 9 दिनों में पूरा कर सकते हैं जबकि $3X$ महिलाएं उसी काम को 5 दिनों में पूरा कर सकती हैं। यदि किसी पुरुष की कार्यक्षमता 20% बढ़ जाए तो $(5Y + 6)$ पुरुष उसी कार्य को 3 दिन में पूरा कर सकते हैं, जबकि $(Y + 4)$ महिलाएं उसी कार्य को समान दक्षता से 8 दिनों में पूरा कर सकती हैं। $(3X + 2Y)$ का मान ज्ञात कीजिए।

- a) 12
- b) 18
- c) 21
- d) 10
- e) None of these

75) A, B and C individually can do a certain piece of work in 70, 84 and 48 days resp. They started working together but after 'x' days A and C left the job and the remaining work is completed by B with 25% more efficiency. If the whole work is completed in $228/5$ days then find the value of x.

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ए, बी और सी व्यक्तिगत रूप से एक निश्चित कार्य को क्रमशः 70, 84 और 48 दिनों में कर सकते हैं। उन्होंने एक साथ काम करना शुरू किया लेकिन 'x' दिनों के बाद A और C ने काम छोड़ दिया और शेष कार्य B द्वारा 25% अधिक दक्षता के साथ पूरा किया गया। यदि पूरा कार्य $228/5$ दिन में पूरा हो जाए तो x का मान ज्ञात कीजिए।

- a) 7
- b) 6
- c) 10
- d) 13
- e) 15

76) Four plumbers had to repair 364 taps. At the end of 3 days working 7 hours per day they had repaired 168 taps. In order to speed up the work, 3 more plumbers were added who also worked 7 hours daily. If the efficiency of each of the plumbers is same, how many additional days would it take to complete the task?
चार प्लंबरों को 364 नलों की मरम्मत करनी थी। प्रतिदिन 7 घंटे काम करके 3 दिनों के अंत में उन्होंने 168 नलों की मरम्मत की। काम में तेजी लाने के लिए 3 और प्लंबर जोड़े गए जो रोजाना 7 घंटे काम करते थे। यदि प्रत्येक प्लंबर की दक्षता समान है, तो कार्य को पूरा करने में कितने अतिरिक्त दिन लगेंगे?

- a) 2 days
- b) 3 days
- c) 4 days
- d) $11/2$ day
- e) None of these

77) (x-5) person can do a work in x days and (x+5) person can do 75 % of the same work in (x-11) days. Then in how many days can (x+10) person finish the work?
(x-5) व्यक्ति किसी कार्य को x दिनों में कर सकता है और (x+5) व्यक्ति उसी कार्य का 75% भाग (x-11) दिनों में कर सकता है। तो (x+10) व्यक्ति कार्य को कितने दिनों में पूरा कर सकता है?

- a) 15 days
- b) 10 days
- c) 20 days
- d) 24 days
- e) None of these

78) A, B, C, and D alone can complete the whole work in 'x' days, 'y' days, 30 days, and 36 days respectively. They all started work together and after 3 days C left work. After another 2 days, A and D left work and the remaining work was completed by B in $6/5$ days. If A is 50% more efficient than B, then find the value of (2x - y).

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A, B, C और D अकेले पूरे काम को क्रमशः 'x' दिन, 'y' दिन, 30 दिन और 36 दिन में पूरा कर सकते हैं। उन सभी ने एक साथ काम शुरू किया और 3 दिन बाद C ने काम छोड़ दिया। अगले 2 दिनों के बाद, A और D ने काम छोड़ दिया और शेष कार्य B द्वारा $6/5$ दिनों में पूरा किया गया। यदि A, B से 50% अधिक कुशल है, तो $(2x - y)$ का मान ज्ञात कीजिए।

- a) 5
- b) 11
- c) 12
- d) 6
- e) 8

- 79) The efficiency of A is 60% less than the efficiency of B, and the efficiency of C is 80 percent more than the efficiency of B. A alone can complete the work in 45 days. A started to work for d days then B joined and they worked for $(d + 1)$ days and then C also joined and they worked for $(d + 3)$ days and completed the work. How many days does B worked?

A की दक्षता B की दक्षता से 60% कम है, और C की दक्षता B की दक्षता से 80 प्रतिशत अधिक है। A अकेले काम को 45 दिनों में पूरा कर सकता है। A ने d दिनों के लिए काम करना शुरू किया, फिर B शामिल हो गया और उन्होंने $(d + 1)$ दिनों तक काम किया और फिर C भी शामिल हो गया और उन्होंने $(d + 3)$ दिनों तक काम किया और काम पूरा किया। B ने कितने दिन काम किया?

- a) 6.8 days
- b) 7.5 days
- c) 5.6 days
- d) 2.4 days
- e) 4.5 days

- 80) The efficiency of a boy is 20% less than the efficiency of a man and the efficiency of a woman is 35% more than the efficiency of a man. 3 men can complete work A in $(2X + 7)$ days while 4 women can complete work A in $(2X + 3)$ days. If the efficiency of a man is decreased by 20%, then 2 men can complete work B in $(5Y + 9)$ days while 9 boys can complete work B in $(Y + 4)$ days with the same efficiency. Find the time taken by X men, 4 women, and $(Y - 14X)$ boys to complete work B with their actual efficiency.

एक लड़के की दक्षता एक पुरुष की दक्षता से 20% कम है और एक महिला की दक्षता एक पुरुष की दक्षता से 35% अधिक है। 3 पुरुष कार्य A को $(2X + 7)$ दिनों में पूरा कर सकते हैं जबकि 4 महिलाएं कार्य A को $(2X + 3)$ दिनों में पूरा कर सकती हैं। यदि एक आदमी की दक्षता 20% कम हो जाती है, तो 2 आदमी कार्य B को $(5Y + 9)$ दिनों में पूरा कर सकते हैं जबकि 9

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लड़के समान दक्षता के साथ कार्य B को $(Y + 4)$ दिनों में पूरा कर सकते हैं। X पुरुषों, 4 महिलाओं और $(Y - 14X)$ लड़कों द्वारा अपनी वास्तविक दक्षता के साथ कार्य B को पूरा करने में लिया गया समय ज्ञात कीजिए।

- a) 30 days
- b) 15.2 days
- c) 21.6 days
- d) 24 days
- e) 16.5 days

- 81) The efficiency of a woman is 25 percent more than the efficiency of a man and the efficiency of a man is 50 percent more than the efficiency of a boy. $(X + 7)$ men and X boys started to work and they worked for 3 days, then 5 men left and 1 boy joined and then they worked for 4 days to complete the work. $(2X + 4)$ women started the same work and worked for 2 days and then 3 women left and the rest of the women worked for 4 days to complete the work. If the value of Y is double that of X, find the time required (in days) to complete the same work by $(X + 4)$ men, 4 women, and $(Y - 1)$ boys together.

एक महिला की दक्षता एक पुरुष की दक्षता से 25 प्रतिशत अधिक है और एक पुरुष की दक्षता एक लड़के की दक्षता से 50 प्रतिशत अधिक है। $(X + 7)$ पुरुषों और X लड़कों ने काम करना शुरू किया और उन्होंने 3 दिनों तक काम किया, फिर 5 आदमी चले गए और 1 लड़का शामिल हो गया और फिर उन्होंने काम पूरा करने के लिए 4 दिनों तक काम किया। $(2X + 4)$ महिलाओं ने वही काम शुरू किया और 2 दिनों तक काम किया और फिर 3 महिलाएं चली गईं और बाकी महिलाओं ने काम पूरा करने के लिए 4 दिनों तक काम किया। यदि Y का मान X के मान से दोगुना है, तो उसी कार्य को $(X + 4)$ पुरुषों, 4 महिलाओं और $(Y - 1)$ लड़कों द्वारा एक साथ पूरा करने के लिए आवश्यक समय (दिनों में) ज्ञात कीजिए।

- a) 7.5
- b) 5
- c) 10.5
- d) 9
- e) 4.5

- 82) A B and C can do a certain piece of work in 40 days. 45 days and 72 days respectively. A and B started working together but after ____ days, both of them left the job and the remaining work is done by C alone in ____ days.
The values given in which of the following options will fill the blanks in the same order in which is it given to make the above statement true:

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A B और C एक निश्चित कार्य 40 दिनों में कर सकते हैं। क्रमशः 45 दिन और 72 दिन। A और B ने एक साथ काम करना शुरू किया लेकिन ___ दिनों के बाद, दोनों ने काम छोड़ दिया और शेष काम C ने अकेले ___ दिनों में पूरा किया।

निम्नलिखित में से किस विकल्प में दिए गए मान रिक्त स्थान को उसी क्रम में भरेंगे जिस क्रम में यह उपरोक्त कथन को सत्य बनाने के लिए दिया गया है:

- (i) 15, 21
- (ii) 19, 7.4
- (iii) 12, 31.2
- a) Only A
- b) Only B
- c) Only C
- d) Only A and C
- e) All A, B and C

Directions ():- Read the data carefully and answer the following questions.

A and C together complete a work in ___(i)___ days. Ratio of efficiency of C and D is ___(ii)___ D takes 8 more days than C and 6 less days than B. C and D together complete the work in ___(iii)___ days when D work with 50% less efficiency. E is 20% more efficient than A. D and E complete work in ___(iv)___ when they work in alternative day starts with D. A is 37.5% less efficient than B. Ratio of efficiency of A and D is 1:2.

निर्देश ():- डेटा को ध्यानपूर्वक पढ़ें और निम्नलिखित प्रश्नों के उत्तर दें।

A और C मिलकर एक काम को ___(i)___ दिनों में पूरा करते हैं। C और D की दक्षता का अनुपात ___(ii)___ है। D को C से 8 दिन अधिक और B से 6 दिन कम लगते हैं। C और D मिलकर काम को ___(iii)___ में पूरा करते हैं। वे दिन जब D 50% कम दक्षता के साथ कार्य करता है। E, A से 20% अधिक कुशल है। D और E ___(iv)___ में काम पूरा करते हैं, जब वे D से शुरू होने वाले वैकल्पिक दिन में काम करते हैं। A, B से 37.5% कम कुशल है। A और D की दक्षता का अनुपात 1:2 है।

83) Find the difference between the value of blank (iv) and (iii).

60) रिक्त स्थान (iv) और (iii) के मान के बीच अंतर ज्ञात करें।

- a) 12
- b) 16
- c) 18
- d) 20
- e) None of these

84) Ratio of efficiency of F and E is same as the value of (ii). Find the number of days taken by F to complete double quantity of work?

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F और E की दक्षता का अनुपात (ii) के मान के समान है। दोगुनी मात्रा का काम पूरा करने में F द्वारा लिए गए दिनों की संख्या ज्ञात कीजिए?

- a) 62.5
- b) 53.33
- c) 57.66
- d) 60
- e) None of these

85) The question consists of two statements numbered "I and II" given below it. You have to decide whether the data provided in the statements are sufficient to answer the question.

'A' and 'B' are hired to complete a job for Rs. 1080. Both of them completed the whole work together in 20 days. Find the share of

Statement I: A takes 9 days less than B to complete the whole work

Statement II: A' can complete the work in 36 days
प्रश्न में नीचे दिए गए दो कथन "I और II" क्रमांकित हैं। आपको यह तय करना होगा कि कथन में दिया गया डेटा प्रश्न का उत्तर देने के लिए पर्याप्त है या नहीं।

'ए' और 'बी' को एक काम पूरा करने के लिए रुपये में काम पर रखा जाता है। 1080. दोनों ने मिलकर पूरा काम 20 दिनों में पूरा किया। का हिस्सा ज्ञात कीजिये

कथन I: A को पूरा काम पूरा करने में B से 9 दिन कम लगते हैं

कथन II: A' पूरा कार्य 36 दिनों में पूरा कर सकता है।

- a) The data in statement I alone are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question.
- b) The data in statement II alone are sufficient to answer the question, while the data in statement I alone are not sufficient to answer the question.
- c) The data either in statement I alone or in statement II alone are sufficient to answer the question.
- d) The data given in both statements I and II together are not sufficient to answer the question.
- e) The data in both statements I and II together are necessary to answer the question.

86) The question consists of two statements numbered "I and II" given below it. You have to decide whether the data provided in the statements are sufficient to answer the question.

A, B and C together can complete a piece of work in 35 days. Find the time taken by A to complete the whole work alone.

Statement I: A is 25% less efficient than C who is twice as efficient as B

Statement II: A and B together can complete the whole work in 63 days while A and C together can complete the whole work in 45 days.

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63) प्रश्न में नीचे दिए गए दो कथन "I और II" क्रमांकित हैं। आपको यह तय करना होगा कि कथन में दिया गया डेटा प्रश्न का उत्तर देने के लिए पर्याप्त है या नहीं।

A, B और C मिलकर एक कार्य को 35 दिनों में पूरा कर सकते हैं। A द्वारा अकेले पूरा कार्य पूरा करने में लिया गया समय ज्ञात कीजिए।

कथन I: A, C से 25% कम कुशल है जो B से दोगुना कुशल है

कथन II: A और B मिलकर पूरा काम 63 दिनों में पूरा कर सकते हैं जबकि A और C मिलकर पूरा काम 45 दिनों में पूरा कर सकते हैं।

- a) The data in statement I alone are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question. A
- b) The data in statement II alone are sufficient to answer the question, while the data in statement I alone are not sufficient to answer the question. B
- c) The data either in statement I alone or in statement II alone are sufficient to answer the question.
- d) The data given in both statements I and II together are not sufficient to answer the question.
- e) The data in both statements I and II together are necessary to answer the question.

87) The question consists of three statements numbered "I, II and III" given below it. You have to decide whether the data provided in the statements are sufficient to answer the question.

There are three persons Harsh, Gourav and Nikhil. Harsh completes a work in 20 days. Find the time taken by Nikhil to complete the same work alone.

Statement I: Harsh is 40% less efficient than Gourav

Statement II: When Gourav works with 50% more efficiency and Nikhil works with thrice efficiency, then Gourav and Nikhil together can complete the work in 4 days.

Statement III: Nikhil takes 4 days more than Harsh to complete the work.

प्रश्न के नीचे तीन कथन "I, II और III" दिए गए हैं। आपको यह तय करना होगा कि कथनों में दिया गया डेटा प्रश्न का उत्तर देने के लिए पर्याप्त है या नहीं।

हर्ष, गौरव और निखिल तीन व्यक्ति हैं। हर्ष एक कार्य को 20 दिनों में पूरा करता है। उसी कार्य को अकेले पूरा करने में निखिल द्वारा लिया गया समय ज्ञात कीजिये।

कथन I: हर्ष, गौरव से 40% कम कुशल है

कथन II: जब गौरव 50% अधिक दक्षता के साथ काम करता है और निखिल तीन गुना दक्षता के साथ काम करता है, तो गौरव और निखिल मिलकर काम को 4 दिनों में पूरा कर सकते हैं।

कथन III: निखिल को कार्य पूरा करने में हर्ष से 4 दिन अधिक लगते हैं।

- a) The data in any two of the statements together is sufficient to answer the question.

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- b) The data in statement I alone is sufficient to answer the question while data in statement II and III together is not sufficient to answer the question.
- c) The data in all the three statements together is necessary to answer the question.
- d) The data either in statements I and II together or data in statement III alone is sufficient to answer the question.
- e) The data in all the three statements together is not sufficient to answer the question.

88) The question consists of three statements numbered I, II and III given below it. You have to decide whether the data provided in the statements are sufficient to answer the question

What is the time taken by 10 men and 6 women to complete a work?

Statement I : 20 men and 12 children can complete the work in 15 days.

Statement II: Ratio of the efficiencies of a woman and 3 children is 2:3 respectively.

Statement III: Ratio of the efficiencies of 10 women and 4 men is 5:3 respectively.

प्रश्न के नीचे तीन कथन संख्या I, II और III दिए गए हैं। आपको यह तय करना होगा कि कथनों में दिया गया डेटा प्रश्न का उत्तर देने के लिए पर्याप्त है या नहीं

एक कार्य को पूरा करने में 10 पुरुषों और 6 महिलाओं को कितना समय लगेगा?

कथन I: 20 पुरुष और 12 बच्चे कार्य को 15 दिनों में पूरा कर सकते हैं।

कथन II: एक महिला और 3 बच्चों की दक्षता का अनुपात क्रमशः 2:3 है।

कथन III: 10 महिलाओं और 4 पुरुषों की दक्षता का अनुपात क्रमशः 5:3 है।

- a) The data in statement I and statement II together is sufficient to answer the question, while the data in statement III alone is not sufficient to answer the question
- b) The data in statement II and statement III together is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question.
- c) The data in statement I and statement III together is sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question.
- d) The data in all statements I, II and III together is necessary to answer the question.
- e) The data in any of the given statements alone is sufficient to answer the question

89) The question consists of two statements numbered "I and II given below it You have to decide whether the data provided in the statements are sufficient to answer the question.

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A alone can complete a work in $(x - 16)$ days while B' alone can complete the work in 'x' days. Find the value of 'x'.

Statement I: B' and C can together complete 75% of the work in 10 days. 'C' takes 10 days less than A' to complete the entire work.

Statement II: If A increases his efficiency by 25% and B decreases his efficiency by 25% then A and B can together complete the work in 8 days less than time taken by B' alone to complete the work.

प्रश्न में नीचे दिए गए दो कथन I और II क्रमांकित हैं। आपको यह तय करना है कि कथन में दिया गया डेटा प्रश्न का उत्तर देने के लिए पर्याप्त है या नहीं।

A अकेले किसी कार्य को $(x - 16)$ दिनों में पूरा कर सकता है जबकि B' अकेले कार्य को 'x' दिनों में पूरा कर सकता है। एक्स का मान ज्ञात करें।

कथन I: B' और C मिलकर 10 दिनों में 75% कार्य पूरा कर सकते हैं। संपूर्ण कार्य पूरा करने में 'C' को A' से 10 दिन कम लगते हैं।

कथन II: यदि A अपनी कार्यकुशलता 25% बढ़ा देता है और B अपनी कार्यकुशलता 25% कम कर देता है, तो A और B मिलकर कार्य को B द्वारा अकेले कार्य पूरा करने में लिए गए समय से 8 दिन कम समय में पूरा कर सकते हैं।

- a) The data in statement I alone are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question.
- b) The data in statement II alone are sufficient to answer the question, while the data in statement I alone are not sufficient to answer the question.
- c) The data either in statement I alone or in statement II alone are sufficient to answer the question.
- d) The data given in both statements I and II together are not sufficient to answer the question.
- e) The data in both statements I and II together are necessary to answer the question.

90) The question consists of two statements numbered "I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the question.

4 men 6 women and 7 children together can build a castle in 6 days. Find the time taken by 20 women and 54 children together to build a castle.

Statement I: 3 men, 7 women and 12 children can build the same castle in 5 days.

Statement II: If a man alone was competing against the team of a women and a child in a castle building competition, no one would win.

प्रश्न में नीचे दिए गए दो कथन I और II क्रमांकित हैं। आपको यह तय करना है कि कथन में दिया गया डेटा प्रश्न का उत्तर देने के लिए पर्याप्त है या नहीं।

4 पुरुष 6 महिलाएं और 7 बच्चे मिलकर 6 दिनों में एक महल बना सकते हैं। एक महल बनाने में 20 महिलाओं और 54 बच्चों द्वारा मिलकर लिया गया समय ज्ञात कीजिए।

कथन I: 3 पुरुष, 7 महिलाएं और 12 बच्चे 5 दिनों में एक ही महल बना सकते हैं।

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कथन II: यदि महल निर्माण प्रतियोगिता में एक पुरुष अकेले महिलाओं और एक बच्चे की टीम के खिलाफ प्रतिस्पर्धा कर रहा हो, तो कोई भी नहीं जीतेगा।

- The data in statement I alone are sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question.
- The data in statement II alone are sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question.
- The data either in statement I alone or in statement II alone is sufficient to answer the question.
- The data given in both statements I and II together is not sufficient to answer the question.
- The data in both statements I and II together is necessary to answer the question.

91) Directions: The question consists of three statements numbered "I, II and III given below it. You have to decide whether the data provided in the statements are sufficient to answer the question.

Sanjay, Nitesh and Anurag are hired to complete a work. Sanjay alone can complete the work in 9 days.

Statement I: Sanjay and Anurag together can complete the work in 6 days.

Statement II: If all of them worked together and the total wage given to all of them was Rs. 1500, out of which Nitesh receives Rs. 500.

Statement III: Sanjay, Nitesh and Anurag together can complete the work in 4 days.

Which of the following option is correct to find the number of days taken by Nitesh alone to complete the work?

निर्देश: प्रश्न में नीचे दिए गए तीन कथन "I, II और III क्रमांकित हैं। आपको यह तय करना होगा कि कथन में दिया गया डेटा प्रश्न का उत्तर देने के लिए पर्याप्त है या नहीं।

संजय, नितेश और अनुराग को एक काम पूरा करने के लिए काम पर रखा गया है। संजय अकेले काम को 9 दिनों में पूरा कर सकता है।

कथन I: संजय और अनुराग मिलकर काम को 6 दिनों में पूरा कर सकते हैं।

कथन II: यदि उन सभी ने एक साथ काम किया और उन सभी को दिया गया कुल वेतन रु। 1500, जिसमें से नितेश को रु. 500.

कथन III: संजय, नितेश और अनुराग मिलकर 4 दिनों में काम पूरा कर सकते हैं।

अकेले नितेश द्वारा कार्य पूरा करने में लगे दिनों की संख्या ज्ञात करने के लिए निम्नलिखित में से कौन सा विकल्प सही है?

- Data in statement I and III together are necessary to answer the question, while data in statement II is not sufficient to answer the question.
- Data in statement II and III together are necessary to answer the question, while data in statement I is not sufficient to answer the question.

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- c) Data in statement I and II together are necessary to answer the question, while data in statement III is not sufficient to answer the question.
- d) Either option a or option c.
- e) Combining any of the two statements together are sufficient to answer the question.

92) A' alone can complete a piece of work in $28\frac{1}{8}$ days while 'A' and B' together can complete the same work in 18 days. B' started the work and after 5 days A joined him. After 6 more days, 'A' left and 'C' joined the work and the whole work was completed in 26 days.

Which one of the following statements is true?

- (i) If 'A' alone worked for 12 days and 'B' alone worked for 16 days then 'C' will complete the remaining work alone in 14.75 days.
- (ii) 'C' can complete the whole work in 23.5 days if he works 150% more efficiently.
- (iii) 'A' and 'C' together can complete 64% of the work in 12 days.
- (iv) If 'A' alone worked on it for 15 days. B' alone worked on it for 5 days then 'C' could have completed the remaining work in 10 days.

A' अकेले एक काम को $28\frac{1}{8}$ दिनों में पूरा कर सकता है जबकि 'A' और B' मिलकर उसी काम को 18 दिनों में पूरा कर सकते हैं। B' ने काम शुरू किया और 5 दिनों के बाद A उसके साथ जुड़ गया। 6 और दिनों के बाद, 'ए' चला गया और 'सी' काम में शामिल हो गया और पूरा काम 26 दिनों में पूरा हो गया।

निम्नलिखित में से कौन सा कथन सत्य है?

- (i) यदि 'ए' अकेले 12 दिन काम करता है और 'बी' अकेले 16 दिन काम करता है तो 'सी' अकेले शेष काम 14.75 दिनों में पूरा करेगा।
- (ii) यदि 'सी' 150% अधिक कुशलता से काम करता है तो वह पूरा काम 23.5 दिनों में पूरा कर सकता है।
- (iii) 'ए' और 'सी' मिलकर 64% काम 12 दिनों में पूरा कर सकते हैं।
- (iv) यदि 'ए' अकेले इस पर 15 दिनों तक काम करता है। B' अकेले इस पर 5 दिनों तक काम करता है तो 'C' शेष कार्य को 10 दिनों में पूरा कर सकता है।

- a) Only I
- b) Only III
- c) Only I and II
- d) Only II and III
- e) Only III and IV

93) Quantity I: Sai is 50 percent more efficient than Ram. They together started the work and $\frac{7}{20}$ th portion of work is left after 20 days. Find in what time Ram alone will complete the whole work.

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Quantity II: L can complete work in the same time in which M and N together complete it. If L and M together could complete it in 20 days and N alone in 30 days, how many days would M takes to complete the work?

मात्रा I: साई राम की तुलना में 50 प्रतिशत अधिक कुशल है। उन्होंने एक साथ काम शुरू किया और 20 दिनों के बाद काम का $7/20$ वां भाग शेष रह गया। ज्ञात कीजिए कि राम अकेले पूरा कार्य कितने समय में पूरा करेगा।

मात्रा II: L उसी कार्य को उतने ही समय में पूरा कर सकता है जितने समय में M और N मिलकर उसे पूरा करते हैं। यदि L और M मिलकर इसे 20 दिनों में और N अकेले 30 दिनों में पूरा कर सकते हैं, तो M को कार्य पूरा करने में कितने दिन लगेंगे?

- a) Quantity I < Quantity II
- b) Quantity I > Quantity II
- c) Quantity I $\leq$ Quantity II
- d) Quantity I $\geq$ Quantity II
- e) Quantity I = Quantity II

94) Quantity I: 15 men can do a piece of work in 20 days. 20 men started working together, left the work in 5 days, 20 women completed the remaining work in 20 days, and then find the number of days required to complete the whole work by 30 women alone?

Quantity II: 20 men can do a piece of work in 20 days. If 10 men started working together, and left the work in 10 days, 20 boys together can complete the remaining work in $x+10$ days, then find the number of days required by 20 boys alone complete the remaining work if one boy is equal to one men in terms of efficiency?

मात्रा I: 15 आदमी एक काम को 20 दिनों में कर सकते हैं। 20 पुरुषों ने एक साथ काम करना शुरू किया, 5 दिनों में काम छोड़ दिया, 20 महिलाओं ने शेष काम 20 दिनों में पूरा किया, और फिर 30 महिलाओं द्वारा अकेले पूरा काम पूरा करने में कितने दिन लगेंगे?

मात्रा II: 20 आदमी एक काम को 20 दिनों में कर सकते हैं। यदि 10 आदमी एक साथ काम करना शुरू करते हैं, और 10 दिनों में काम छोड़ देते हैं, तो 20 लड़के मिलकर शेष कार्य को $x+10$ दिनों में पूरा कर सकते हैं, तो 20 लड़कों द्वारा अकेले शेष कार्य को पूरा करने के लिए आवश्यक दिनों की संख्या ज्ञात कीजिए, यदि एक लड़का बराबर है कार्यकुशलता की दृष्टि से एक व्यक्ति?

- a) Quantity I < Quantity II
- b) Quantity I > Quantity II
- c) Quantity I $\leq$ Quantity II
- d) Quantity I $\geq$ Quantity II
- e) Quantity I = Quantity II

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95) Quantity I: 12 women can complete a piece of work in 5 days whereas 3 women and 9 children together can complete the same piece of work in 10 days. In how many days can 36 children complete the same piece of work?

Quantity II: 15 boys alone can do a piece of work in 20 days. 30 boys and 30 girls working alternatively starting with 30 boys, then find the total number of days taken to complete the whole work if one boy is equal to two girls?

मात्रा I: 12 महिलाएं एक काम को 5 दिनों में पूरा कर सकती हैं जबकि 3 महिलाएं और 9 बच्चे मिलकर उसी काम को 10 दिनों में पूरा कर सकते हैं। 36 बच्चे उसी कार्य को कितने दिनों में पूरा कर सकते हैं?

मात्रा II: 15 लड़के अकेले एक काम को 20 दिनों में कर सकते हैं। 30 लड़के और 30 लड़कियाँ 30 लड़कों के साथ शुरू करके वैकल्पिक रूप से काम करते हैं, यदि एक लड़का दो लड़कियों के बराबर है, तो पूरे काम को पूरा करने में लगने वाले दिनों की कुल संख्या ज्ञात कीजिए?

- a) Quantity I < Quantity II
- b) Quantity I > Quantity II
- c) Quantity I $\leq$ Quantity II
- d) Quantity I $\geq$ Quantity II
- e) Quantity I = Quantity II

96) Quantity I: X men can complete a work in 12 days, $1.6X$ women can complete same work in 15 days, if 5 men and 5 women together can complete the same work in 8 days, then find the time taken by $(X+5)$ men to complete the same work?

Quantity II: A can do a work in 16 days and B is 25% more efficient than A. B started from nothing and worked for 8 days and left. Then how many days does it take for A to complete the remaining work?

मात्रा I: X पुरुष एक काम को 12 दिनों में पूरा कर सकते हैं, $1.6X$ महिलाएं उसी काम को 15 दिनों में पूरा कर सकती हैं, यदि 5 पुरुष और 5 महिलाएं मिलकर उसी कार्य को 8 दिनों में पूरा कर सकते हैं, तो $(X+5)$ पुरुषों द्वारा उसी कार्य को पूरा करने में लिया गया समय ज्ञात कीजिए?

मात्रा II: A एक काम को 16 दिनों में कर सकता है और B, A से 25% अधिक कुशल है। B ने शून्य से शुरुआत की और 8 दिनों तक काम किया और छोड़ दिया। तो A को शेष कार्य पूरा करने में कितने दिन लगेंगे?

- a) Quantity I < Quantity II
- b) Quantity I > Quantity II
- c) Quantity I $\leq$ Quantity II
- d) Quantity I $\geq$ Quantity II
- e) Quantity I = Quantity II

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97) Quantity I: A, B and C can do a piece of work in 32 days, 48 days and 24 days respectively. They work together for 4 days, then A left. After this B and C worked for 2 days and the B left the work and remaining work is done by C. if they received Rs.6840 for whole work then the share of C?

Quantity II: The efficiency of A is 10% less than that of B, and the efficiency of C is 80% of that of B. A started the work, and B and C joined him after 4 days. They completed the work in two more days. If the total wage of A, B, and C is Rs. 22500, then find the wage of B.

मात्रा I: A, B और C एक काम को क्रमशः 32 दिन, 48 दिन और 24 दिन में कर सकते हैं। वे 4 दिनों तक एक साथ काम करते हैं, फिर A चला जाता है। इसके बाद B और C ने 2 दिनों तक काम किया और B ने काम छोड़ दिया और शेष काम C द्वारा किया गया। यदि उन्हें पूरे काम के लिए 6840 रुपये मिले तो C का हिस्सा क्या होगा?

मात्रा II: A की दक्षता B की तुलना में 10% कम है, और C की दक्षता B की तुलना में 80% है। A ने काम शुरू किया, और B और C 4 दिनों के बाद उसके साथ जुड़ गए। उन्होंने दो और दिनों में काम पूरा कर लिया। यदि A, B और C का कुल वेतन रु. 22500 है, तो B का वेतन ज्ञात कीजिये।

- a) Quantity I < Quantity II
- b) Quantity I > Quantity II
- c) Quantity I ≤ Quantity II
- d) Quantity I ≥ Quantity II
- e) Quantity I = Quantity II

98) Quantity I: 12 men, 8 women, and 12 children can finish a work in 20 days. While 10 men, 12 women, and 24 children can complete the same work in 15 days. If the efficiency of children is half that of men, then find the time required by 48 children to complete the whole work.

Quantity II: A and B together can complete the whole work in $60/7$ days. B and C together can complete the whole work in $120/11$ days. A and B worked together for 5 days and the remaining work was done by C in 10 days. How many days C alone can complete the total work?

Quantity III: 40 men can complete the whole work in 20 days. 30 women can complete the same work in 15 days. The efficiency of children is 25% more than that of women. How many days will it take 10 men, 15 women, and 6 children to complete the whole work?

मात्रा I: 12 पुरुष, 8 महिलाएं और 12 बच्चे एक काम को 20 दिनों में पूरा कर सकते हैं। जबकि 10 पुरुष, 12 महिलाएं और 24 बच्चे उसी कार्य को 15 दिनों में पूरा कर सकते हैं। यदि बच्चों की दक्षता पुरुषों की तुलना में आधी है, तो पूरे कार्य को पूरा करने के लिए 48 बच्चों द्वारा आवश्यक समय ज्ञात कीजिए।

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मात्रा II: A और B मिलकर पूरा काम 60/7 दिनों में पूरा कर सकते हैं। B और C मिलकर पूरा काम 120/11 दिन में पूरा कर सकते हैं। A और B ने 5 दिनों तक एक साथ काम किया और शेष कार्य C ने 10 दिनों में पूरा किया। C अकेले कुल कार्य कितने दिनों में पूरा कर सकता है?

मात्रा III: 40 आदमी पूरे काम को 20 दिनों में पूरा कर सकते हैं। 30 महिलाएँ उसी कार्य को 15 दिनों में पूरा कर सकती हैं। बच्चों की कार्यक्षमता महिलाओं की तुलना में 25% अधिक होती है। पूरे कार्य को पूरा करने में 10 पुरुषों, 15 महिलाओं और 6 बच्चों को कितने दिन लगेंगे?

- a) Quantity I < Quantity II < Quantity III
- b) Quantity I > Quantity II > Quantity III
- c) Quantity I ≤ Quantity II < Quantity III
- d) Quantity I ≥ Quantity II < Quantity III
- e) Quantity I = Quantity II = Quantity III

99) Quantity I: A, B, and C can complete a work in 20 days, 40 days, and 30 days, respectively. A, B, and C started the work together, after two days B left the work, after 3 more days A left the work and D joined. If the remaining work is completed by C and D together in 4 days, then find the time D will take to complete the whole work alone.

Quantity II: A, B, and C alone can complete the whole work in 45/2 days, 18 days, and 15/2 days, respectively. A started the work for some days, then left; after that, B only did the work for 6 days, then left; and after that, C only did the work for 2 days. How many days did A work?

Quantity III: A alone can complete the work in 30 days and the efficiency of A is 33.33% less than the efficiency of B and the efficiency of C is 16.66% less than the efficiency of A. If B worked for 4 days and left, then find the time taken by A and C to complete the remaining work.

79) मात्रा I: A, B और C एक कार्य को क्रमशः 20 दिन, 40 दिन और 30 दिन में पूरा कर सकते हैं। A, B और C ने एक साथ काम शुरू किया, दो दिन बाद B ने काम छोड़ दिया, 3 और दिनों के बाद A ने काम छोड़ दिया और D शामिल हो गया। यदि शेष कार्य C और D द्वारा मिलकर 4 दिनों में पूरा किया जाता है, तो D को अकेले पूरा कार्य पूरा करने में कितना समय लगेगा।

मात्रा II: A, B और C अकेले पूरे कार्य को क्रमशः 45/2 दिन, 18 दिन और 15/2 दिन में पूरा कर सकते हैं। A ने कुछ दिनों तक काम शुरू किया, फिर छोड़ दिया; उसके बाद, B ने केवल 6 दिनों तक काम किया, फिर छोड़ दिया; और उसके बाद, C ने केवल 2 दिनों के लिए कार्य किया। A ने कितने दिन काम किया?

मात्रा III: A अकेले काम को 30 दिनों में पूरा कर सकता है और A की दक्षता B की दक्षता से 33.33% कम है और C की दक्षता A की दक्षता से 16.66% कम है। यदि B ने 4 दिन काम किया और छोड़ दिया, तो शेष कार्य को पूरा करने में A और C द्वारा लिया गया समय ज्ञात कीजिये।

- a) Quantity I < Quantity II < Quantity III
- b) Quantity I > Quantity II > Quantity III

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- c) Quantity I < Quantity II > Quantity III
- d) Quantity I > Quantity II < Quantity III
- e) Quantity I = Quantity II = Quantity III

100) Quantity I: A, B, and C have equal rates of doing the work. A works 10 hours for 15 days, B works 'x' hours for 'x' days and C works 'y' hours for 'y' days. A can complete the work in 27.5 days working for the respective hours. If the value of 'x' is five more than that of 'y', then find the time taken to complete the work while working together with their respective hours in a day.

Quantity II: 18 persons can complete the work by working 6 hours a day in 10 days and x persons can do the same work by working 3 hours a day in 30 days, Then the value of x is how much more or less than 18.

Quantity III: A and B together can do a piece of work in 12 days, and B and C together can do it in 18 days. A and B started the work and A left after 4 days. Then C joined with B, and B left at the end of the 6th day of the work. Then the remaining work was completed by C. If C totally worked for 17 days, then in how many days can A alone finish the work?

मात्रा I: A, B और C की कार्य करने की दर समान है। A 15 दिनों के लिए 10 घंटे काम करता है, B 'x' दिनों के लिए 'x' घंटे काम करता है और C 'y' दिनों के लिए 'y' घंटे काम करता है। A संबंधित घंटों तक काम करके 27.5 दिनों में काम पूरा कर सकता है। यदि 'x' का मान 'y' के मान से पांच अधिक है, तो एक दिन में उनके संबंधित घंटों के साथ काम करते हुए काम पूरा करने में लगने वाला समय ज्ञात करें।

मात्रा II: 18 व्यक्ति प्रति दिन 6 घंटे काम करके 10 दिनों में कार्य पूरा कर सकते हैं और x व्यक्ति उसी कार्य को 30 दिनों में प्रति दिन 3 घंटे काम करके पूरा कर सकते हैं, तो x का मान 18 से कितना अधिक या कम है।

मात्रा III: A और B मिलकर एक काम को 12 दिनों में कर सकते हैं, और B और C मिलकर इसे 18 दिनों में कर सकते हैं। A और B ने काम शुरू किया और A ने 4 दिन बाद काम छोड़ दिया। फिर C, B के साथ जुड़ गया और B काम के छठे दिन के अंत में चला गया। फिर शेष कार्य C द्वारा पूरा किया गया। यदि C ने पूरी तरह से 17 दिनों तक कार्य किया, तो A अकेले कितने दिनों में कार्य पूरा कर सकता है?

- a) Quantity I < Quantity II < Quantity III
- b) Quantity I > Quantity II > Quantity III
- c) Quantity I < Quantity II > Quantity III
- d) Quantity I > Quantity II < Quantity III
- e) Quantity I = Quantity II = Quantity III

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SOLUTION

1) Answer C

Let, efficiency of each man and each boy be 'M' and 'B', respectively.

$$\text{So, } (6M + 8B) \times 40 = (9M + 4B) \times 30$$

$$\text{Or, } 24M + 32B = 27M + 12B$$

$$\text{Or, } 3M = 20B$$

$$\text{Total work} = (6M + 8B) \times 40 = (48 \times 40B) \text{ units}$$

$$(3M + 40B) = (20B + 40B) = 60B$$

$$\text{Required time taken} = (48 \times 40B) / 60B = 32 \text{ days}$$

2) Answer C

Let, efficiencies of Ganesh and Ramesh is $8x$ units/day and $11x$ units/day, respectively.

$$\text{Total work} = 8x \times 30 = 240 \text{ units}$$

$$\text{Required time taken} = 11/12 \times 240/11 = 20 \text{ days}$$

3) Answer D

$$10 \text{ women} = 6 \text{ men}$$

Let, 10 women can complete the work in X days.

According to the question,

$$20 \times 6 = (12 \times 5) + (X \times 6)$$

$$120 = 60 + 6x$$

$$120 - 60 = 6x$$

$$6x = 60$$

$$x = 10 \text{ days}$$

4) Answer A

$$\text{Total work} = 180 \text{ units (L.C.M of } 90 \& 60)$$

$$\text{Efficiency of A} = 180/90 = 2 \text{ units/day}$$

$$\text{Efficiency of B} = 180/60 = 3 \text{ units/day}$$

$$\text{Time taken by A and B to complete the work with their normal efficiencies} = 180/5 = 36 \text{ days}$$

$$\text{Required number of days to complete the work} = 18 \text{ days}$$

$$\text{Required efficiency} = 180/18 = 10 \text{ units/day}$$

$$\text{Increased efficiency of A} = 10 - 3 = 7 \text{ units/day}$$





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Required percentage = $\{(7-2)/2\} \times 100 = 250\%$

5) Answer C

Let, the total amount of work = LCM of 10 and 18 = 180 units

Amount of work done by Ankit and Monika in a day = $180/10 = 18$ units

Amount of work done by Ankit in a day = $180/18 = 10$ units

Amount of work done by Monika in a day = $18 - 10 = 8$ units

Monika can do the work in $180/8 = 22.5$ days

6) Answer C

Let the efficiency of A and B be 15 units per day and 17 units per day, respectively.

So, total amount of work = $(15 + 17) \times 15 = 480$ units

Time taken by A alone to complete the whole work = $480/15 = 32$ days

7) Answer A

It is given,

16 men = 20 women (based on their efficiency)

10 women + 4 men = 10 women + $4 \times (10/8)$ Women

= $(10 + 5)$ women = 15 women

So, $w_1 \times t_1 = w_2 \times t_2$

$20 \times 21 = 15 \times t_2$

$t_2 = 28$ days.

8) Answer 0

Subham can complete one-fifth of the work in 8 days.

Therefore, he can complete the whole work in $(8 \times 5) = 40$ days

Lucky can complete the work in 32 days.

Let the total work be L.C.M of $(40, 32) = 160$ units

Efficiency of Subham = $160/40 = 4$ units/day

Efficiency of Lucky = $160/32 = 5$ units/day

But, Lucky worked with only 20% of his efficiency = $0.20 \times 5 = 1$ unit/day

Time taken to complete the work = $160/(4 + 1) = 160/5 = 32$ days

9) Answer A

Let total amount of work = 72 units (LCM of 36 and 18)

Amount of work completed by 'A' in one day $72/36 = 2$ units

Amount of work completed by 'A' and 'B' together in one day = $72/18 = 4$ units

Amount of work completed by 'B' in one day = $4 - 2 = 2$ units

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Amount of work done by 'A' and 'B' together in 10 days = $4 \times 10 = 40$ units

Remaining work = $72 - 40 = 32$

Desired time = $32/2 = 16$

10) Answer C

Time taken by 'A' to complete the work alone = $24/40 \times 100 = 60$ days

Let total amount of work = 420 units (LCM of 60 and 35)

Amount of work completed by 'A' alone in one day = $420/60 = 7$ units

Amount of work completed by 'A' and 'B' together in one day = $420/35 = 12$ units

Amount of work completed by 'B' alone in one day = $12 - 7 = 5$ units

Amount of work done by 'A' and 'B' together in 10 days = $10 \times 12 = 120$ units

Remaining work = $420 - 120 = 300$ units

Desired time = $300/5 = 60$ days

11) Answer C

Let, efficiencies of 'A' and 'B' be ' $4x$ ' units/day and ' x ' units/day, respectively.

So, total units of work = $5x \times 24 = 120x$ units

Efficiency of 'C' = $(4x + x)/2 = '2.5x'$ units/day

Required time taken = $120x/2.5x = 48$ days

12) Answer B

Let the total work = 180 units

Efficiency of (A + B) = $180/10 = 18$ units/day

Efficiency of 'A' = $180/18 = 10$ unit/day

Therefore, efficiency of 'B' = $18 - 10 = 8$ units/day

Efficiency of 'C' = $1.5 \times 8 = 12$ units/day

Required time taken = $180/12 = 15$ days

13) Answer B

Let total amount of work = 850 units (LCM of 34 and 50)

Amount of work done by 'A' in one day = $850/50 = 17$ units

Amount of work done by 'A' and 'B' together in one day = $850/34 = 25$ units

Amount of work done by 'B' alone in one day = $25 - 17 = 8$ units

Amount of work done by 'B' in 35 days = $35 \times 8 = 280$ units

Remaining work = $850 - 280 = 570$ units

Time taken by 'A' and 'B' together to complete 570 units of work = $570/25 = 22.8$ days

Desired time = $35 + 22.8 = 57.8$ days

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14) Answer A

Let efficiency of each man is 1 unit per day.

Total work = $36 \times 80 = 2880$ units

According to question;

$2880 = x \times 30 + 56 \times 30$

Or, $x \times 30 = 2880 - 1680 = 1200$

Or, $x = 40$

15) Answer D

Let total amount of work = 210 units {LCM of 70, 42 and 30}

Amount of work done by 'A' alone in one day = $210/70 = 3$ units

Amount of work done by 'A' and 'B' together in one day = $210/42 = 5$ units

Amount of work done by 'B' alone in one day = $5 - 3 = 2$ units

Amount of work done by 'A' and 'C' together in one day = $210/30 = 7$ units

Amount of work done by 'C' alone in one day = $7 - 3 = 4$ units

Amount of work done by 'A' alone in 40 days = $3 \times 40 = 120$ units

Amount of work left = $210 - 120 = 90$ units

Desired time = $90/(2 + 4) = 15$ days

16) Answer A

Given, Chirag can complete the work in 16 days.

Since, he worked for 12 days only, therefore, percentage of the work completed by him

= $(12/16) \times 100 = 75\%$

Therefore, Hemant completed 25% of the work in 5 days

Time taken by Hemant to complete the whole work = $5/0.25 = 20$ days

Let total work = 80 units

Efficiency of Chirag = $80/16 = 5$ units/day

Efficiency of Hemant = $80/20 = 4$ units/day

Therefore, total time taken to complete the work = $\{80/(5 + 4)\} = (80/9)$ days

17) Answer A

Let total amount of work = 144 units (LCM of 16 and 9)

Amount of work done by 'A' in one day = $144/16 = 9$ units

Amount of work done by 'A' and 'B' together in one day = $144/9 = 16$ units

Amount of work done by 'B' in one day = $16 - 9 = 7$ units

Amount of work done by 'B' in 18 days = $18 \times 7 = 126$ units





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$$\text{Desired time} = (144-126)/9 = 2 \text{ days}$$

18) Answer C

Let total amount of work be 300 units (LCM of 50 and 60)

Efficiency of 'A' = $300/50 = 6$ units/day

Efficiency of 'B' = $300/60 = 5$ units/day

Let efficiency of 'C' be 'x' units/day.

Amount of work done by 'A' and 'B' together in 10 days = $(6 + 5) \times 10 = 110$ units

Remaining work = $300 - 110 = 190$ units

ATQ,

$$10 = 190/(6 + 5 + x)$$

$$\text{Or, } 110 + 10x = 190$$

$$\text{Or, } x = 8$$

$$\text{Required time} = 300/8 = 37.5 \text{ days}$$

19) Answer D

Let total work = 240 units

Efficiency of 'A' = $240/40 = 6$ units/day

Efficiency of 'B' = $240/48 = 5$ units/day

Amount of work completed by 'B' and 'C', together = $240 - 18 \times (6 + 5) = 42$ units

Let the efficiency of 'C' be 'x' units/day

According to the question,

$$5 + x = 42/6$$

$$\text{Or, } x = 7 - 5 = 2$$

Efficiency of 'C' = 2 units/day

Therefore, required time taken = $240/2 = 120$ days.

20) Answer A

Time taken by 'B' to complete the whole work = $24/0.6 = 40$ days

Time taken by 'A' to complete the whole work = $24/0.4 = 60$ days

Let total work = 120 units

Efficiency of 'B' = $120/40 = 3$ units/day

Efficiency of 'A' = $120/60 = 2$ units/day

Required time taken = $120/5 = 24$ days

21) Answer A

Let the efficiency of 'B' be 'x' units/day

Therefore, efficiency of 'A' = $1.7x$ units/day

Total work = $(70/9) \times (x + 1.7x) = 21x$ units

Amount of work completed by 'A' in 10 days = $10 \times 1.7x = 17x$ units

Time taken by 'B' to complete the remaining work = $\{(21x - 17x)/x\} = 4$ days

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Therefore, total time taken = $10 + 4 = 14$ days

22) Answer A

According to the question,

4 men = 6 boys

Or, $2M = 3B$

Therefore, 6 men + 7 boys = 9 boys + 7 boys = 16 boys

6 boys can complete a work in 36 days

Therefore, 16 boys will take = $(36 \times 6) / 16 = 13.5$ days

23) Answer C

Let, total amount of work = 720 units (LCM of 80, 144 and 90)

Efficiency of 'A' = $720 / 80 = 9$ units/day

Efficiency of 'B' = $720 / 144 = 5$ units/day

Efficiency of 'C' = $720 / 90 = 8$ units/day

Let, the whole work is completed in 'x' days

So, 'A' worked for 'x-40' days

And, 'C' worked for $x - 40 + 10 = 'x - 30'$ days

So, $9(x - 40) + 8(x - 30) + 5x = 720$

Or, $22x = 1320$

Or, $x = 60$

24) Answer D

Amount earned by (8 men + 5 boys) in 1 day = $3240 / 3 = \text{Rs. } 1080 \dots (1)$

Amount earned by (3 men + 2 boys) in 1 day = $2484 / 6 = 414 \dots (2)$

On solving equation (1) and (2), we get

Amount earned by a man in 1 day = Rs. 90

Amount earned by a boy in 1 day = Rs. 72

Therefore, number of days taken by (5 men + 4 boys) to earn Rs. 1476 =
 $= 1476 / \{(5 \times 90 + 4 \times 72)\} = 2$ days

25) Answer C

According to question;

$1/(x + 32) + 1/90 = 1/x$

Or, $x^2 + 122x = 90x + 2880$

Or, $x^2 + 32x - 2880 = 0$

Or, $x^2 + 72x - 40x - 2880 = 0$

Or, $x(x + 72) - 40(x + 72) = 0$

Or, $(x - 40)(x + 72) = 0$

So, $x = 40, -72$

Number of days can't be negative, so $x = 40$

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Time taken by A to complete the work = $40 + 32$
= 72 days

Total work = 360 units (LCM of 40 and 72)

Efficiency of A = $360/72 = 5$ units per day

Efficiency of C = $1.20 \times 5 = 6$ units per day

Desired time = $360/6 = 60$ days

26) Answer A

Let total amount of work = 630 units (LCM of 90, 70 and 126)

Amount of work done by A in one day = $630/90 = 7$ units

Amount of work done by B in one day = $630/70 = 9$ units

Amount of work done by C in one day = $630/126 = 5$ units

Let total time taken to complete the whole work is 'x' days

According to question;

$$5X + 9 \times (X - 28) + 7 \times (X - 48) = 630$$

$$\text{Or, } 21x = 630 + 336 + 252$$

$$\text{Or, } 21x = 1218$$

$$\text{Or, } x = 58$$

27) Answer B

Time taken by 'A' to complete the whole work = $8 \times 3 = 24$ days

Time taken by 'B' to complete the whole work = $16 \times (3/2) = 24$ days

Time taken by C to complete the whole work = $0.75 \times 24 = 18$ days

Let the total work be 72 units

Efficiency of 'A' = $72/24 = 3$ units/day

Efficiency of 'B' = $72/24 = 3$ units/day

Efficiency of 'C' = $72/18 = 4$ units/day

Required time taken = $(5/6) \times (72/10) = 6$ days

28) Answer E

According to the question,

$$48x = (x - 7) \times 90$$

$$\text{Or, } 42x = 630$$

$$\text{Or, } x = 15$$

Hence, option e.

29) Answer B

Time taken by 'A' to complete the whole work = $48/0.40 = 120$ days

Time taken by 'A' and 'B' together to complete the whole work = $45/0.625 = 72$ days

Time taken by 'A, B' and 'C' together to complete the whole work = $27/0.5 = 54$ days

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Let total amount of work = 1080 units (LCM of 120, 72 and 54)

Amount of work done by A in one day = $1080/120 = 9$ units

Amount of work done by 'A' and 'B' together in one day = $1080/72 = 15$ units

Amount of work done by 'A', 'B' and 'C' together in one day = $1080/54 = 20$ units

Amount of work done by 'C' in one day = $20 - 15 = 5$ units

Amount of work done by 'B' in one day = $15 - 9 = 6$ units

Desired Percentage = $[(6-5)/5] \times 100 = 20\%$

30) Answer E

Using ('a' men x d' days)/work = (b' men x 'c days)/work

According to the question,

$(60 \times 120) = (n \times 120)$

Or, $n = 60$

Therefore, no extra man has to be employed

31) Answer C

Let assume total work = 30 units

A's one day work = 3 units

R's one day work = 2 units

If they together work for 5 days = $5 \times 3 + 5 \times 2 = 25$ units

The remaining work = $30 - 25 = 5$ units

5 units work completed by 'N' in two days

So, N's one day work = 2.5 units

If they get Rs. 1500 for 30 units

So, 1 units wage = $1500/30 = 50$ units

'R' and 'N' one day work = $2 + 2.5 = 4.5$ units

So, sum of the daily wages of 'R' and 'N' is $4.5 \times 50 = \text{Rs. } 225$.

32) Answer A

SINCE THE TOTAL WORK IS SAME, $(a+10) \times (a-11) = (a+5) \times (a-10)$

Solve the equation we get,

$4a = 60$

$a = 15$

Total work = $(a+10) \times (a-11)$

$= 25 \times 4$

$= 100$

Time taken by (a-13) person to complete $1/10$ the of the same work, = {

$(1/10) \times 100 \} / (a-13)$

$= 10/2$

$= 5$ days

Time taken is 5 days.

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33) Answer B

Let the efficiencies (work done per day) of (AA+BB), (BB+CC) AND (CC+AA) be a, b, and c respectively

Let total work be 1 unit.

Efficiency of (AA+BB) is $a \times 100 = 1$; $\Rightarrow a = 1/100$; similarly (BB+CC) is $b \Rightarrow 1/250$; and (CC+AA) is $c = 1/500$

Time taken by (AA+BB), (BB+CC), (CC+AA) to complete the whole work :

$(AA+BB) + (BB+CC) + (CC+AA) = 8$

$(AA+BB+CC) = 4$

Time taken by AA, BB, CC to complete the whole work : $500/4 \Rightarrow 250/2 \Rightarrow 125$

So, AA, BB, CC together complete the whole work in 125 days.

34) Answer A

Total work = time for A to complete the work * efficiency of A

Total work = $30 \times 5 = 150$ units

Work done by A in 1 hour = $150/30 = 5$ units

Work done by B in 1 hour = $150/25 = 6$ units

Work done in two days = $5 + 6 = 11$ units

Work done in first 27 i.e. (26+1) hours = $11 \times 13 + 5 = 148$

Time for remaining 2 units work = $2/6 = 1/3 = 0.33$ hours

Total time = $27 + 0.33 = 27.33$ hours.

35) Answer D

A and C PERFORM 12/19 OF THE WORK CONTRACTED.

SO PART OF WORK DONE BY B = $1 - 12/19 = 7/19$

Total contract for the work = Rs. 4465 lakhs

B is to be paid for $7/19$ of work = $4465 \times 7/19 = 1645$ lakhs

36) Answer E

Total work is same in both conditions, (21 boys + 13 girls) * 1 day = (9 boys + 18 girls) * 1 day

$21 \text{ boys} + 13 \text{ girls} = 9 \text{ boys} + 18 \text{ girls}$

$12 \text{ boys} = 5 \text{ girls}$

Ratio of the efficiency of boy to girl is 5:12

Total work = (21 boys + 13 girls) * 1 day

$= 21(5) + 13(12)$

$= 261$

Three girls can finish the job in, $= 261 \text{ work} / 3 \text{ girls} = 261 / (3 \times 12) = 29/4$ days.

37) Answer D

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A'S ONE DAY WORK = $1/40$

B'S ONE DAY WORK = $1/60$

(A's + B's) one day work = $(1/40) + (1/60) = 1/24$

(A's + B's) 4 days work = $1/6$

Remaining work = $1 - 1/6 = 5/6$

38) Answer E

Let the percentage of part of car is = 100%

S can build that part of car in 4 days

So S can build PART OF CAR IN 1 DAY IS = $(100/4)\% = 25\%$

R can build the same part of car in 10 days

So R can build part of car in 1 day = $(100/10)\% = 10\%$

Together they can work in 1 day = $25\% + 10\% = 35\%$

Now increment of work percentage = $6 \times 100\% = 600\%$

So the total work will become = $600\% + 100\% = 700\%$

Total work can be done in = $700/35 = 20$ days.

39) Answer A

T takes 25 days to complete the work

U takes 50 days

V takes 100 days

Now 70% work is left part of work they can complete in 1 day = $(1/25 + 1/50 + 1/100) = 7/100$

Hence, the remaining work (70% of total work) is completed in = $(70/100) \times (100/7) = 10$ days.

40) Answer - D

(2 men + 1 women)'s 1 day work = $1/12$

(2*12men + 1*12women) can finish the work in 1 day

Also, (4 women + 2 men)'s one day work = $1/8$

(4*8women + 2*8) can finish the whole work in 1 day.

So, (24men + 12 women) = (32women + 16men)

=> 8 men = 20 women

Now, 8 men get = $8 \times 75 = \text{Rs } 600$ per day

20 women will get = Rs 600 per day

1 women will get = $\text{Rs } 600/20$

=> Rs 30 per day

The value of x = 30

(30*30 + 100)

1000

So the value is 1000

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41) Answer – B

Let, the efficiency of Men, Women, and Children be m , w , and c

$$(2m + 5w) * 8 = (6w + c) * 10$$

$$8m + 20w = 30w + 5c$$

$$8m = 10w + 5c \text{ -----(i)}$$

$$(6w + c) * 10 = (6m + 5c) * 5$$

$$12w + 2c = 6m + 5c$$

$$6m = 12w - 3c \text{ -----(ii)}$$

Multiplying equation (i) and equation (ii) with 3 and 4 respectively we have,

$$30w + 15c = 48w - 12c$$

$$18w = 27c$$

$$w / c = 3 / 2$$

Let, the efficiency of women and children be $3x$ and $2x$ respectively.

So, the efficiency of the men = $m = (30x + 10x) / 8 = 5x$ (from equation (i))

$$\text{Total work} = \{(2 * 5x) + (5 * 3x)\} * 8 = 200x$$

$$\text{Required time} = 100x / \{4 * (5x + 3x + 2x)\} = 100x / 40x = 2.5 \text{ days}$$

42) Answer – E

A, B, and C can complete a work in $(2x - 4)$ days, $(x+2)$ days, and $(x+5)$ days

$$\text{Efficiency of B} = 1/(x+2)$$

$$\text{Efficiency of C} = 1 / (x+5)$$

30% of the work is done by B and C in 2 days

$$100\% \text{ will be done in } 2 * 100 / 30 = 20/3 \text{ days}$$

$$\text{Efficiency of B and C together} = 3/20$$

$$1/(x+2) + 1/(x+5) = 3/20$$

$$(x + 5 + x + 2)/(x^2 + 7x + 10) = 3/20$$

$$(2x + 7) * 20 = 3x^2 + 21x + 30$$

$$3x^2 + 21x + 30 - 40x - 140 = 0$$

$$3x^2 - 19x - 110 = 0$$

$$3x^2 - 30x + 11x - 110 = 0$$

$$3x(x - 10) + 11(x - 10) = 0$$

$$(3x + 11)(x - 10) = 0$$

$$x = 10 \text{ or } -11/3 \text{ (x cannot be negative)}$$

$$\text{So, } x = 10$$

A, B, and C can complete a work in $(2 * 10 - 4) = 16$ days, $(10 + 2) = 12$ days and $(10 + 5) = 15$ days

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Let the total work be LCM of 16, 12, and 15 which is 240

Efficiency of A, B and C = 15, 20 and 16

The efficiency of D = y

$$35 * 2 + (51 + y) * 2 + 3.1y = 240$$

$$70 + 102 + 2y + 3.1y = 240$$

$$5.1y = 68$$

$$y = 40/3$$

Efficiency of D = 40/3

$$\text{Time taken by D} = 240 / (40/3) = 18 \text{ days} = (2x - 2) \text{ days}$$

43) Answer - E

From the man-day formula:

$$N_1 M_1 D_1 T_1 / W_1 = N_2 M_2 D_2 T_2 / W_2$$

$$(5 * M * 2 * 4) / 5\% = ((5 + X) * M * 12 * 8) / 60\%$$

$$X = 0$$

So, no extra men are required to complete the job in given time.

44) Answer - A

A can type 72 pages in 24 hours. So, the efficiency of A = $72 / 24 = 3$ pages/hr

In 12 hours A can type = $12 * 3 = 36$ pages. So, B types = $96 - 36 = 60$ pages in 12 hours. So, the efficiency of B = $60 / 12 = 5$ pages/hr.

If B left before 2 hours of completion, the number of pages A has typed in 2 hours = $2 * 3 = 6$ pages

So, they together typed = $166 - 6 = 160$ pages

Time taken by them to type 160 pages together = $160 / (5 + 3) = 160 / 8 = 20$ hours

So, B alone has typed = $20 * 5 = 100$ pages and A alone has typed = $22 * 3 = 66$ pages

The ratio of their wages = $66:100 = 33:50$

The difference between their wages = $(50 - 33) / (50 + 33) * 2490 = 17 / 83 * 2490 = \text{Rs. } 510$

45) Answer - D

Total units of work = 264

(Sunil + Aman + Ankit)'s one hour's work = 66 units

Sunil's one hour's work = 22 units

Aman = 12 units

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Ankit's one hour's work = $66 - 34 = 32$ units

Ankit's efficiency to do the work alone = $3 \times 32/4$

Time reqd. to complete the work = $264 \times 4 / (32 \times 3)$
= 11 hours

46) Answer B

Let total amount of work be 360 units (LCM of 24, 30 and 36)

Efficiency of A = $360/24 = 15$ units per day

Efficiency of B = $360/30 = 12$ units per day

Efficiency of C = $360/36 = 10$ units per day

Amount of money received by A and C together = $25/37 \times 1184 = \text{Rs. } 800$

Amount received by B = $1184 - 800 = \text{Rs. } 384$

Desired difference = $800 - 384 = \text{Rs. } 416$

47) Answer D

Let total amount of work = 168 units (LCM of 21, 24 and 14)

Amount of work completed by A, B and C together in one day = $168/14 = 12$ units

Amount of work completed by A and C together in one day = $168/21 = 8$ units

Amount of work completed by B and C together in one day = $168/24 = 7$ units

Efficiency of B = $12 - 8 = 4$ units per day

Efficiency of C = $7 - 4 = 3$ units per day

Desired time = $1/4 \times 168/3 = 14$ days

48) Answer C

Time taken by A and B together to complete the whole work = $54/0.60 = 90$ days

Time taken by B alone to complete the whole work = $1.60 \times 90 = 144$ days

Total work = 720 units (LCM of 144 and 90)

Amount of work done by A and B together in one day = $720/90 = 8$ units

Amount of work done by B alone in one day = $720/144 = 5$ units

Amount of work done by A alone in one day = $8 - 5 = 3$ units

Desired time = $(0.40 \times 720)/3 = 96$ days

49) Answer C

According to the question,

$25 \times 24 = 16x + (16 + 6) \times 20$

Or, $16x = 600 - 440$

Or, $x = 160/16 = 10$

50) Answer 0

$1/24 + 1/32 + 1/C = 3/32$

$1/C = (3/32) - (1/24 + 1/32)$

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$$(1/C) = (3/32) - (7/96) = 1/48$$

C can do it in 48 days.

$$\text{Efficiency of A, B and C} = (1/24) : (1/32) : (1/48) \\ = 4:3:2$$

$$9's = 4500$$

$$1's = 500$$

$$\text{The share of C} = \text{Rs } 1000$$

51) Answer D

$$\text{Total work} = \text{men} \times \text{days}$$

$$\text{Total units of work} = 25 \times 32 = 800 \text{ units}$$

$$\text{Work done in 8 days} = 25 \times 8 = 200 \text{ units}$$

$$\text{Remaining work} = 800 - 200 = 600 \text{ units}$$

Let the number of men left after 8 days be x,

According to the question,

$$600/30 = 25 - x$$

$$\text{Or, } x = 5$$

After 8 days from the start of the work, 5 men left the job.

52) Answer B

$$\text{Time required by Q and R to complete the whole work} \\ = 5 \times 28 / 2 = 70 \text{ days}$$

$$\text{Time taken by P, Q and R} = 40 \text{ days}$$

$$\text{Total units of work} = 280 \text{ units}$$

$$\text{Q and R one day work} = 4 \text{ units}$$

$$\text{P, Q and R one day work} = 7 \text{ units}$$

$$\text{P's one day work} = 7 - 4 = 3 \text{ units}$$

$$\text{Units of work done by P} = 3 \times 280 / 5 = 168 \text{ units}$$

$$\text{Required value of } x = 168 / 3 = 56 \text{ days}$$

53) Answer D

$$\text{Time taken by P} = 16 \times 3 / 2 = 24 \text{ days}$$

$$Q = 7 \times 6 = 42 \text{ days}$$

$$R = 16 \times 5 / 4 = 20 \text{ days}$$

$$\text{Total units of work} = 840 \text{ units}$$

$$\text{P's one day work} = 35 \text{ units}$$

$$\text{Q's one day work} = 20 \text{ units}$$

$$\text{R's one day work} = 42 \text{ units}$$

$$\text{Work done in 6 days} = 97 \times 6 = 582 \text{ units}$$

$$\text{Time required by P to complete the remaining work}$$

$$\Rightarrow (840 - 582) / 35 = 258 / 35 \text{ days.}$$

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54) Answer C

Total units of work = $10 \times 16 = 160$ units

Work done in 5 days = $10 \times 5 = 50$ units

Remaining work = $160 - 50 = 110$ units

Let the number of men left after 5 days be x

So, $10 - x$ men completed the remaining 110 units in $55/3$ days

So,

$$(10 - x) \times 55/3 = 110$$

$$550 - 55x = 330$$

$$55x = 220$$

$$x = 4 \text{ men.}$$

55) Answer A

$$(1/30) + (1/24) + 1 \text{ boy} = (1/10)$$

$$1 \text{ boy} = (12 - 4 - 5)/120 = 1/40$$

$$\text{The share of the men and a boy} = 1/30 : 1/24 : 1/40$$

$$\Rightarrow 4 : 5 : 3$$

$$12 \text{ men} = 6000$$

$$1 \text{ boy} = 500$$

$$\text{The share of boy} = \text{RS. } 1500$$

56) Answer D

$$\text{The ratio of Efficiency of Kartik and Raju} = 140 : 100 = 7 : 5$$

$$\text{Then work ratio of Kartik and Raju} = 5 : 7$$

$$\text{Now given that, Kartik's one day's work} = 1/20$$

$$\text{Then Raju's one day's work} = 1/28$$

$$\text{Thus in two days, the work done by them} = 1/20 + 1/28 = 12/140$$

$$\text{Then work done by them in 22 days} = (12 \times 11)/140 = 132/140$$

$$\text{Work done by them in 23 days} = (132/140) + (1/40) = 139/140$$

$$\text{Remaining work} = 1 - (139/140) = 1/140$$

$$\text{Remaining work done by Raju in} = (1/140) \times 28 = (1/5) \text{ days}$$

$$\text{Thus total days to complete the work} = 23 + (1/5) = 23 \frac{1}{5} \text{ days}$$

57) Answer - B

$$70\% \text{ of work completed by D in 7 days}$$

$$100\% \text{ of work completed by D in 10 days}$$

Let

$$\text{Time taken to B and C together to complete the work} = (1/3x) + (1/6x) = 3/6x =$$

$$1/2x$$

$$1/2x = 1/10$$

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$$1/x = 1/5$$

$$x = 5$$

A can complete the whole work in 5 days, B can complete the whole work in 15 days and C can complete the whole work in 30 days

Then

$$(1/5) + (1/15) + (1/30) + (1/10) = (6 + 2 + 1 + 3)/30 = 12/30 = 1/2.5$$

Time taken by A, B, C, and D together to complete the work is 2.5 days.

58) Answer B

A=x days, B= x+10 days

$$A+B= 1/x+1/(x+10) = 1/(75/8)$$

$$1/x+ 1/(x+10) = 8/75$$

$$(x+x+10)/(x^2+10x) = 8/75$$

$$(2x+10)*75 = 8(x^2+10x)$$

$$75x+375=4x^2+40x$$

$$4x^2-35x-375 = 0$$

Simplify the above equation, we get $x = 60/4, -25/4 = 15, -25/4$ (Eliminate the -ve value)

X=15 days

A= 15 and B= 25 days

Efficiency ratio of A and B = $1/15: 1/25 = 5:3$

59) Answer - B

Let's assume Men, Women and children as M, W and C respectively

The ratio of time taken by a man, woman and a child to complete the work in 2: 5: 8

So, M: W: C = 2: 5: 8

Total work = LCM of 2, 5 and 8 = 40

Efficiency of Man = $40/2 = 20$ units

Efficiency of woman = $40/5 = 8$ units

Efficiency of child = $40/8 = 5$ units

Let

Work done by 12 men = $20 * 12 = 240$ units

Work done by 15 women = $8 * 15 = 120$ units

Work done by 20 children = $5 * 20 = 100$ units

Then,

Wage ratio of M: W: C = $240: 120: 100 = 12: 6: 5$

Wages of 12 men = $(10580/23) * 12 = \text{Rs.}5520$

Wages of 1 man = $5520/12 = \text{Rs.}460$

Wages of 15 women = $(10580/23) * 6 = \text{Rs.}2760$

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Wages of 1 woman = $2760/15 = \text{Rs.}184$

Wages of 20 children = $(10580/23) * 5 = \text{Rs.}2300$

Wages of 1 child = $2300/20 = \text{Rs.}115$

So

x number of men get a wage of Rs.6900

WKT, 1 men wage is Rs.460

Then

$x = 6900/460 = 15$

Let

Total wages of 'x' = 15 men, 'x - 3' = 12 women and 'x - 5' = 10 children

$= (15 * 460) + [12 * 184] + [10 * 115]$

$= 6900 + 2208 + 1150$

$= \text{Rs.}10258$

60) Answer C

Total share of A, B and C = 1800

Share of C = $1800 - 1400 = 400$

Share of B = $1800 - 400 - 800 = 600$

Efficiency ratio of A, B and C = $800 : 600 : 400 = 4 : 3 : 2$

61) Answer D

Manu and Ragav's one day work = $1/30 + 1/36 = 11/180$

Manu and Ragav 12 day work = $(11/180) * 12 = 11/15$

Remanning work $4/15$ done by Manu and Ranjni.

Manu and Ranjni finished it in $8/3$ days

$(4/15) * (\text{Manu} + \text{Ranjni})$'s whole work = $(8/3) * (15/4) = 10$ days

Ranjni's one day work = $(1/10) - (1/30) = 1/15$

Ranjni alone can complete the work in 15 days .

62) Answer A

Let initially the no of men be x,

A piece of work has to be completed in 30 days

According to the question,

Men	Days	Work
X	20	$1/2$
(x+30)	10	$1/2$

Work = Men * DAYS

$20X / (1/2) = (X + 30) * 10 / (1/2)$

X = 30 Men

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63) Answer C

The ratio of efficiency of aman and neha = 6:5

The ratio of number of days taken by aman and neha = 5:6

The ratio of number of days taken by pranav and neha = 3:2

Ratio of number of days taken by aman: neha: pranav= 5:6:9

According to the question,

Neha-aman=8 days

$6s - 5s = 3$

$1s = 3$

Number of days taken to finish the whole work,

Aman= 15 days. Neha = 18 days, pranav = 27 days

Work done by pranav and neha in one days, $(1/18) + (1/27) = 5/54$

Pranav and neha's 3 days work = $5/18$

Rest of the work $13/18$

The number of days taken by aman to finish the remaning work is,
number of days= $(13/18) * 15 = 10(5/6)$

64) Answer B

LCM of 12,6,18 = 36

A= 3 units

B = 6 units

C = 2 units

B+C=8 units

A+C = 5 units

A+B=9 units

1 cycle(3 days) = $8+5+9=22$ units

Remaning $36-22=14$ then bc completed the work by 8 unit

then remaning the $14-8= 6$ units then remaning the work done by AC is 5 units
, then remaning the work $6-5=1$ done by AB IS $1/9$ DAYS

So, whole work completed by $(5+(1/9))=46/9$ days

65) Answer D

The ratio of efficiency Anu and chintu = 5:3

The ratio of number of days taken by anu and chintu = 3:5 the ratio of number of days taken by chintu and banu = 2:3 ratio of number of days taken by anu: chintu: banu= 6;10:15

According to the question, chintu-anu = 16 days





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$$10's - 6's = 16$$

$$4's = 16$$

$$1's = 4$$

Number of days taken to finish the whole work, anu = 24 days, chintu = 40 days, banu = 60 days

Work done by banu and chintu in two days, $(4/40) + (4/60) = 25/(10 \times 15) = 1/6$

Rest of work = $5/6$

The number of days taken by anu to finish the remaining work is,

Number of days = $(5/6) \times 24 = 20$ days

66) Answer A

A's 1 day work = $1/20$

B's 1 day work = $1/30$

(A+B)'s 1 day work = $(1/20 + 1/30) = 50/60$

Remaining work = $1 - (50/60) = 10/60$ work

$(10/60) \times C$'s whole work = 4

C's whole work = 24

C's one day work = $1/24$ days

A:B:C = $1/20 : 1/30 : 1/24 = 6:4:5$

$$15'S = 6000$$

$$1'S = 400$$

The daily wages of a, b and c is Rs 2400, Rs 1600, and Rs 2000 respectively.

67) Answer B

A's one day output = $1/40$

B's one day output = $1/50$

C's one day = $1/60$

Given $5/50 + 2/40 + 1/60 = (60 + 30 + 10)/600 = 1/6$

Remaining = $1 - (1/6) = 5/6$

$(5/6) \times D$'s whole work = 10

D's whole work = 12 days

68) Answer C

A, B, C together can do a piece of work in $2(2/3)$ days

(a+b+c)'s one day work = $3/8$

A and B TOGETHER CAN do the same work in $3(3/7)$ days

(A+B)'s one day work = $7/24$

C's one day work = $(3/8) - (7/24) = 1/12$

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B takes 2 days more than A

$B = A + 2$ (Let us assume $A = x, B = x + 2$)

$= 1/x + 1/(x+2) = 7/24$

$X = 6$

A's days to complete the work = 6, B's = 8, C's = 12

total work = 24 (eff A=4, B= 3, C=2)

in three days = 23 in three days

on fourth day A and B work then

Required number of days $3(1/7)$

69) Answer E

LCM of 24, 36 and 72 = 72

Shamli's per day work = $1/24 = 3$ units

Sahiba's per day work = $1/36 = 2$ units

Ahkil's per day work = $1/72 = 1$ unit

1st day = 2 unit

2nd day = 5 unit

3rd day = 6 unit

At the end of 3rd day 13 units completed out of 72

Required no of days = 17

70) Answer C

A's one day work with 100% efficiency = $1/30$

C's one day work = $1/30$

B and C's one day work = $1/12$

B's one one day work = $(1/12) - (1/30) = 1/20$

Required no of days = 16 days

71) Answer E

Given,

$2(A+B+C) = 1/4 + 7/24 + 5/24 = 3/4$

$A+B+C = 3/8$

A's one day work $(3/8 - 5/24) = 1/6$, B's work $(3/8 - 7/24) = 1/12$, C's work $(3/8 - 7/24) = 1/8$

Every two days $2*(1/6 + 1/12) = 1/2$; remaning work $1 - (1/2) = 1/2$

them they work for one day all together = $3/8$

$1/2 + 3/8 = 7/8$

$1/8 + 1/4 = 1/2$

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then the total time to complete the work = 3.5 days

72) Answer D

$$A+B = 8 \text{ DAYS}$$

$$B+C=6 \text{ days}$$

$$\text{So, } (1/8)*2+(1/6)+(x/12)=1$$

$$X=7$$

$$\text{the total time for C work} = x+1=8 \text{ days}$$

73) Answer - C

Since a man is 20% more efficient than a woman and a woman is 25% more efficient than a child, then

Let the work done by a man in a day be $6x$ units

Let the work done by a woman in a day be $5x$ units

Let the work done by a child in a day be $4x$ units

Work done by men = work done by women = work done by children

$$6x * m * 10 = 5x * 18 * n = 4x * p * 18$$

$$60m = 90n = 72p$$

$$20m = 30n = 24p$$

$$10m = 15n = 12p$$

$$\text{Now, } 10m = 15n$$

$$m: n = 3:2$$

$$\text{And } 15n = 12p$$

$$n: p = 4:5$$

$$\text{So, } m: n: p = 6:4:5$$

Now, p is 2 more than n

$$\text{So Let } p = 5y \text{ and } n = 4y$$

$$5y - 4y = 2$$

$$y = 2$$

$$\text{So, } m = 12, n = 8 \text{ and } p = 10$$

$2n$ women and p children are now working together.

$$\text{work done in a single day} = (5x * 16) + (4x * 10) = 120x \text{ units}$$

$$\text{Total work to done} = 6x * m * 10 = 720x \text{ units}$$

$$\text{Time taken} = 720x/120x = 6 \text{ days}$$

74) Answer - B

Let, the efficiency of a man be M and the efficiency of a woman be W

$$M: W = 100: 135 = 20: 27$$

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According to the question,

$$(X + 5) * 20 * 9 = 3X * 27 * 5$$

$$4(X + 5) = 9X$$

$$4X + 20 = 9X$$

$$5X = 20$$

$$X = 4$$

The efficiency of a man is increased by 20% and the efficiency of a woman remains the same

$$(5Y + 6) * (20 * 120/100) * 3 = (Y + 4) * 27 * 8$$

$$(5Y + 6) * 72 = (Y + 4) * 27 * 8$$

$$(5Y + 6) = 3(Y + 4)$$

$$5Y - 3Y = 12 - 6$$

$$2Y = 6$$

$$Y = 3$$

$$\text{The required value} = (3X + 2Y) = (3*4 + 2*3) = 12 + 6 = 18$$

75) Answer C

$$\text{Total work (LCM of 70,84,48)}=1680$$

$$\text{Amount of work done by A,B,C} = 24, 20, \text{AND } 35 \text{ units}$$

$$\text{Amount of work done by A,B,C together in 'x' days} = (24+20+35)*x=79x \text{ units}$$

$$\text{Amount of work done by B alone in one day after increased efficiency} = 20*1.25=25 \text{ units}$$

$$\text{Amount of work done by B alone in } \{(228/5)-x\} \text{ days}$$

$$\text{With increased efficiency} = \{(228/5)-x\}*25$$

$$X=10$$

76) Answer A

$$(3*7*4)/168=(7*7*X)/(364-168)$$

$$X=2 \text{ DAYS}$$

77) Answer B

$$(X-5) \text{ PERSON can do whole work in } x \text{ days}$$

$$(x+5) \text{ person can do } (75/100)*\text{work in } =(x-11)$$

$$\text{Whole work} = (x-11)*(4/3)$$

$$\text{Solve this } x=-11, 20 \text{ (neglected negative number)}$$

$$\text{So, } x = 20$$

Person	days
--------	------

15	20
----	----

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$$30 \quad ?$$
$$(15 \times 20) = 30 \times y$$
$$Y = 10 \text{ days}$$

78) Answer - E

A alone can complete the whole work = x days

B alone can complete the whole work = y days

C alone can complete the whole work = 30 days

D alone can complete the whole work = 36 days

Total work = 360 units

The efficiency of A = 3a units/day

The efficiency of B = 2a units/day

The efficiency of C = 12 units/day

The efficiency of D = 10 units/day

$$(3a + 2a + 12 + 10) \times 3 + (3a + 2a + 10) \times 2 + 2a \times \frac{6}{5} = 360$$

$$(5a + 22) \times 3 + (5a + 10) \times 2 + 12a/5 = 360$$

$$15a + 66 + 10a + 20 + 12a/5 = 360$$

$$25a + 12a/5 = 274$$

$$125a + 12a = 274 \times 5$$

$$137a = 1370$$

$$a = 10$$

The efficiency of A = 30 units/day

The efficiency of B = 20 units/day

A alone can complete the whole work = $360/30 = 12$ days

B alone can complete the whole work = $360/20 = 18$ days

$$\text{Required value} = 2 \times 12 - 18 = 24 - 18 = 6$$

79) Answer - E

Efficiency of A = 40% of Efficiency of B

The ratio of efficiency of A and B = 40: 100 = 2: 5

Efficiency of C = 180% of Efficiency of B

The ratio of efficiency of B and C = 100: 180 = 5: 9

The ratio of efficiency of A, B, and C = 2: 5: 9

The total work = $45 \times 2 = 90$ units

A alone worked for d days, (A + B) worked for (d + 1) days, and (A + B + C) worked for (d + 3) days.

$$2d + (2 + 5)(d + 1) + (2 + 5 + 9)(d + 3) = 90$$

$$2d + 7(d + 1) + 16(d + 3) = 90$$

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$$25d = 90 - 7 - 48$$

$$25d = 35$$

$$d = 35/25 = 1.4$$

$$\text{Number of days B worked} = (d + 1) + (d + 3) = (2d + 4)$$

$$= 2.8 + 4 = 6.8 \text{ days}$$

80) Answer - E

Let, the efficiency of a man be M, the efficiency of a woman be W and the efficiency of a boy be B

$$M: W = 100: 135 = 20: 27$$

$$M: B = 100: 80 = 5: 4 = 20: 16$$

$$M: W: B = 20: 27: 16$$

According to the question,

$$(2X + 7) * 20 * 3 = (2X + 3) * 27 * 4$$

$$(2X + 7) * 5 = (2X + 3) * 9$$

$$5(2X + 7) = 9(2X + 3)$$

$$18X - 10X = 35 - 27$$

$$X = 1$$

If the efficiency of a man is decreased by 20%,

$$(5Y + 9) * (20 * 80/100) * 2 = (Y + 4) * 16 * 9$$

$$(5Y + 9) * 16 * 2 = (Y + 4) * 16 * 9$$

$$2(5Y + 9) = 9(Y + 4)$$

$$10Y - 9Y = 36 - 18$$

$$Y = 18$$

$$Y - 14X = 18 - 14 = 4$$

The time required to complete the work B by 1 man, 4 women, and 4 boys

$$= [(18 + 4) * 16 * 9] / [(20 + (27 * 4) + (4 * 16))] \text{ days}$$

$$= 3168/192$$

$$= 16.5 \text{ days}$$

81) Answer - E

Let us assume, the efficiency of a man is M, the efficiency of a woman is W and the efficiency of a boy is B

$$M: W = 100: 125 = 4: 5 = 12: 15$$

$$M: B = 150: 100 = 3: 2 = 12: 8$$

$$M: W: B = 12: 15: 8$$

According to the question,

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$$[(X + 7) * 12 + 8X] * 3 + [(X + 2) * 12 + 8(X + 1)] * 4 = [(2X + 4) * 15] * 2 + [(2X + 1) * 15] * 4$$

$$[12X + 84 + 8X] * 3 + [12X + 24 + 8X + 8] * 4 = (2X + 4) * 30 + (2X + 1) * 60$$

$$[20X + 84] * 3 + [20X + 32] * 4 = 180X + 180$$

$$140X + 380 = 180X + 180$$

$$40X = 200$$

$$X = 5$$

$$Y = 2X = (2 * 5) = 10$$

$$\text{The total work} = [(140 * 5) + 380] \text{ unit} = 1080 \text{ unit}$$

The time required to complete the same work by $(X + 4)$ men, 4 women, and $(Y - 1)$ boys

$$= 1080 / [(9 * 12) + (4 * 15) + (9 * 8)]$$

$$= 1080 / 240$$

$$= 4.5 \text{ days}$$

82) Answer - E

Let the total work 360 units (LCM of 40, 45 and 72)

Amount of work done by A alone in one day = $360/40 = 9$ units

Amount of work done by B alone in one day $360/45 = 8$ units

Amount of work done by C alone in one day $360/72 = 5$ units

Amount of work done by A and B together in one day = $9 + 8 = 17$ units

For option A

Amount of work done by A and B together in 15 * days = $15 * 17 = 255$ units

Amount of work done by C in 21 days = $21 * 5 = 105$ units

Total work done = $255 + 105 = 360$ units

So, option A can be the answer.

For option B:

Amount of work done by A and B together in 19 days = $19 * 17 = 323$ units

Amount of work done by C in 7.4 days = $7.4 * 5 = 37$ units

Total work done = $323 + 37 = 360$ units

So, option B can be the answer.

For option

Amount of work done by A and B together in 12 days = $12 * 17 = 204$ units

Amount of work done by C in 2 days = $2 * 5 = 10$ units

Total work done = $204 + 156 = 360$ units

So, option C can be the answer.

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()

A is 37.5% less efficient than B.

So, the ratio of efficiency between A and B is 5:8.

So, if A takes $8x$ days, then B takes $5x$ days.

D takes $5x - 6$ days.

C takes $5x - 6 - 8 = 5x - 14$

$5x - 6/8x = 1/2$

$10x - 12 = 8x$

$2x = 12/2 = 6$

So, A takes 48 days, and B takes $5 \times 6 = 30$ days.

C takes $30 - 14 = 16$ days D takes $30 - 6 = 24$ days.

E takes $48 \times 5 / 6 = 40$ days.

So, total work - LCM of 48, 30, 16, 24 and 40 = 240

So, the efficiency of A, B, C, D, and E is 5, 8, 15, 10, and 6, respectively.

So, the value of (i) = $240/20 = 12$.

Value of (ii) = $15:10 = 3:2$

Value of (iii) = $[240/(15 + (10/2))] = 12$

Value of (iv)

D and E complete in 2 days = $10 + 6 = 16$ units.

So, in 30 days, they complete $16 \times 15 = 240$ units.

So, total work completed in = 30 days.

83) Answer - C

Req. difference = $30 - 12 = 18$

84) Answer - B

Efficiency of F = $6 \times 3/2 = 9$

Req. days = $480/9 = 53.33$ days

85) Answer - C

Statement I

Let time taken by A' is x days

So, time taken by B' = $x + 9$ days

$1/x + 1/(x + 9) = 1/20$

$20(2x + 9) = x(x + 9)$

$40x + 180 = x^2 + 9x$

$x^2 - 31x - 180 = 0$

$x^2 - 36x + 5x - 180 = 0$

$= (x + 5)(x - 36) = 0$

$x = 36$

Time taken by B' to complete the whole work = $36 \times 9 = 45$

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Ratio of efficiency of A to B ratio of profit share of A to B = $45.36 = 5.4$

So profit share of B' = $4/9 * 1080 = \text{Rs } 480$

So, data in statement I alone is sufficient to answer the question.

Statement II.

Let total amount of work 180 units (LCM of 36 and 20) Efficiency of A and B together = $180/20 = 9$ units per day

Efficiency of A = $180/36 = 5$ units per day

Efficiency of B = $9 - 5 = 4$ units per day

So share of = B = $4/9 * 1080 = \text{Rs } 480$

So data in statement II alone is sufficient to answer the question.

So, data either in statement I alone or in statement II alone are sufficient to answer the question.

86) Answer - C

Statement I:

Let efficiency of C be $4x$ units per day

Efficiency of 'A' = $0.75 * 4x = 3x$ units per day

Efficiency of 'B' = $4x / 2 = 2x$ units per day

Total work $(4x + 3x + 2x) * 35 = 315x$

Desired time = $(315x / 3 * x) = 105$ days

So, data in statement I alone is sufficient to answer the question.

Statement II:

Let total amount of work = 315 units (LCM of 63, 35 and 45)

Efficiency of (A + B) = $315/63 = 5$ units per day

Efficiency of (AC) = $315/45 = 7$ units per day

Efficiency of (ABC) = $315/35 = 9$ units per day

Efficiency of 'C' = $9 - 5 = 4$ units per day

Efficiency of 'A' = $7 - 4 = 3$ units per day

Desired time * $315/3 = 105$ days

So, data in statement II alone is sufficient to answer the question.

87) Answer - D

Statement I

Time taken by Gourav to complete the work = $0.6 * 20 = 12$ days

So, data in statement I alone is not sufficient to answer the question.

Statement II:

When Gourav works with 50% more efficiency and Nikhil works with thrice efficiency then the work gets completed in 4 days.

So, data in statement II alone is not sufficient to answer the question.

Statement III:

Time taken by Nikhil to complete the work = $20 + 4 = 24$ days

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So, data in statement III alone is sufficient to answer the question.

Combining statement I and II:

Let Nikhil takes 'x' days to complete the work

Let the total work = $12x$

Efficiency of Gourav = $12x/12 = x$ units/day

Efficiency of Nikhil = $12x/x = 12$ units/day

According to the question,

$$\{1.5x + 3 * 12\} * 4 = 12x$$

$$1.5x + 36 = 3x$$

$$x = 36/1.5 = 24 \text{ days}$$

Therefore, data either in statements I and II together or data in statement III alone is sufficient to answer the question.

88) Answer - D

20 men and 12 children can complete the work in 15 days.

So, data in statement I alone is not sufficient to answer the question

Statement II:

Ratio of the efficiencies of a woman and 3 children is 2:3 respectively

So, data in statement II alone is not sufficient to answer the question

Statement III

Ratio of the efficiencies of 10 women and 4 men is 5:3 respectively

So, data in statement III alone is not sufficient to answer the question.

Combining statements I, II and III:

Let efficiency of a woman = $2x$ units/day

Efficiency of 3 children = $3x$ units/day

Efficiency of a child - x units/day

Again, Efficiency of 4 men $2x * 10 * 3 / 5 = 12x$ units/day

Efficiency of each man = $12x / 4 = 3x$ units/day

Total work $(20 * 3x * 12x) * 15 = 1080x$ unit

Required time for 10 men and 6 women to complete the work

$$= 1080x / (30x + 12x)$$

$$= 25.71 \text{ days}$$

89) Answer - C

Efficiency of A = $[1 / (x + 16)]$ units/day

Efficiency of B' = $(1 / x)$ units/day

Statement I

Number of days taken by C alone to complete the work $k = x + 16 - 10 - (x * 6)$ days

So efficiency of C alone - $[1 / (x - 6)]$ units/day

Since B and C can together complete 75% of the work in 10 days, so combined efficiency of B and C' = $(1 * 0.75) / (10) = (3/40)$ units/day

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So.. $(1/x)(1/(x+6)) = ((x-6) * x) / (x^2 * 6x) - (3/40)$ Or $80x + 240 - 3x^2 - 18x$ Or $0.3x^2 - 62x - 240 = 0$ Or $3x^2 - 72x + 10x - 240 + 0$ Or $3x(x-24) * 10(x-24) + 0$ Or $(3x-10)(x-24) + 0$ So. $x - (-10/3) * \text{or} * 24$

Since value of x cannot be negative. So. $x = 24$.

Therefore data in statement I alone is sufficient to answer the question.

Statement II

Increased efficiency of 'A' = $\{1.25 / (x-16)\}$ units/day

Decreased efficiency of B' = $(0.75 / x)$ units/day

Time taken by A and B together to complete the work $k = (x-8)$ days

So combined efficiency of A and B = $\{1 / (x-8)\}$ units/day So. $\{1.25 / (x+16)\} + (0.75 / x) - \{1 / (x-8)\}$ Or. $(1.25x + 0.75x + 12) / (x^2 + 16x) = (1 / (x-8))$ Or $2x^2 + 12x - 16x - 96 - (x^2 - 16x)$ Or $x^2 - 20x - 96 + 0$ Or $x^2 + 4x - 96 - 0$ Or. $x(x-4) - 24(x-4) = 0$ Or. $(x-24)(x-4) + 0$

So $x = 24$ or -4

Since number of days cannot be negative, so $x = 24$

Therefore data in statement II alone is sufficient to answer the question.

The data either in statement I alone or in statement II alone are sufficient to answer the question.

90) Answer - A

Let the efficiency of each man, woman and child be M' units/day. W units/ day and C units/ day respectively

Statement I

Let the total work be 30 units. (LCM (6 and 5)) $4M + 6W + 7C = (30/6) = 5$(I) Also, $3M + 7W + 12C = (30/5) = 6$(II)

On subtracting $(3 \times \text{equation (I)})$ from $(4 \times \text{equation (II)})$, we have

$10W + 27C = 9$ So, Required time = $(30 / (2 * 9)) \text{ sim } 1.67$ days

So, data in statement I alone is sufficient to answer the question.

Statement II:

Let total work be $6d$ units.. So. $M = WC$ So, $4M + 6W + 7C = (6 * d / 6) = d$ Or $0.4(WC) * 6W + 7C = d$ Or $4W + 4C + 6W + 7C = d$ Or $10W + 11C = d$ So, data in statement II alone is not sufficient to answer the question.

Therefore, the data in statement I alone are sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question.

91) Answer - E

Statement I

We don't know the efficiency of Nitesh with respect to Sanjay or Anurag.

So statement I alone is not sufficient to answer the question.

Statement II

We don't know the efficiency of Nitesh with respect to Sanjay or Anurag.

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So, statement II alone is not sufficient to answer the question

Statement III

We don't know the efficiency of Nitesh with respect to Sanjay or Anurag.

So statement III alone is not sufficient to answer the question.

Combining statement I and statement II

Let the total work be LCM of (9 and 6) 18 units

Number of units of work done by Sanjay alone in one day - $18/9 = 2$ units

Number of units of work done by Sanjay and Anurag together in one day $18/6 = 3$ units

So, number of units of work done by Anurag alone in one day - $3 - 2 = 1$ unit

Since the ratio of the efficiency of Sanjay: Anurag - 2:1

Total wage received by all of them for the completion of work - Rs. 1500

Wage received by Nitesh - Rs 500

So the wage received by Sanjay - $(2/3) \times (1500 - 500) = \text{Rs } 1000/3$

Wage received by Anurag - Rs. $1000/3$

So, the ratio of efficiency of Sanjay Nitesh. 4:3

So, the ratio of the time taken by Sanjay Nitesh - 3:4

So time taken by Nitesh alone to complete the work $(4/3) \times 9 = 12$ days

So, statement I and statement II together are sufficient to answer the question.

Combining statement I and statement III:

Time taken by Sanjay alone to complete the work. 9 days

Time taken by Sanjay and Anurag together to complete the work - 6 days

Time taken by Sanjay Nitesh and Anurag together to complete the work - 4 days

So, time taken by Nitesh alone to complete the work = $1 / (1/9 + 1/6) = 12$ days

So, statement I and statement III together are sufficient to answer the question

Combining statement II and statement III:

Total work LCM of 9 and 4 = 36 units

Total units of work done by all three together $36/4 = 9$ units

Wage of Nitesh Rs 500, which is $(1/3)$ rd of total wage.

So. Nitesh will do 3 units of work in a day

Required time taken = $36/3 = 12$ days

92) Answer - B

Time taken by 'A' to complete the whole work $225/8$ days

Let total amount of work = 450 units (LCM of 225 and 18)

So, efficiency of A = $450 / (225/8) = 16$ units per day

Efficiency of A' + B' = $450/16 = 28.125$ units per day

Efficiency of B' = $28.125 - 16 = 12.125$ units per day

Amount of work done by 'B' in 5 days $12.125 \times 5 = 60.625$ units

Amount of work done by 'A' and 'B' together in 6 days = $6 \times 28.125 = 168.75$ units

Amount of work done by 'B' and 'C' together in 15 days = $450 - 168.75 = 281.25$ units

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So, efficiency of B and C together = $255/15 = 17$ units per day

Efficiency of 'C' = $17 - 9 = 8$ units per day

For option (a):

Amount of work done by 'A' in 12 days = $12 \times 16 = 192$ units

Amount of work done by 'B' in 16 days = $16 \times 9 = 144$ units

Amount of work done by 'C' in $14\frac{3}{4}$ days = $59\frac{3}{4} \times 8 = 478$ units

Total work done by A, B and 'C' together = $192 + 144 + 478 = 794$ units

So, option (a) cannot be the answer.

For option (b):

Efficiency of 'C' = $25 \times 8 = 200$ units per day

Time taken by 'C' to complete the whole work = $450/20 = 22.5$ days

So, option (b) cannot be the answer.

For option (c):

Desired time = $(0.64 \times 450)/(168) = 1.71$ days

93) Answer - A

Quantity I:

Capacity ratio of Sai to Ram = 3:2

Portion of work completed = $1 - (7/20) = (13/20)$

Work done in 20 days = $5 \times 20 = 100$

Work done in full 20th portion = $100 \times (20/13) = 2000/13$

Time taken by Ram = $2000/(13 \times 2) = 1000/13$

Quantity II:

One day work of L = One day work of (M+N)

(L+M)'s 1 day work = $1/20$

N's 1 day work = $1/30$

So, $L+M+N = (1/20) + (1/30) = 1/12$

$2M+2N = 1/12$

$2M+2 \times 1/30 = 1/12$

$M = 120$

94) Answer - B

Quantity I:

For men total work = $15 \times 20 = 300$

For 5 days = $5 \times 20 = 100$

In 5 days $1/3$ of the work is completed so remaining work is done by 20 women in 20 days

$1/3$ work = 20 days

Whole work by 20 women = 30 days

we need for 30 women so days

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$$(20 \times 30) = (30 \times x)$$

$$X = 20 \text{ days}$$

Quantity II:

$$20 \text{ men complete the whole work in 20 days} = 20 \times 20 = 400$$

$$\text{So 10 men in 10 days} = 100$$

$$\text{Work completed} = 100/400 = 1/4^{\text{th}}$$

$$20 \text{ boys can complete the } 3/4^{\text{th}} \text{ of the job in } (x+10) \text{ days}$$

$$300 = 20 \times (x+10)$$

Solving this we get

$$X = 5 \text{ days}$$

95) Answer - A

Quantity I:

$$12 \text{ women can complete the work in 5 days}$$

$$1 \text{ women 1 day work} = 1/(5 \times 12) = 1/60$$

$$\text{So 3 women in 10 days} = (3 \times 10)/60 = 1/2$$

$$\text{So 9 children complete } 1/2 \text{ of the job in 10 days}$$

For whole job they take 20 days

In question they have asked the number of days required for 36 children

$$\text{So time} = 5 \text{ days}$$

Quantity II:

$$\text{Boys one day work} = 30/(15 \times 20) = 1/10$$

$$\text{Girls one day work} = 30/(30 \times 20) = 1/20$$

$$\text{So in two days} = 1/10 + 1/20 = 3/20$$

Per cycle they complete 3 work out of 20 work

In 6 cycle they complete 18 work out of 20

Remaining 2/20 work is next done by boys so

$$\text{Days} = (2/20)/(1/10) = 1 \text{ days}$$

$$\text{Total days} = (6 \times 2) + 1 = 13 \text{ days}$$

96) Answer - E

Quantity I:

$$\text{Efficiency of a man} = a$$

$$\text{Efficiency of a women} = b$$

$$X \times a \times 12 = 1.6X \times b \times 15$$

$$a = 2b$$

$$a/b = 2/1$$

$$5 \text{ men} + 5 \text{ women} = 15 \text{ women complete works in 8 days}$$

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So, total work be 120 units

X men complete 120 units in 12 days

Efficiency of man = 2

$$X \times 2 \times 12 = 120$$

$$X = 5$$

Time taken by (X+5) men complete the same work = $120 / ((5+5) \times 2) = 6$ days

Quantity II:

A completes in 16 days.

Let Total work is 160 units

A completes in one day is 10 units.

B completes in one day is 25% more than A = 12.5 units completes in one day

B completes in 8 days = $12.5 \times 8 = 100$

Remaining work = $160 - 100 = 60$

Remaining work is completed by A in = $60 / 10 = 6$ days.

97) Answer - B

Quantity I:

LCM of 32, 48 & 24 = 96

Efficiency

A = 3 B = 3 C = 4

A worked for only 4 days so work completed = $3 \times 4 = 12$ work

B worked for only 6 days so work completed = $6 \times 2 = 12$ work

Remaining is done by C = $96 - 12 - 12 = 72$

Then the share of C = $6840 \times (72/96) = 5130$

Quantity II:

Let the efficiency of B is 10.

The efficiency of A and C is 9 and 8 respectively.

A worked for 6 days and work done by him = $6 \times 9 = 54$

B and C worked for 2 days and the work done by them is 20 and 16 respectively.

Now, the wage of B = $\{22500 / (54 + 20 + 16)\} \times 20 = \text{Rs. } 5000$

98) Answer - B

Quantity I:

The efficiency of the children is half of the efficiency of the Men.

If the efficiency of the children is x then the efficiency of men is 2x.

Let the efficiency of Women be y.

12 men, 8 women, and 12 children can finish a work in 20 days

Total work done = $[(12 \times 2x) + (8 \times y) + (12 \times x)] \times 20$

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10 men, 12 women, and 24 children can complete the same work in 15 days.

$$\text{Total work done} = [(10 * 2x) + (12 * y) + (24 * x)] * 15$$

$$(36x + 8y) * 20 = (44x + 12y) * 15$$

$$144x + 32y = 132x + 36y$$

$$12x = 4y$$

$$x : y = 1 : 3$$

So the ratio of the efficiency of Men, Women, and Children = 2 : 3 : 1

$$\text{Total work done} = (12 * 2 + 8 * 3 + 12 * 1) * 20 = 60 * 20 = 1200 \text{ units}$$

$$\text{Required time} = 1200 / 48 = 25 \text{ days}$$

Quantity II:

A and B together can complete the whole work = 60/7 days

B and C together can complete the whole work = 120/11 days

$$\text{Total work} = 120$$

The efficiency of A and B = 14 units/day

The efficiency of B and C = 11 units/day

The efficiency of C = x units/day

$$14 * 5 + x * 10 = 120$$

$$70 + 10x = 120$$

$$10x = 50$$

$$x = 5$$

The efficiency of C = 5 units/day

C alone can complete the whole work = 120/5 = 24 days

Quantity III:

$$40 * M * 20 = 30 * W * 15$$

$$16M = 9W$$

$$M/W = 9/16$$

$$\text{The total work} = 40 * 9 * 20 = 7200$$

$$\text{The efficiency of children} = 16 * 5 / 4 = 20$$

$$(10 * 9 + 15 * 16 + 6 * 20) * d = 7200$$

$$(90 + 240 + 120) * d = 7200$$

$$450 * d = 7200$$

$$d = 16 \text{ days}$$

99) Answer - D

Quantity I:

Let the total amount of work be the LCM of 20, 40 and 30 = 120 units.

So the efficiency of A, B, and C is 6, 3, and 4 respectively

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They all worked for 2 days. The work done by them in 2 days = $(6 + 3 + 4) * 2 = 26$ units

A and C worked 3 days. The work done by them in 3 days = $(6 + 4) * 3 = 30$ units

The left over work = $120 - (30 + 26) = 64$ units

64 units of work are done by C and D together in 4 days. The efficiency of C and D working together = $64 / 4 = 16$

The efficiency of D = $(16 - 4) = 12$

So the required time = $120 / 12 = 10$ days

Quantity II:

A alone complete the whole work = $45/2$ days

B alone can complete the whole work = 18 days

C alone can complete the whole work = $15/2$ days

Total work = 90 units

The efficiency of A = 4 units/day

The efficiency of B = 5 units/day

The efficiency of C = 12 units/day

$$4*x + 5*6 + 12*2 = 90$$

$$4x + 30 + 24 = 90$$

$$4x = 90 - 54$$

$$4x = 36$$

$$x = 9$$

Quantity III:

The time taken by A to complete the work = 30 days

The efficiency of A is 33.33% less than the efficiency of B

The efficiency ratio of A to B = 2: 3

The time ratio of A to B = 3: 2

Time taken by B = $30/3 * 2 = 20$ days

The efficiency of C is 16.66% less than the efficiency of A

The efficiency ratio of C to A = 5: 6

The time ratio of C to A = 6: 5

Time taken by C to complete the work = $30 * (6/5) = 36$ days

Total work = LCM of (30, 20, 36) = 180 units

Efficiency of A = $180/30 = 6$ units/day

Efficiency of B = $180/20 = 9$ units/day

Efficiency of C = $180/36 = 5$ units/day

Work done B in 4 days = $4 * 9 = 36$ units

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Time taken by A and C to complete the remaining work = $(180 - 36)/(6 + 5)$
= $(13 \frac{1}{11})$ days

100) Answer - D

Quantity I:

It is given that the rate of doing the work by A, B and C is equal, this means that their efficiency of doing work is same.

Let A, B and C alone can do 'p' units of work respectively in an hour.

A works for 15 hours for 10 days + B works for 'x' hours for 'x' days + C works for 'y' hours for 'y' days = Total Work (1)

Work done by A for working 10 hours for 15 days = $p * 10 * 15 = 150p$ units
..... (2)

Work done by B for working 'x' hours for 'x' days = $p * x * x = x^2p$ units (3)

Work done by C for working 'y' hours for 'y' days = $p * y * y = y^2p$ units (4)

And A can complete total work in 27.5 days for working 15 hours a day.

Total work = $p * 10 * 27.5 = 275p$ units (5)

Put (2), (3), (4), (5) in (1)

$$150p + x^2p + y^2p = 275p$$

$$x^2p + y^2p = 125p$$

$$x^2 + y^2 = 125 \text{ (6)}$$

Now, $x - y = 5$, put $x = y + 5$ in (6)

$$(y + 5)^2 + y^2 = 125$$

$$y^2 + 25 + 10y + y^2 = 125$$

$$2y^2 + 10y - 100 = 0$$

$$y^2 + 5y - 50 = 0$$

$$y^2 + 10y - 5y - 50 = 0$$

$$y(y + 10) - 5(y + 10) = 0$$

$$(y - 5)(y + 10) = 0$$

$$y = 5 \text{ and } y = -10$$

Neglecting the negative value, we have, $y = 5$

And $x = 10$

Time taken to complete whole work

$$= 275p / (10p + xp + yp)$$

$$= 275p / (10p + 10p + 5p)$$

$$= 275/25$$

$$= 11 \text{ days}$$

Quantity II:

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$$M * D * H = (18 * 6 * 10)$$

If x persons do the same work for 3 hours for 30 days

$$(18 * 6 * 10) = (x * 3 * 30)$$

$$x = (6 * 2)$$

$$x = 12$$

$$\text{Required difference} = 18 - 12 = 6$$

Quantity III:

C worked alone for 15 days (because for the first 2 days, he worked with B), A and B worked together for 4 days and B and C together worked for 2 days.

$$\text{So total work done by A and B in 4 days} = 4 * (1/12) = 1/3$$

$$\text{So total work done by B and C in 2 days} = 2 * (1/18) = 1/9$$

$$\text{So the total work done by C in 15 days} = (1 - 1/3 - 1/9) = 5/9$$

$$\text{So work done by C in 1 day} = (5/9)/15 = 1/27$$

$$\text{Time taken by C alone to complete the whole work} = 27 \text{ days}$$

It is given that B and C together can do the work in 18 days

$$\text{Hence work done by B in one day} = 1/18 - 1/27 = 1/54$$

$$\text{Time taken by B alone to complete the whole work} = 54 \text{ days}$$

Similarly, A and B can finish the work in 12 days

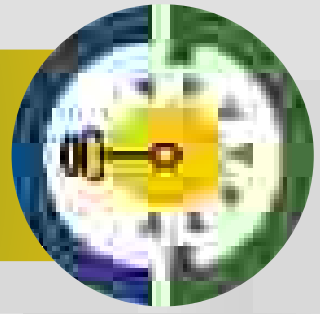
$$\text{So work done by A in one day} = 1/12 - 1/54 = 7/108$$

$$\text{Time taken by A alone to complete the whole work} = 108/7 \text{ days}$$





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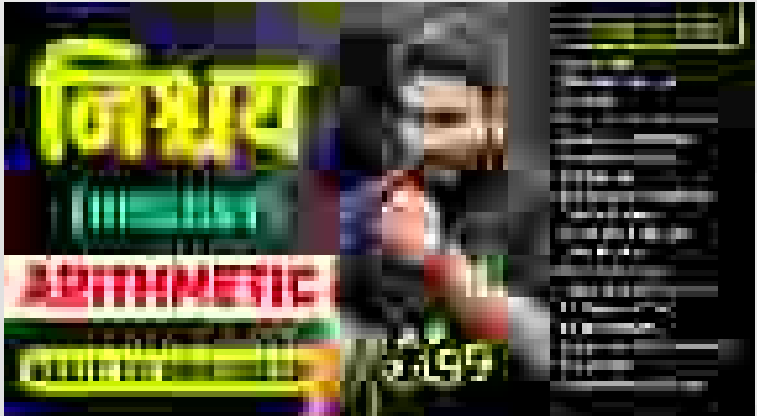


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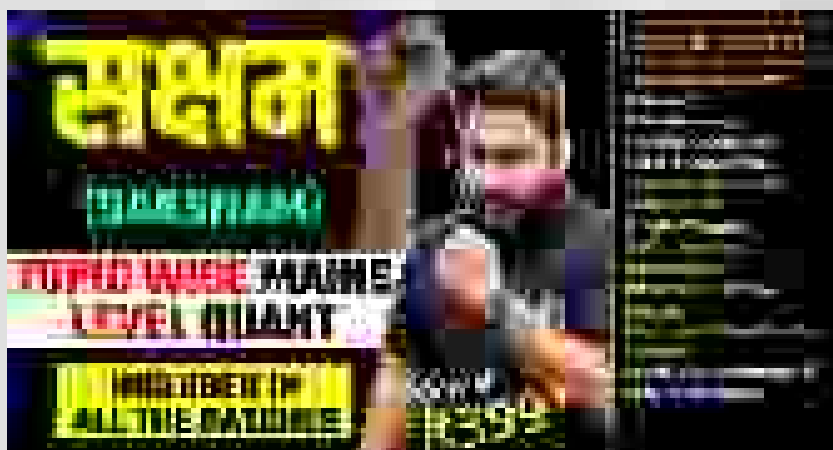
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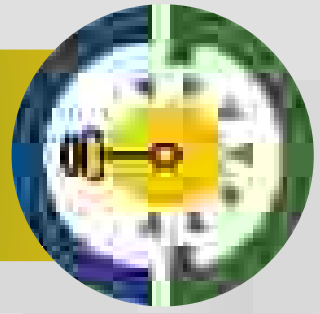
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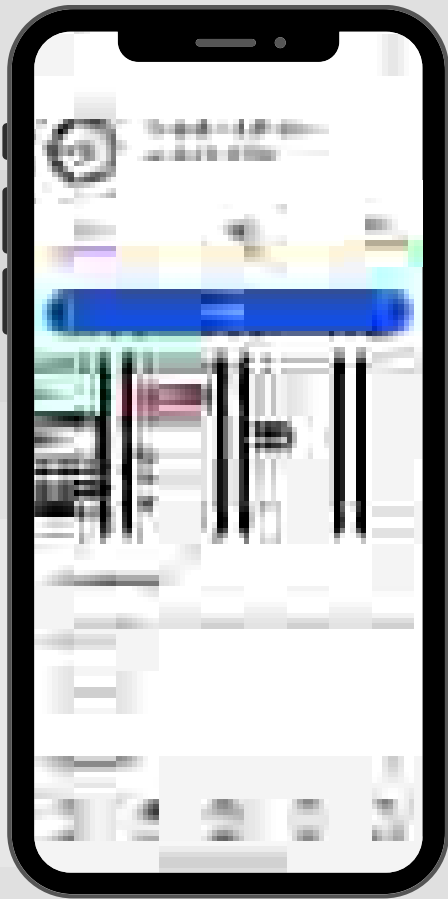


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